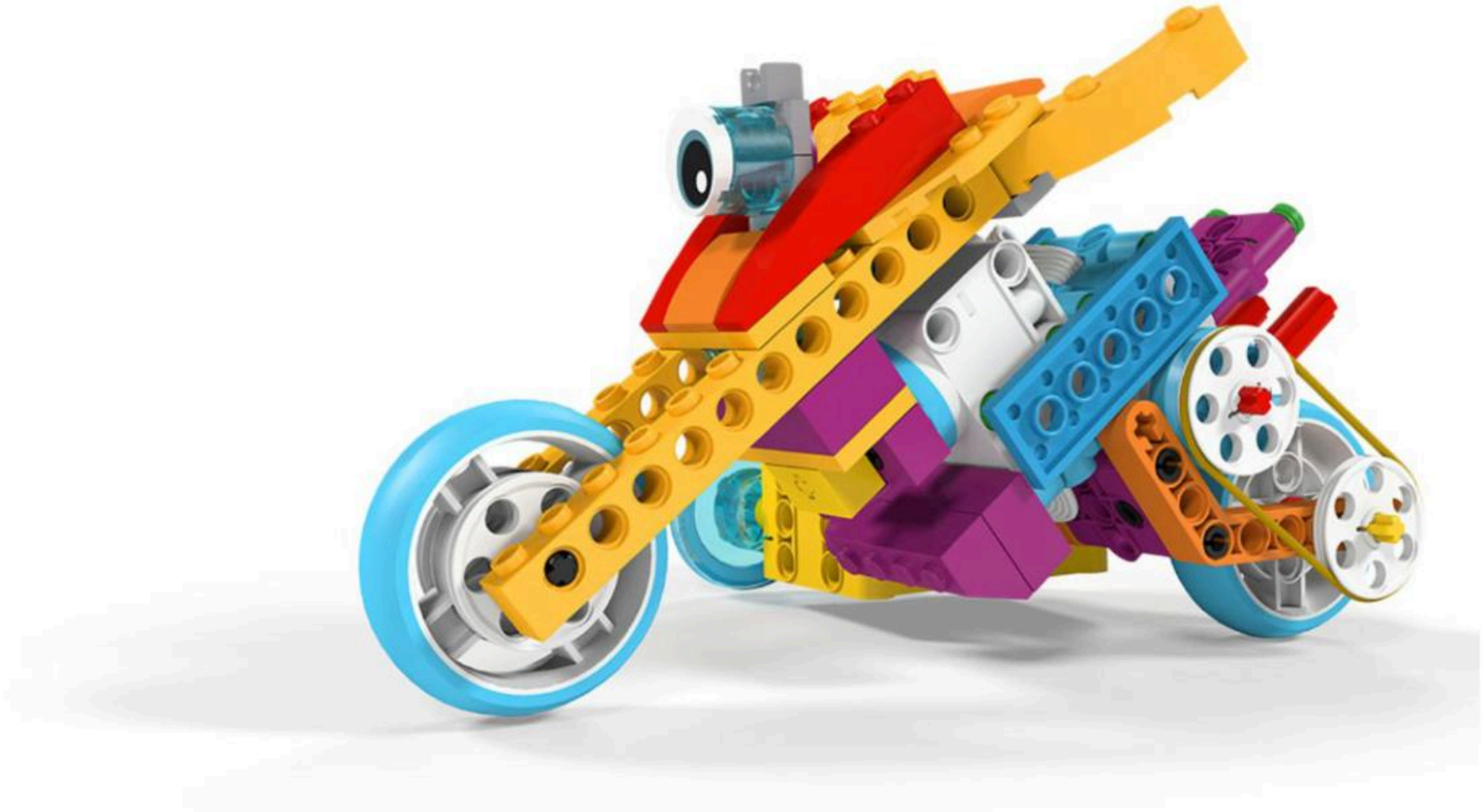


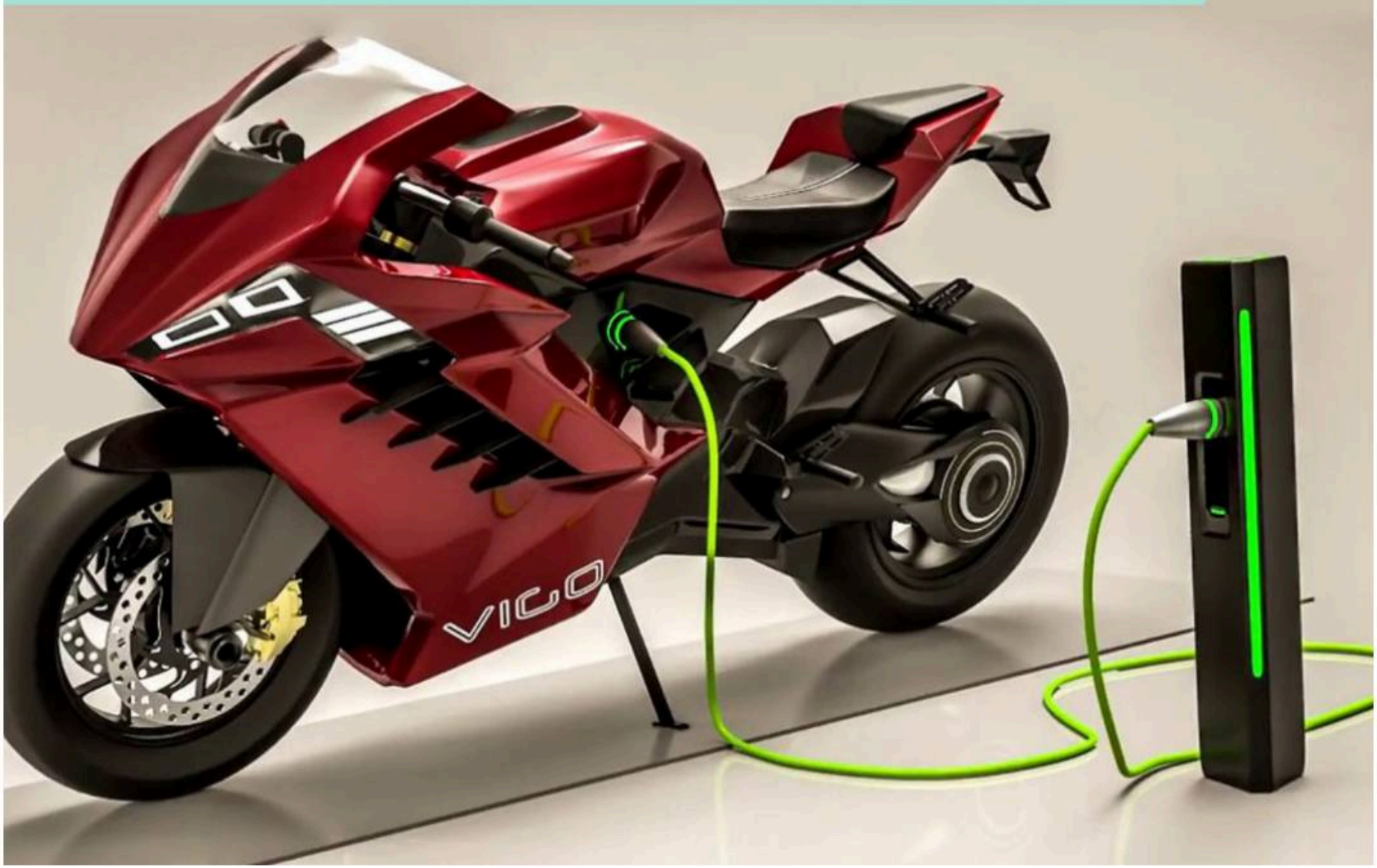
I paid attention to how polluted the air is in cities. Especially those located in the lowlands and not blown by the winds, like Mexico City



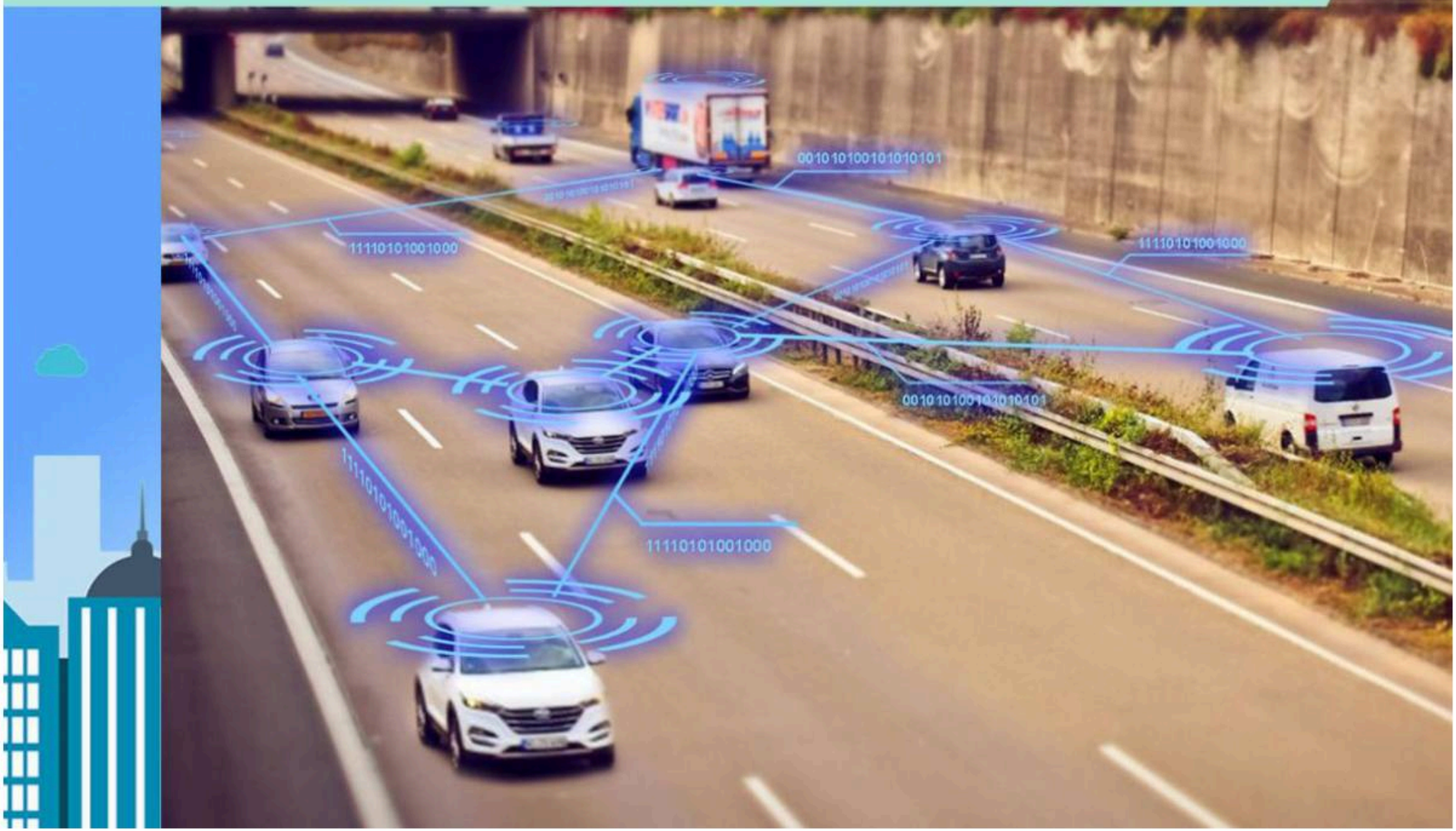
The main thing is not to forget that electricity for powering transport must also be produced without harmful emissions.



Hello! I suggest starting with the development of electric motorcycles. They are very popular in countries with a warm climate, but have a lot of disadvantages.



And by adding the possibility of data exchange between different vehicles, we can create a network in which each car or motorcycle will know exactly the position and direction of everyone around



The robot has two motors which allows it to drive and maneuver. One is a drive motor, and the second is a steering motor.



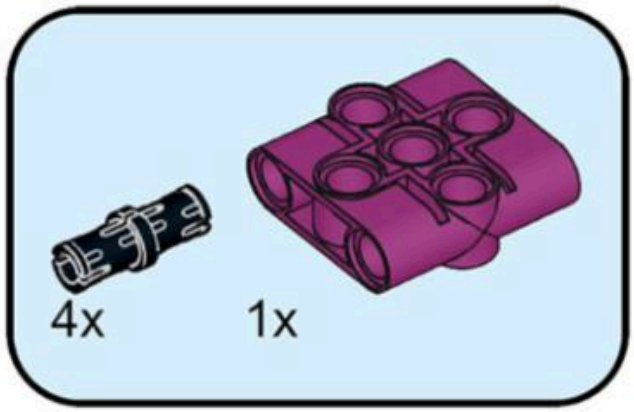
Find:

- Smarthub
- Drive motor
- Steering motor



A belt transmission is used to transfer rotation from the motor to the drive rear wheel. Due to the fact that it can slide through, when the robot is blocked, the motor is not blocked. This protects the motor and the model from damage.

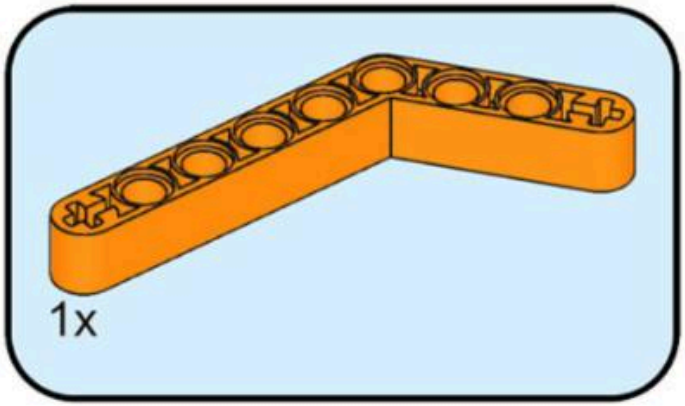




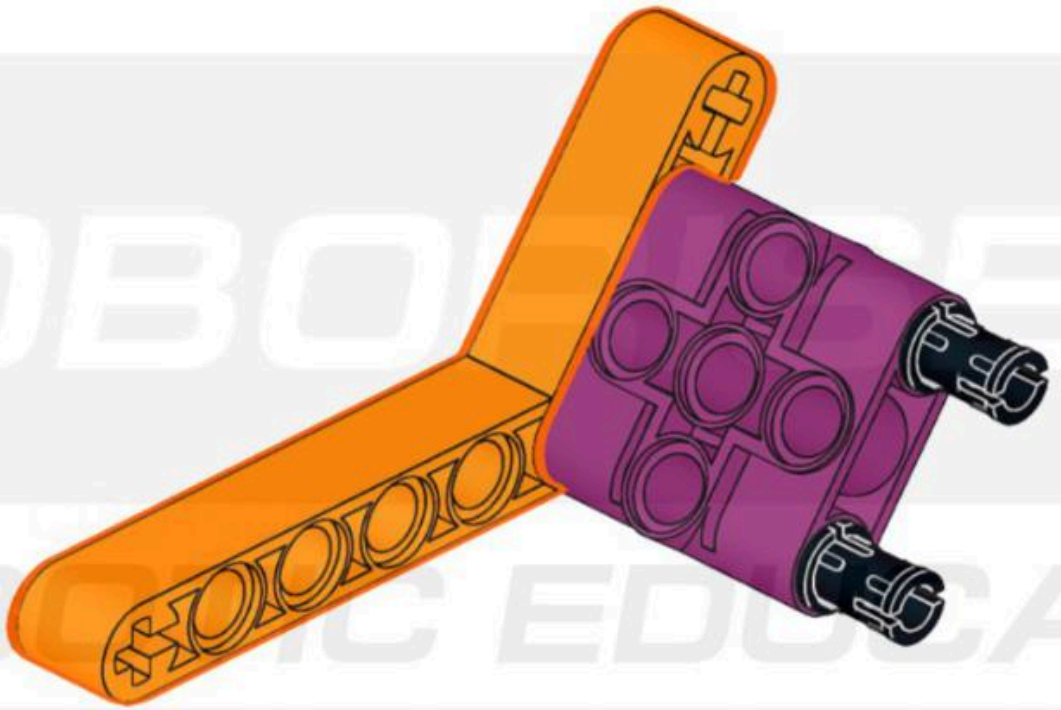
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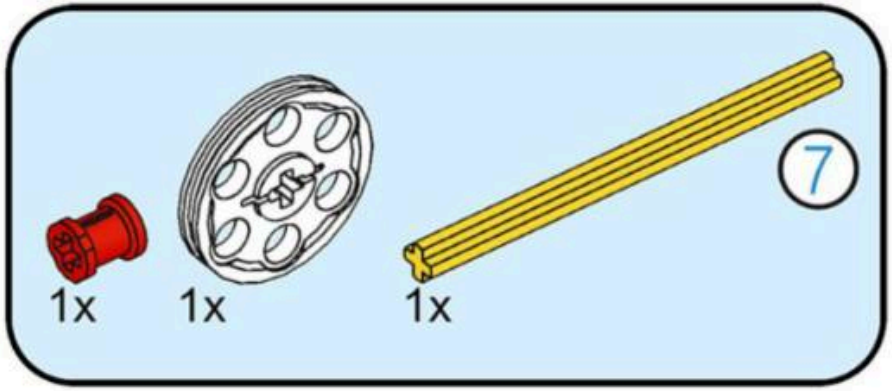
ROBOTIC EDUCA



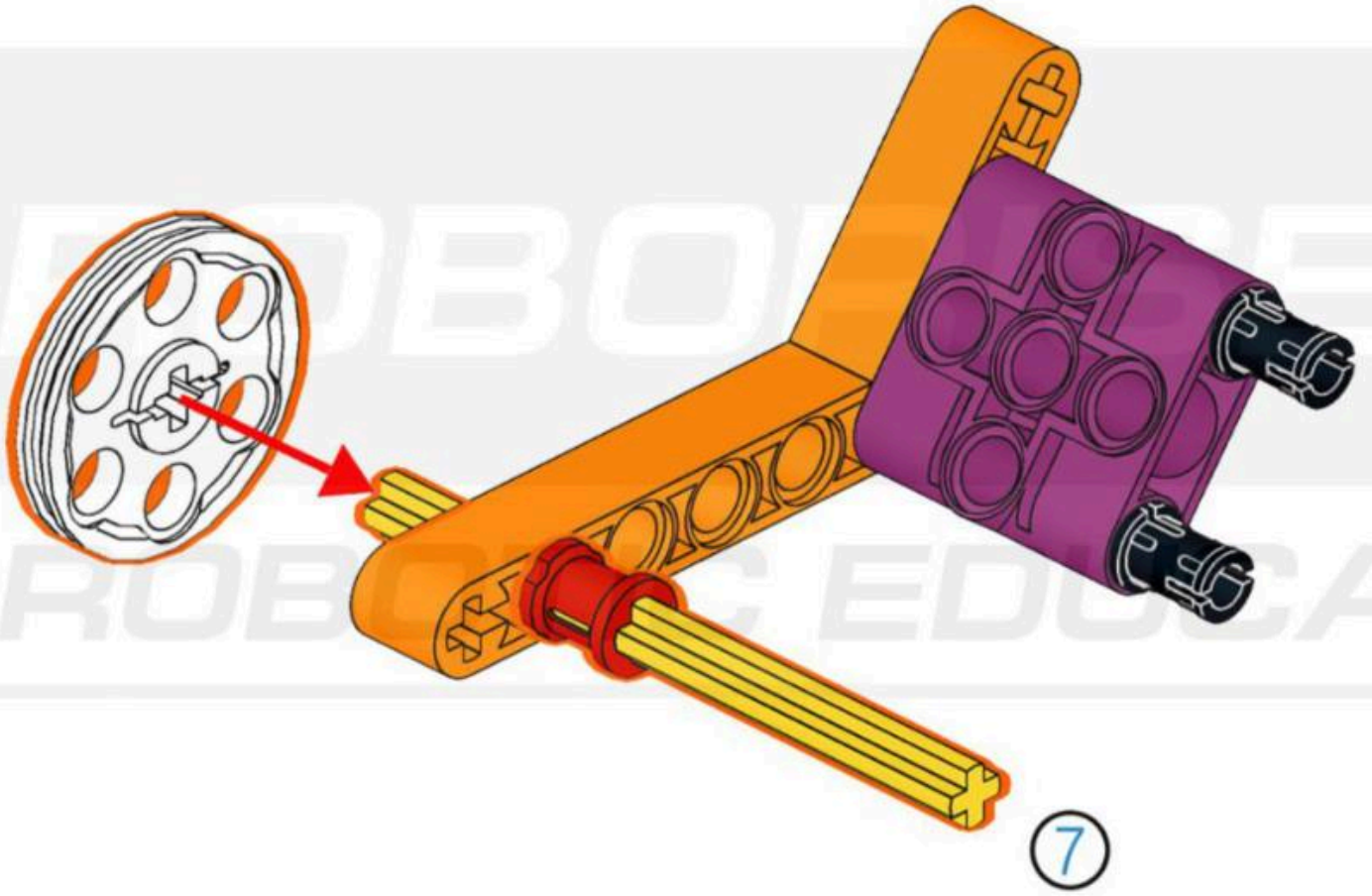
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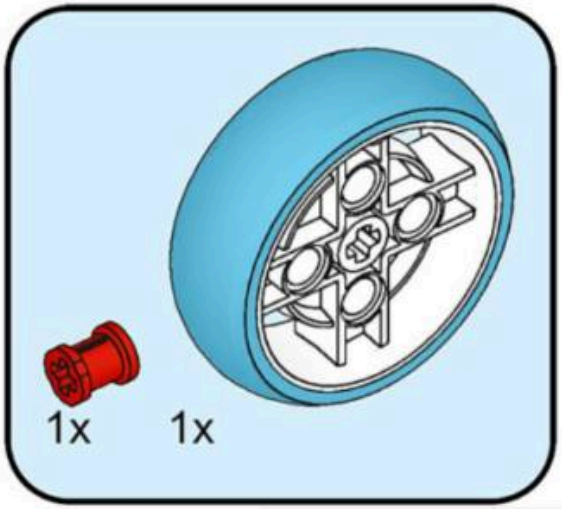


ROBOTICS EDUCAT

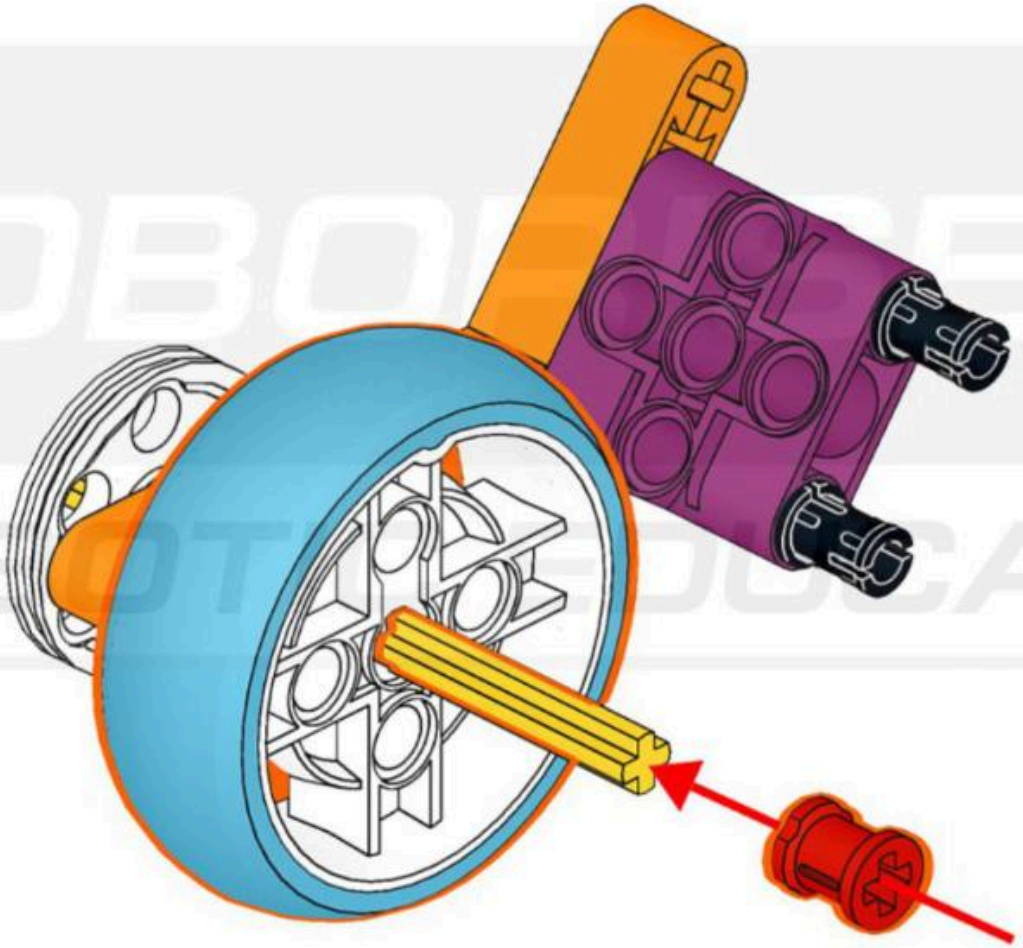


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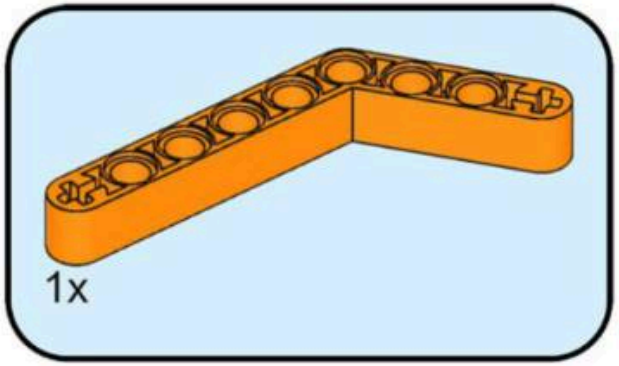




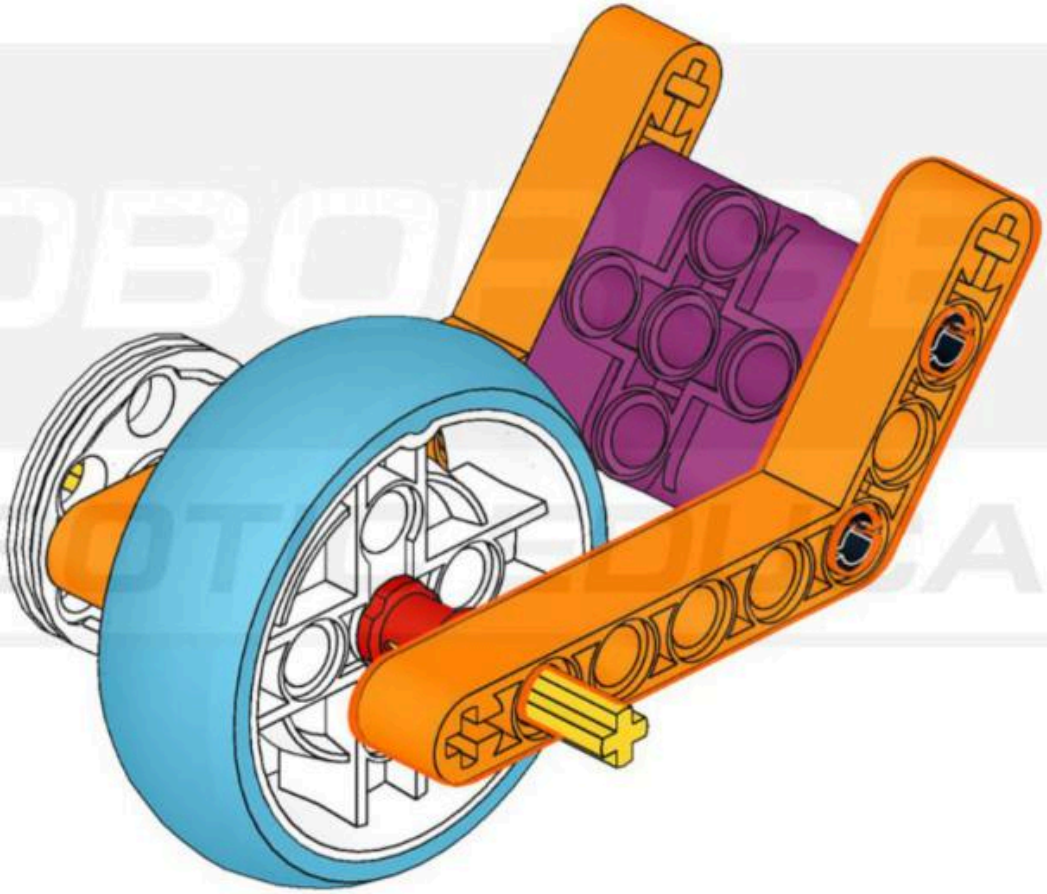
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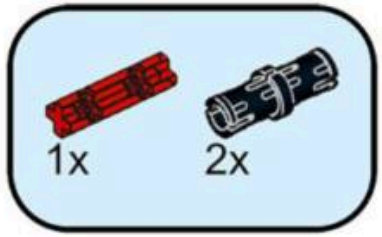


ROBOTSHOP EDUCAT

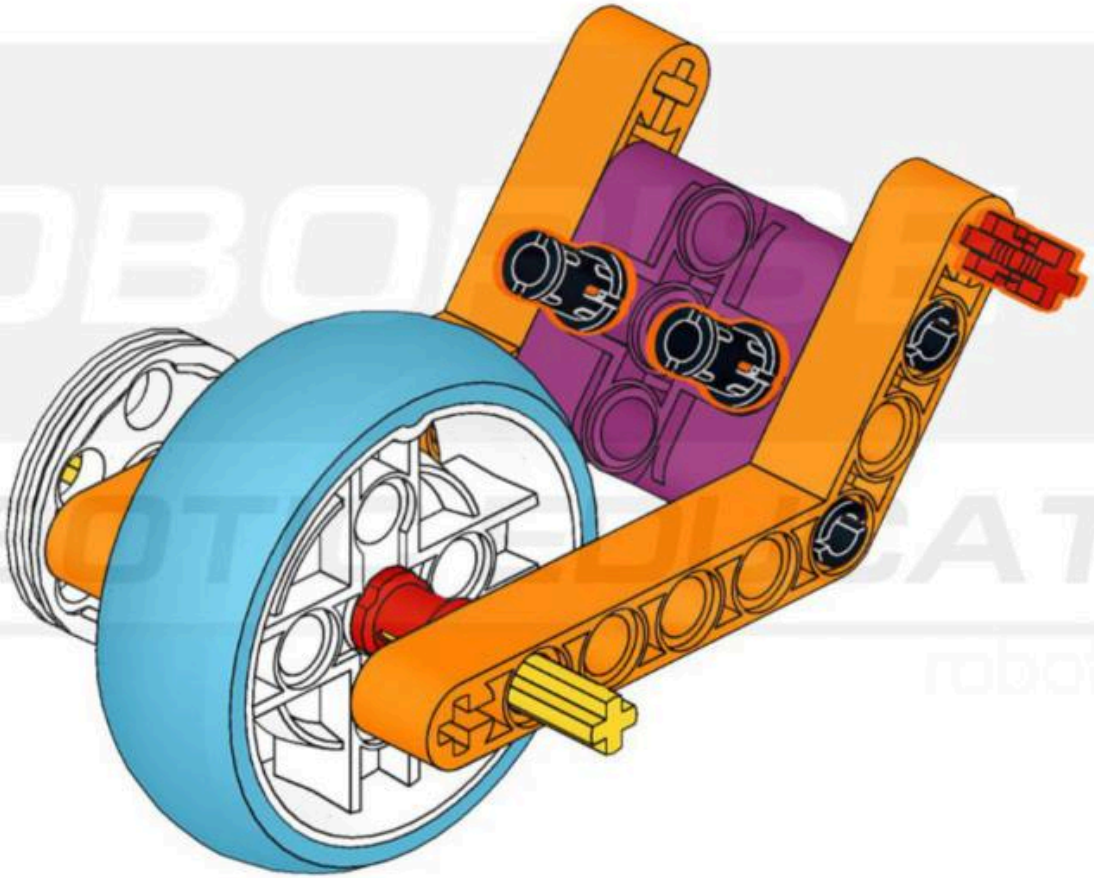


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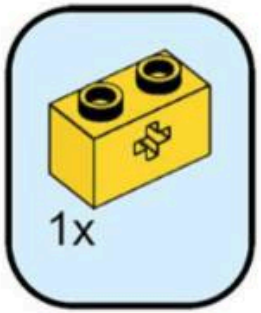




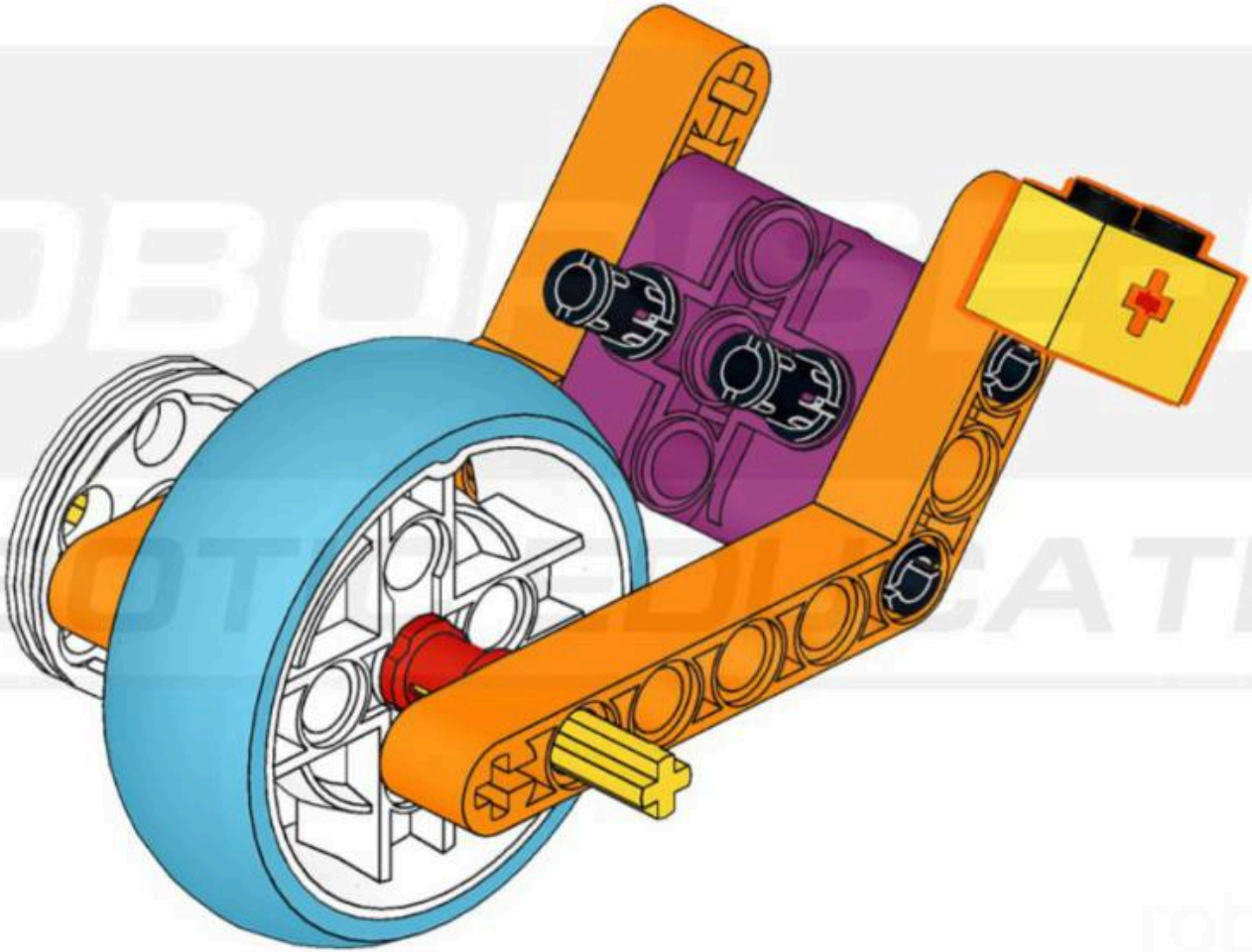
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ROBOTSHOP
ROBOTSHOP EDUCAT
robot

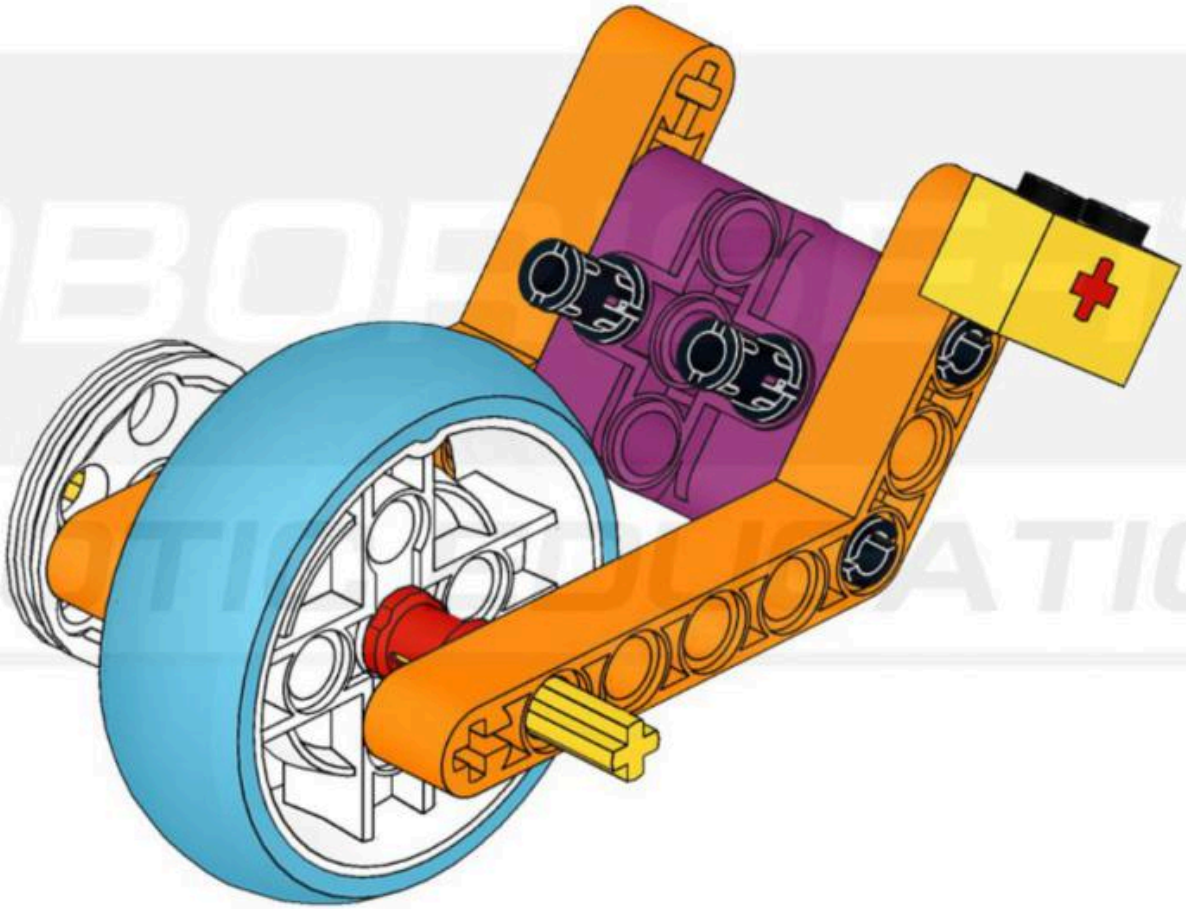


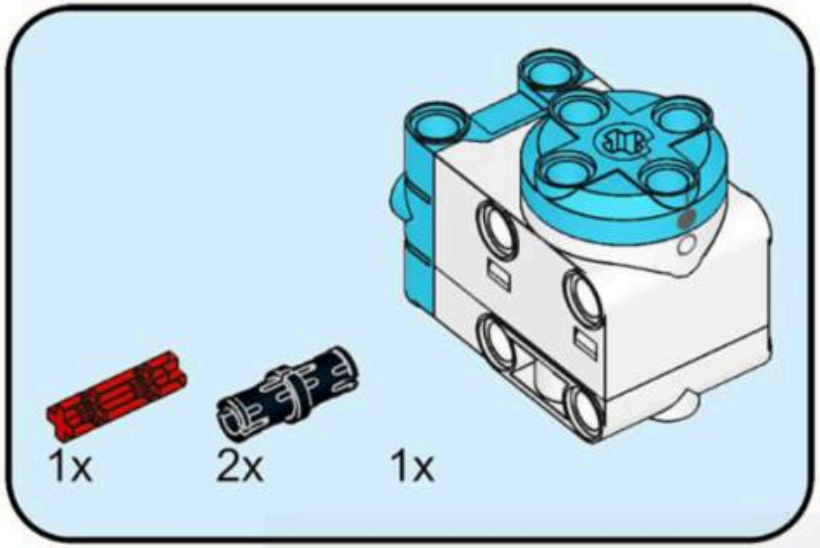
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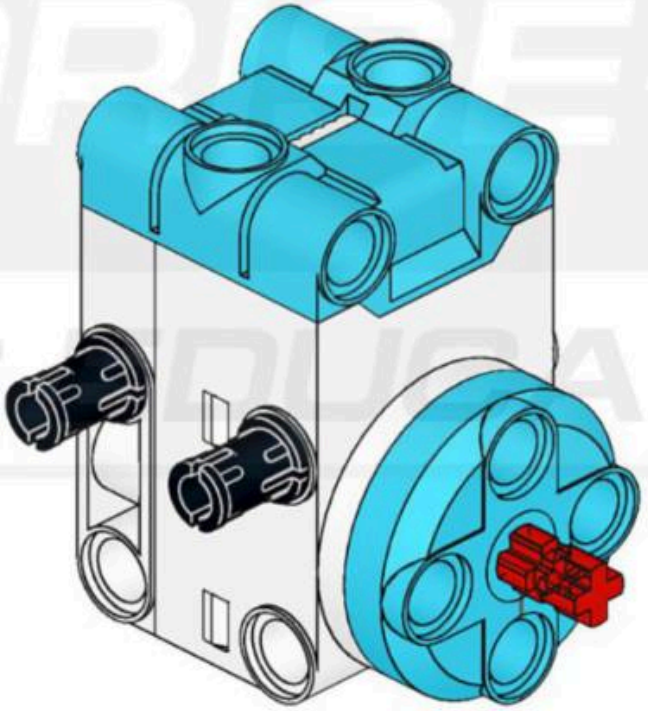
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ROBOTSHOP EDUCATION

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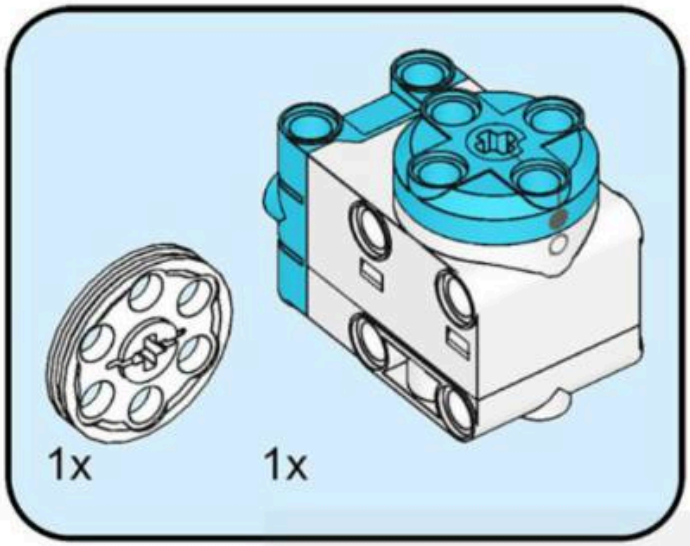




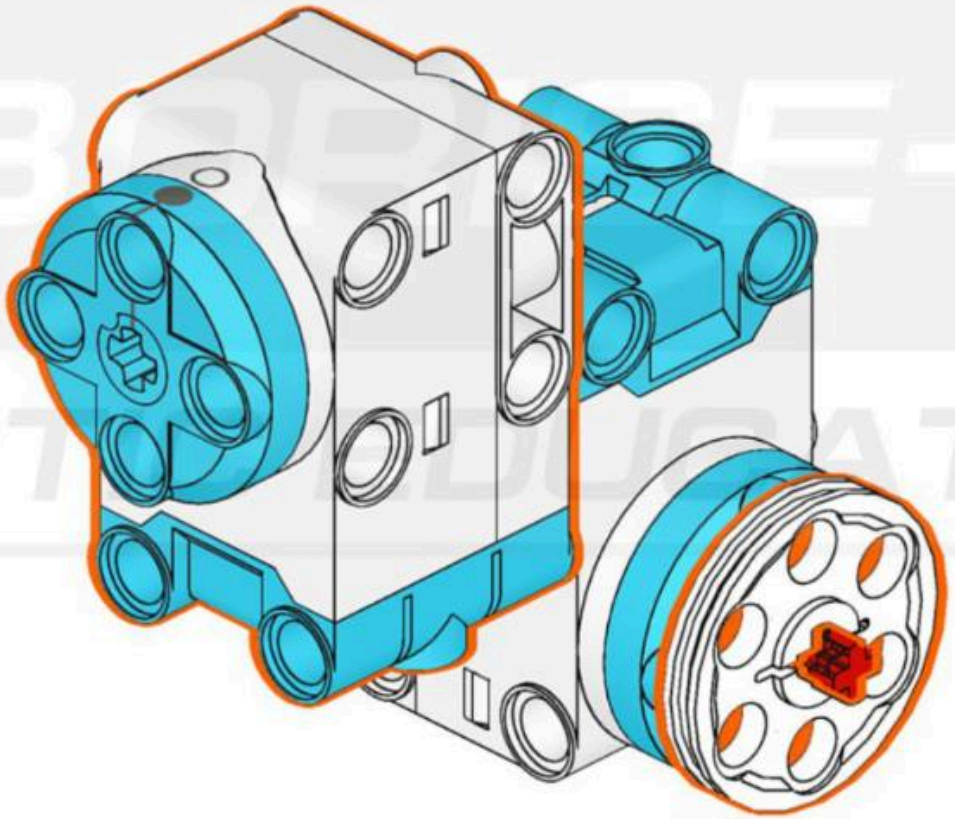
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ROBOTIC INDUSTRIES



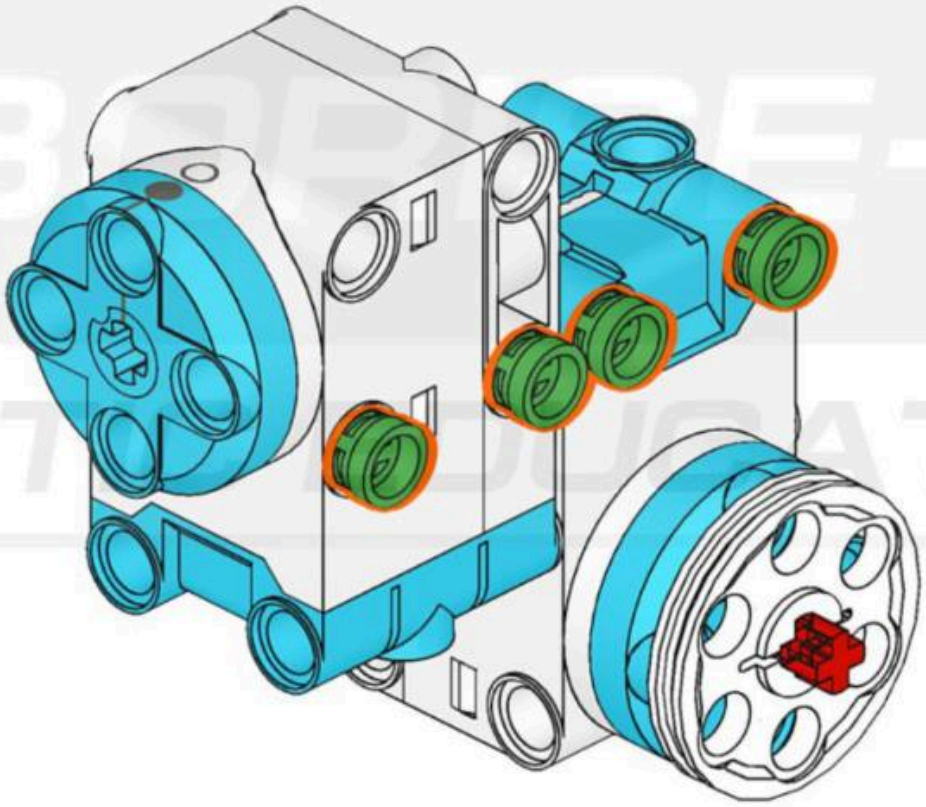
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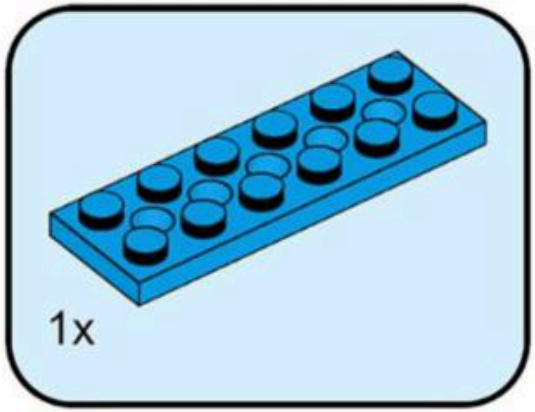
robotics@littlealpha.
ROBOTIC EDUCAT



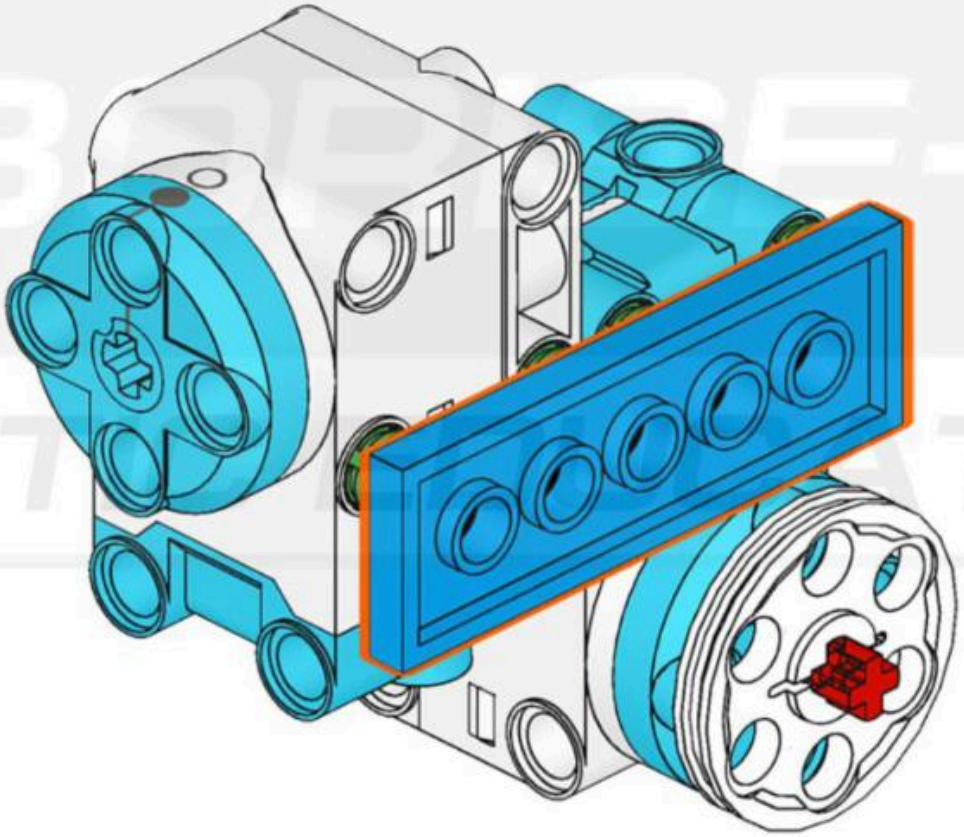
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ROBOTSHOP



12

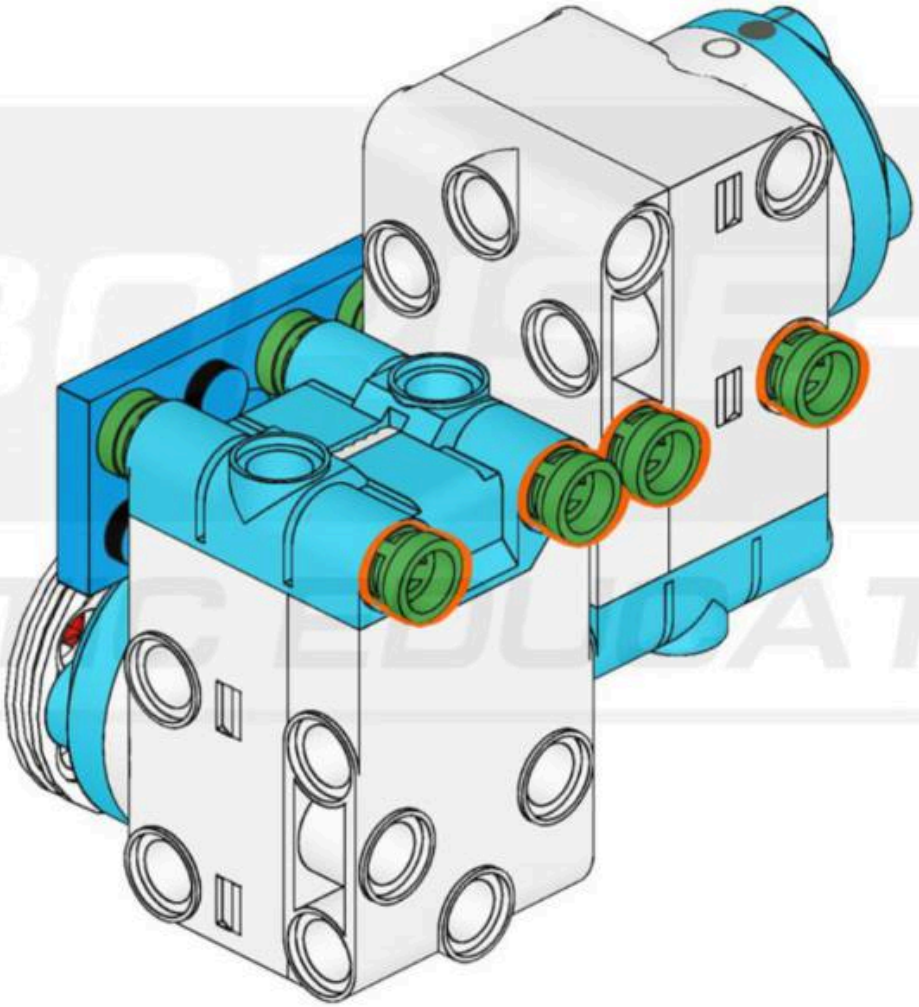


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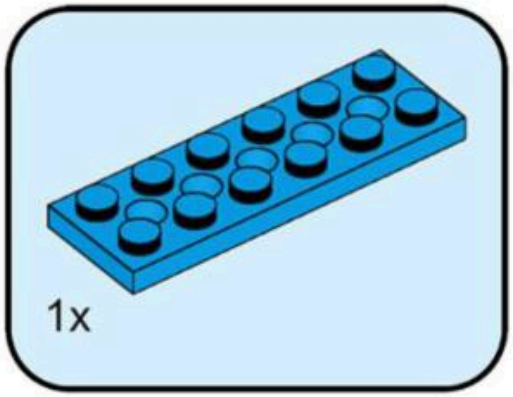


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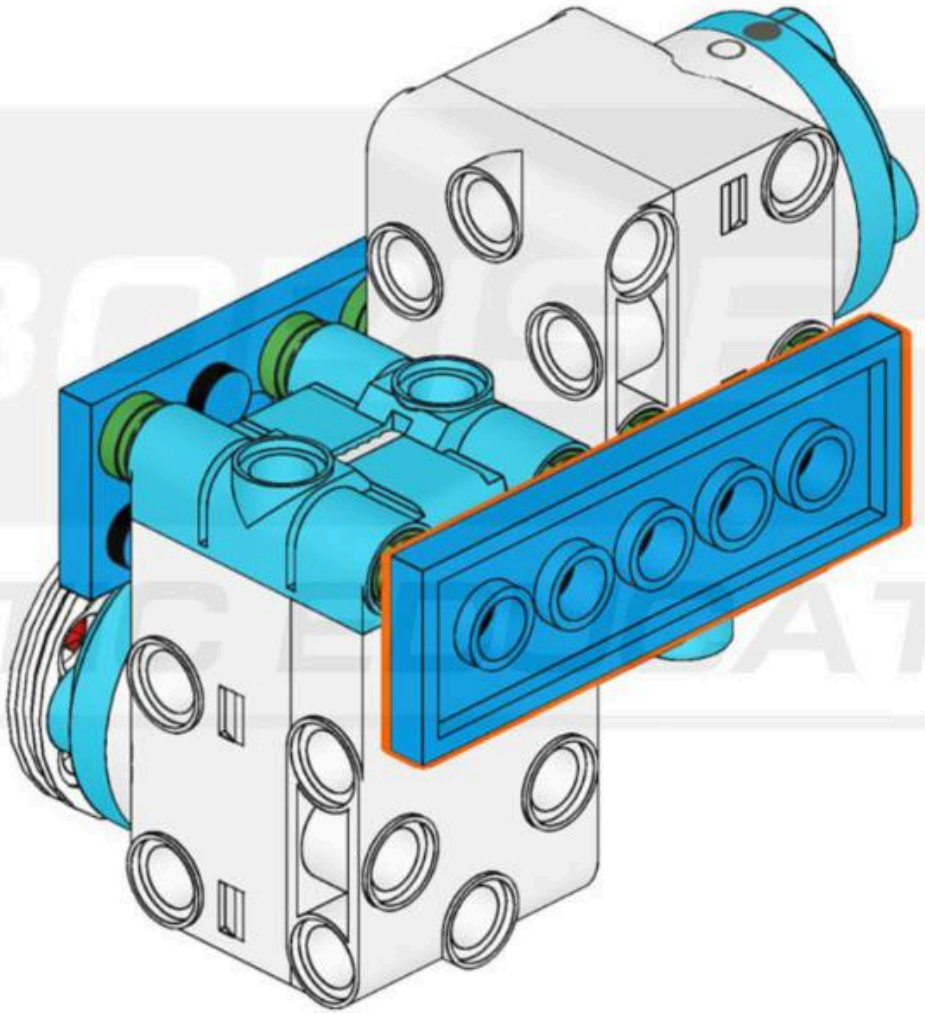




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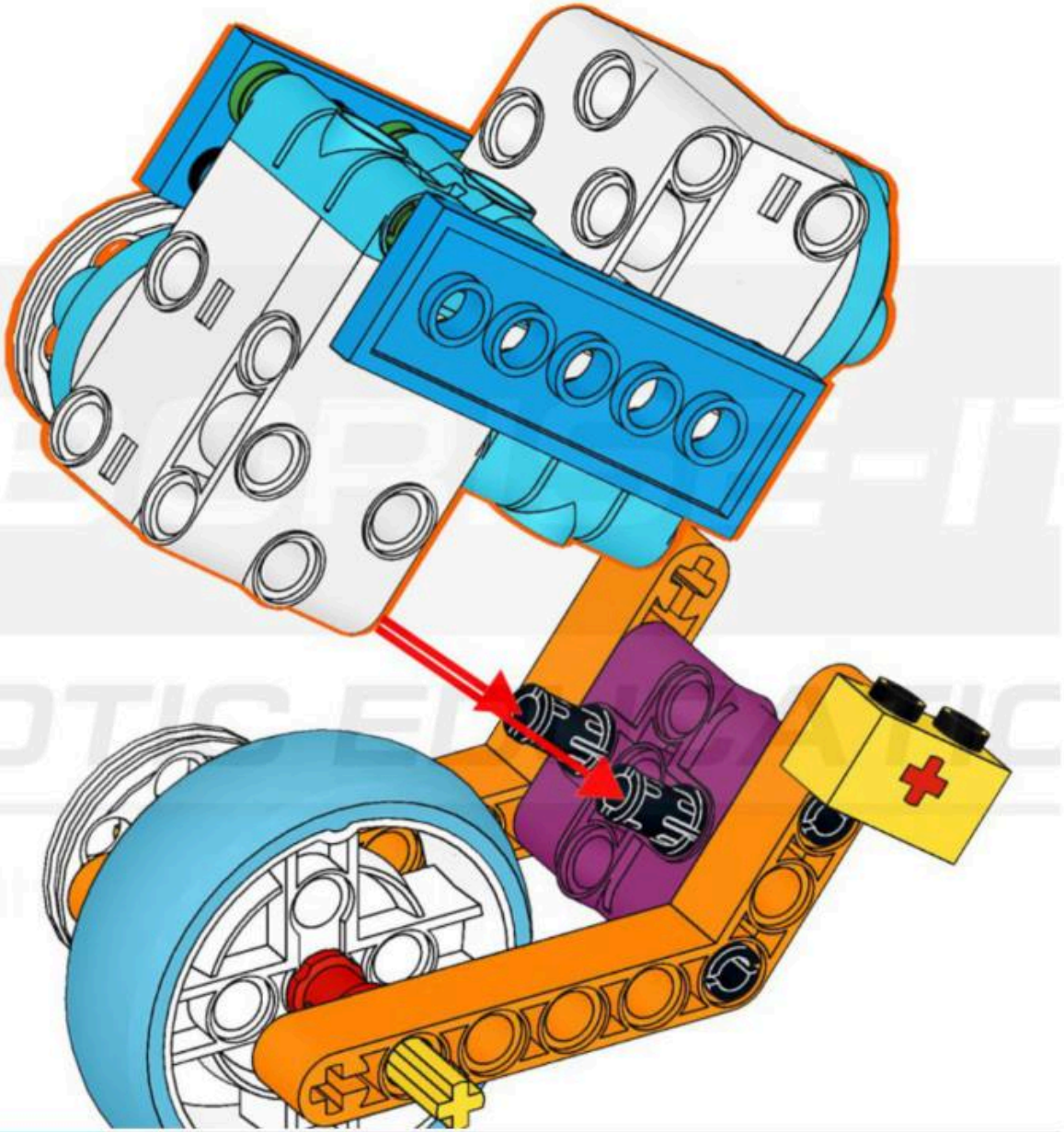


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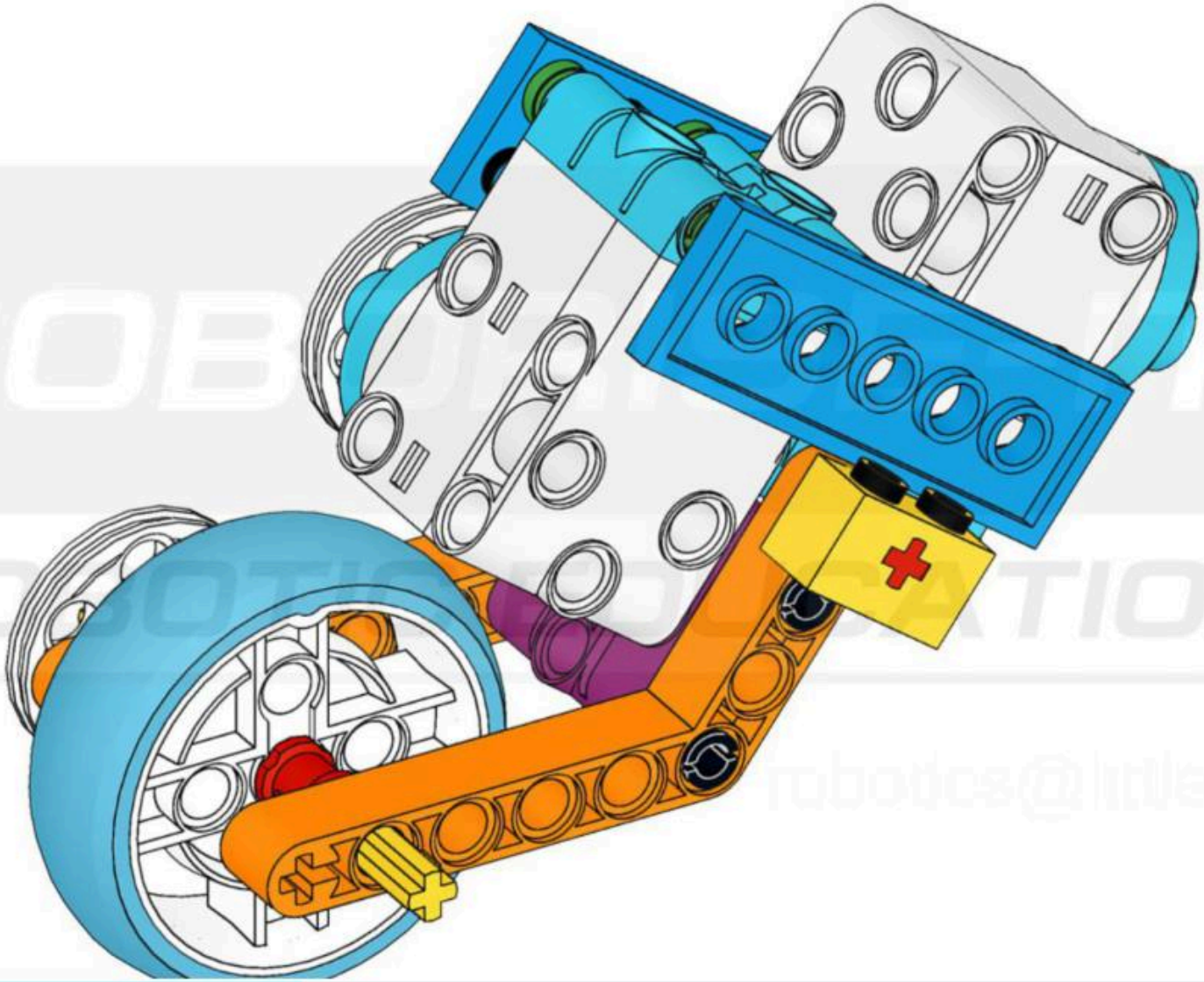


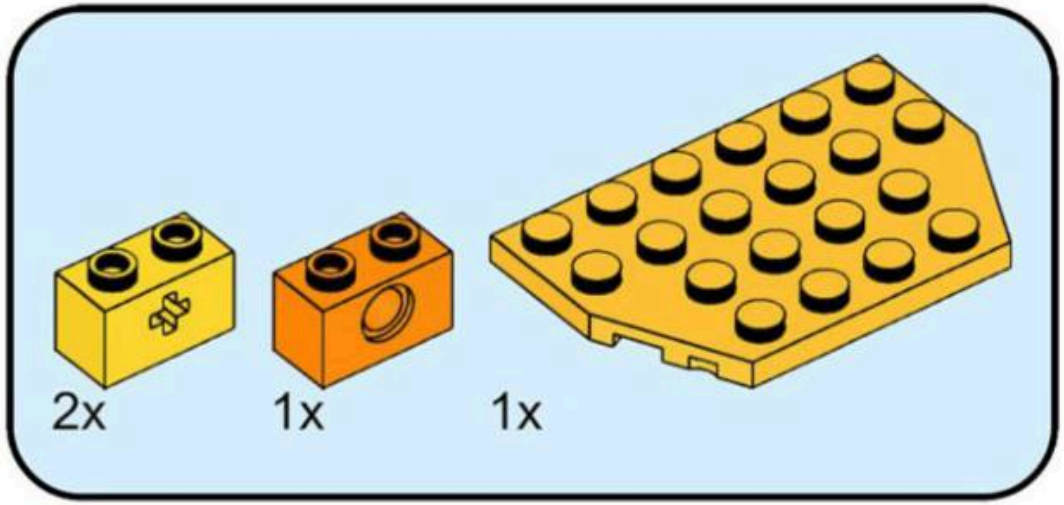
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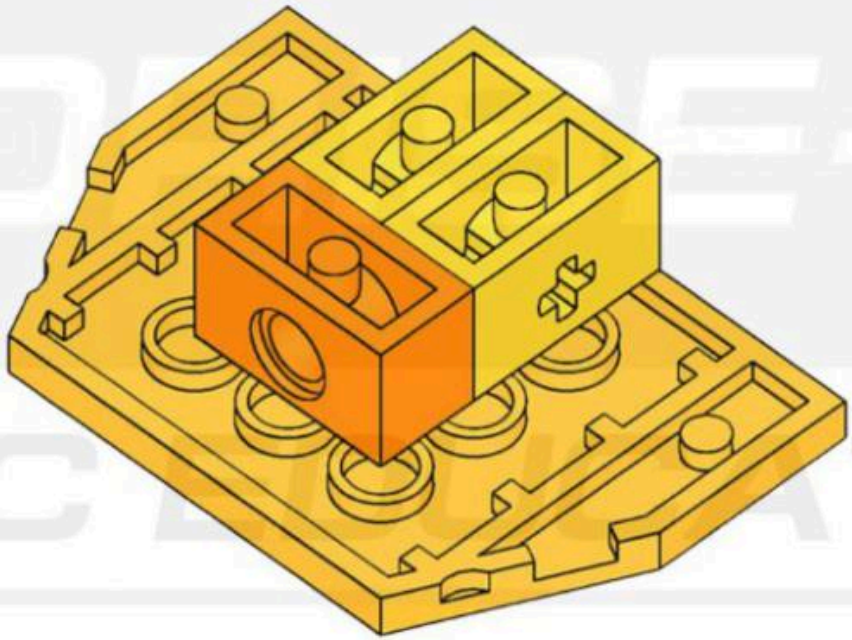


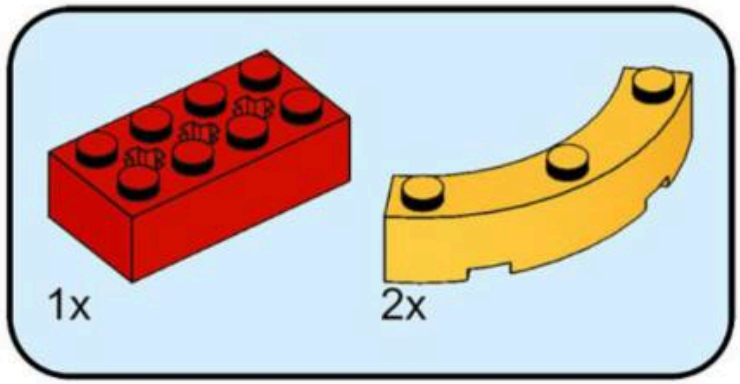
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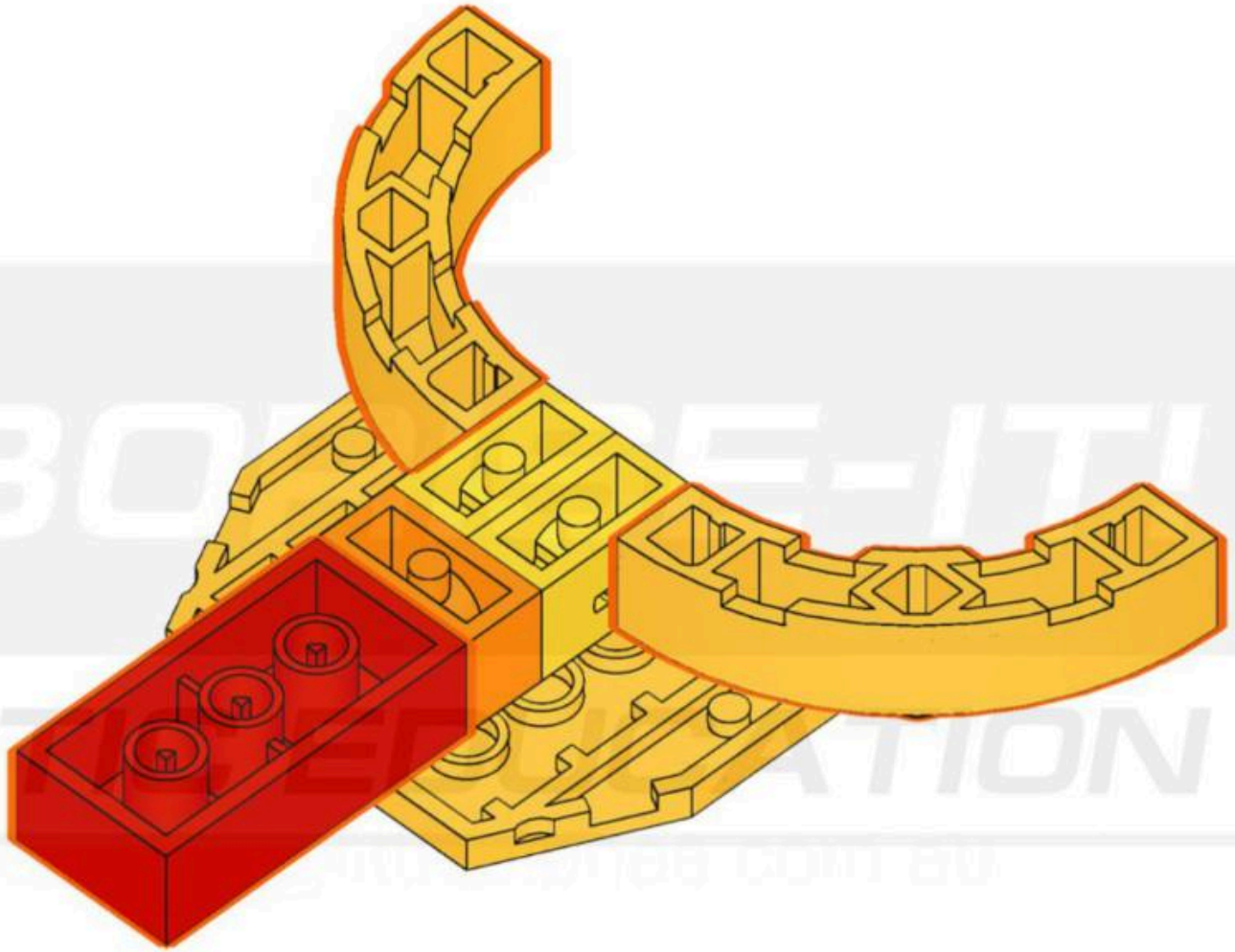


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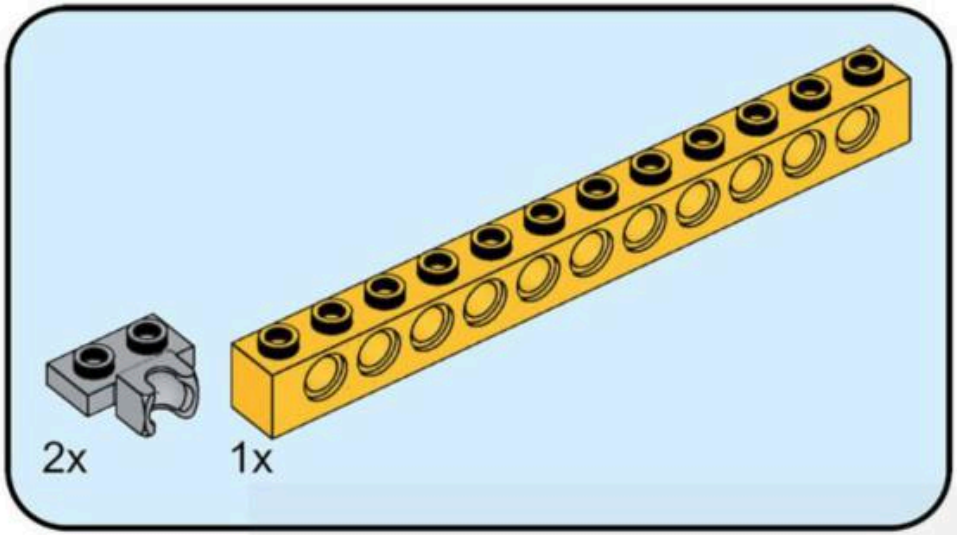




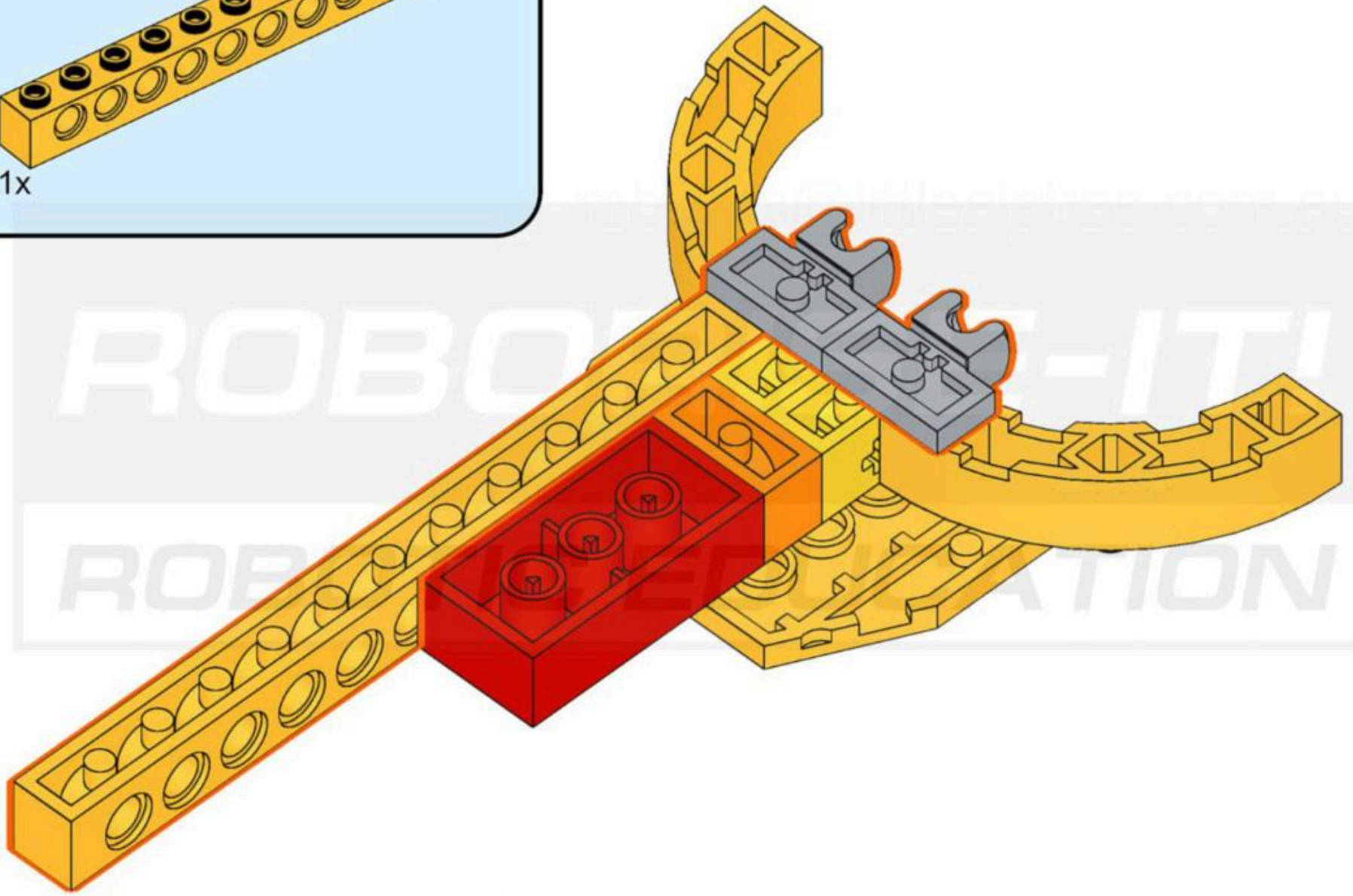
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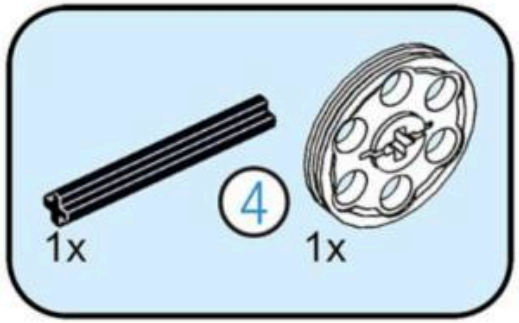


ROBOTSHOP-IT!
ROBOTSHOP EDUCATION

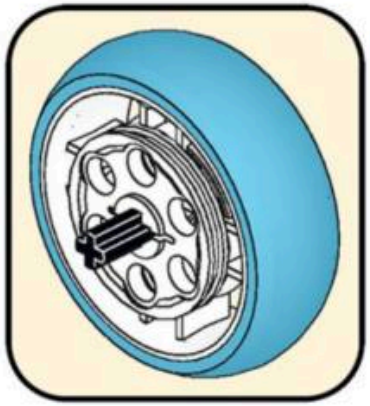


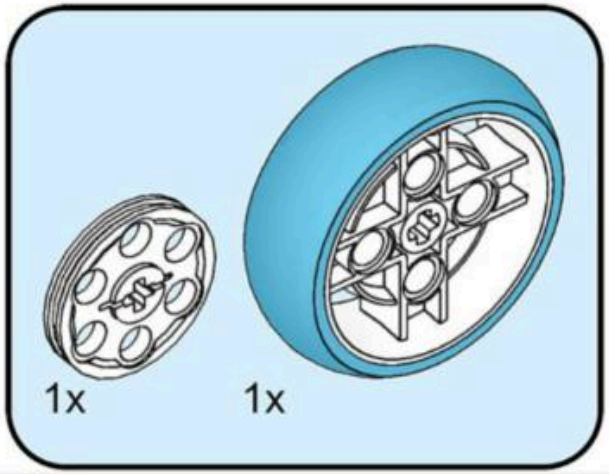
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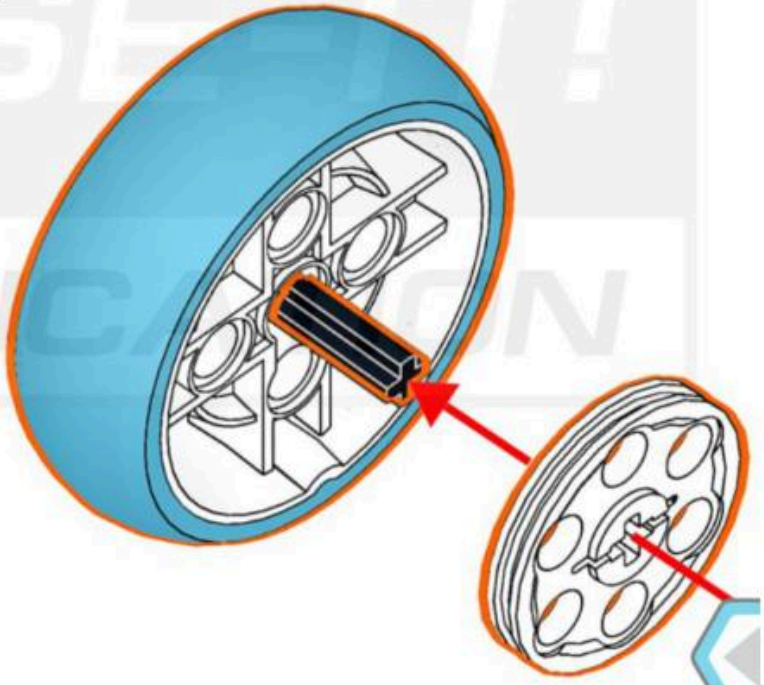
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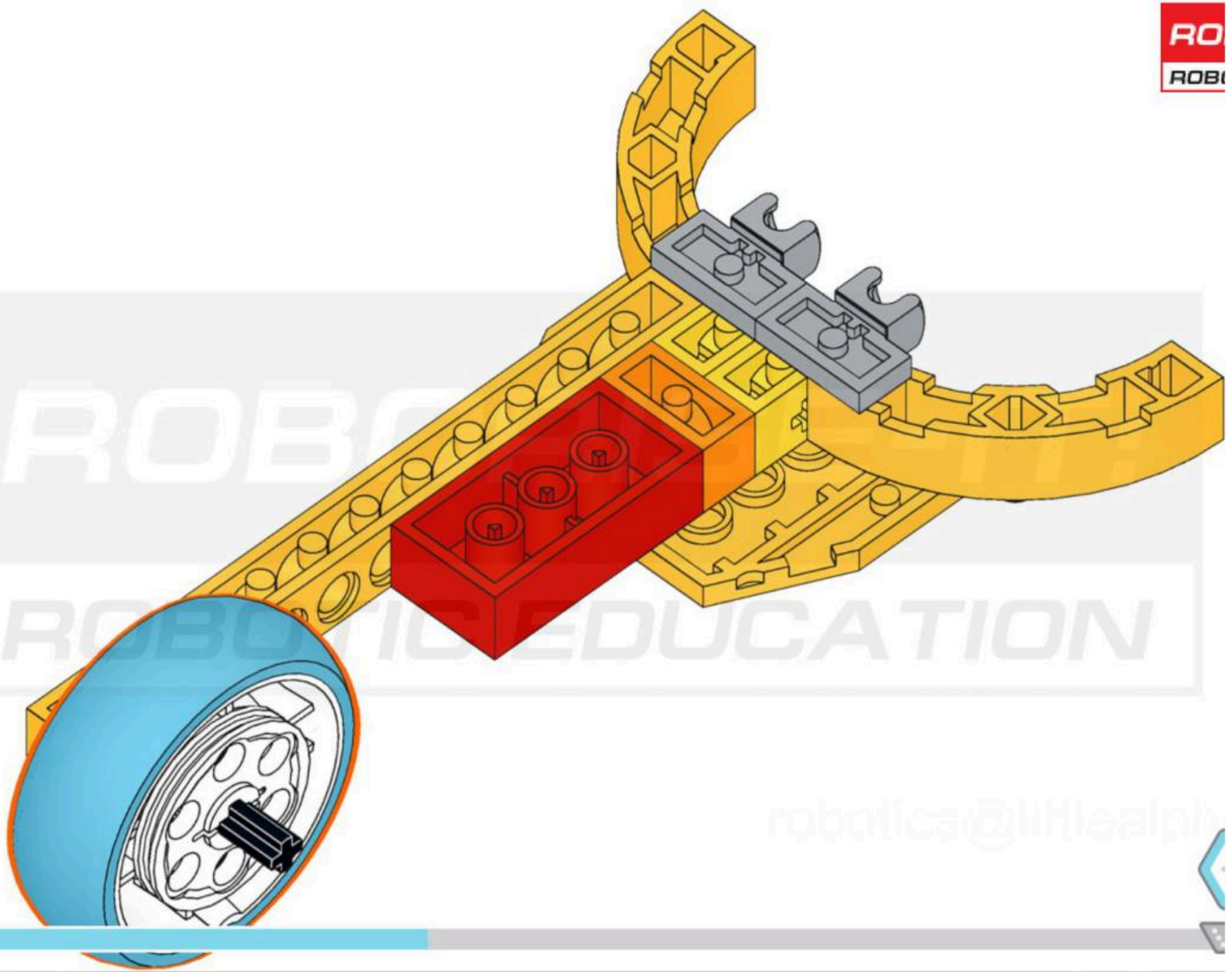
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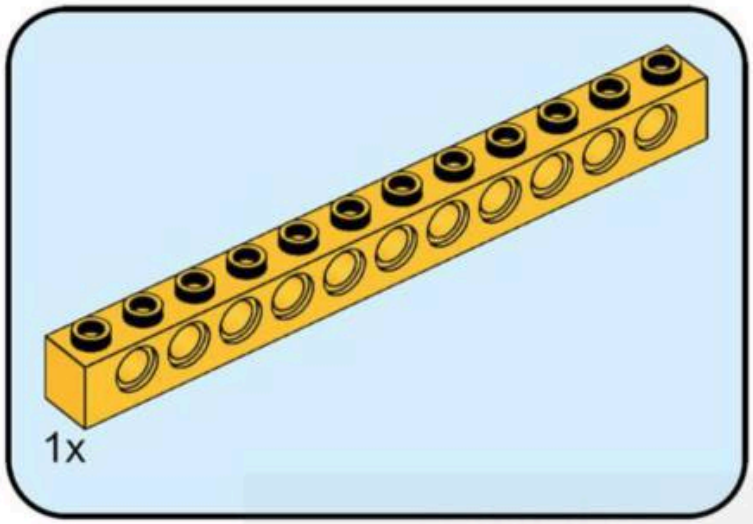




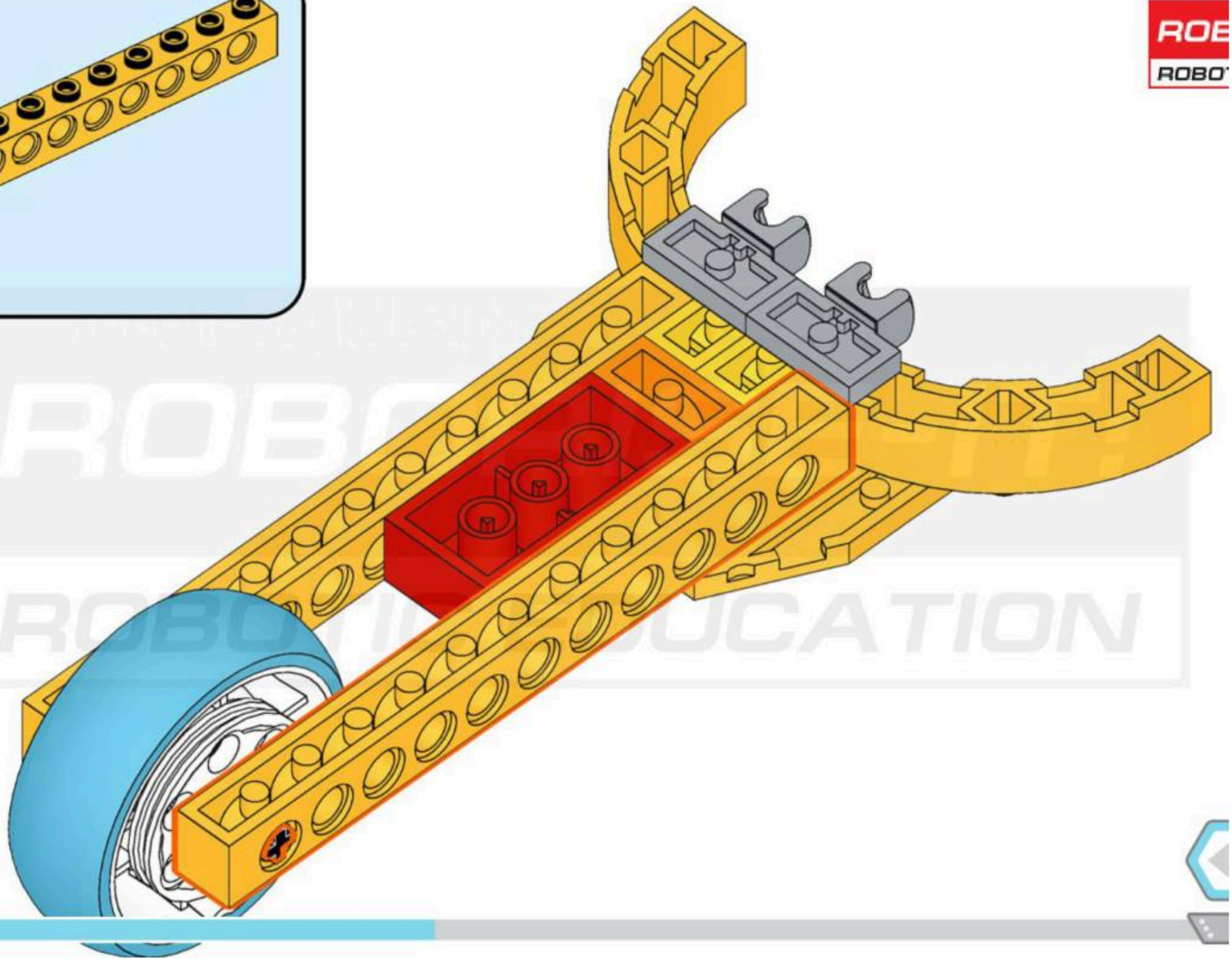
ROBORISE IT!
ROBOTIC EDUCATION

22

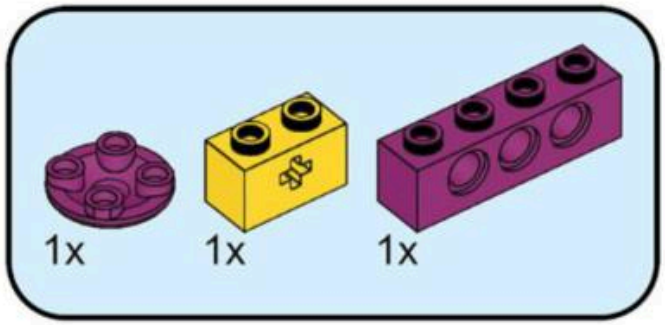


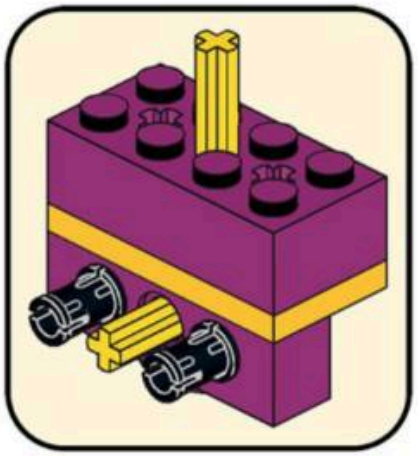


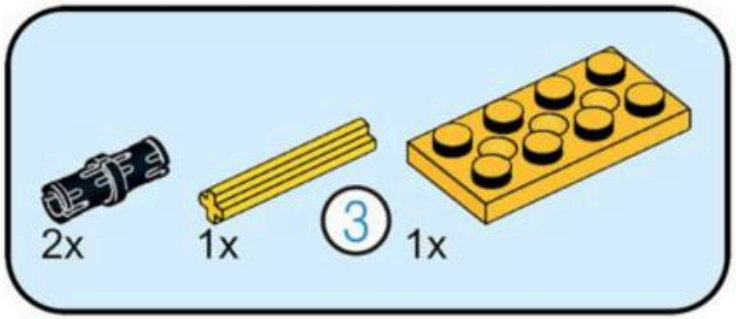
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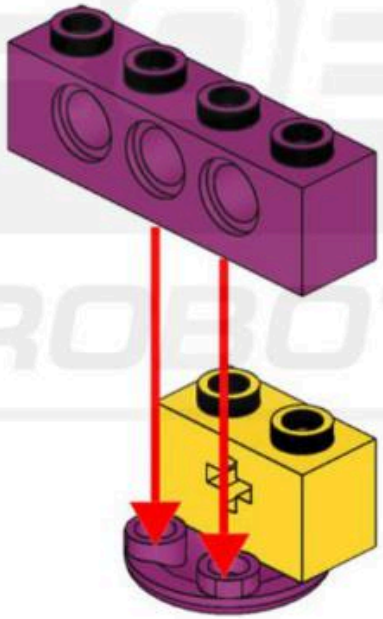
ROBOTIKA EDUCATION



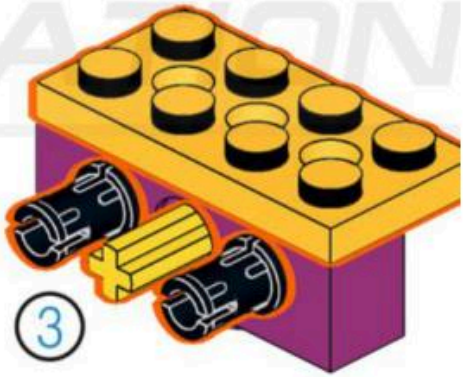




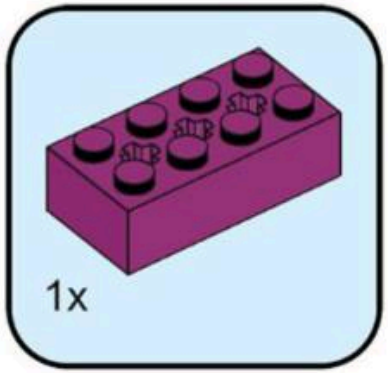
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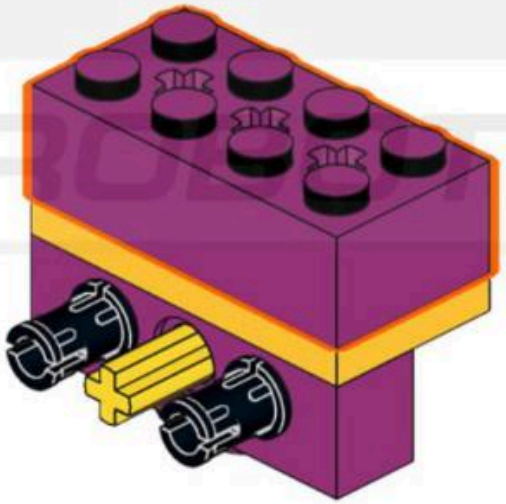
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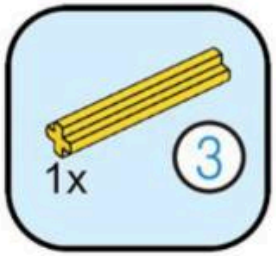


ROBORISE-IT!
ROBOTIC EDUCATION

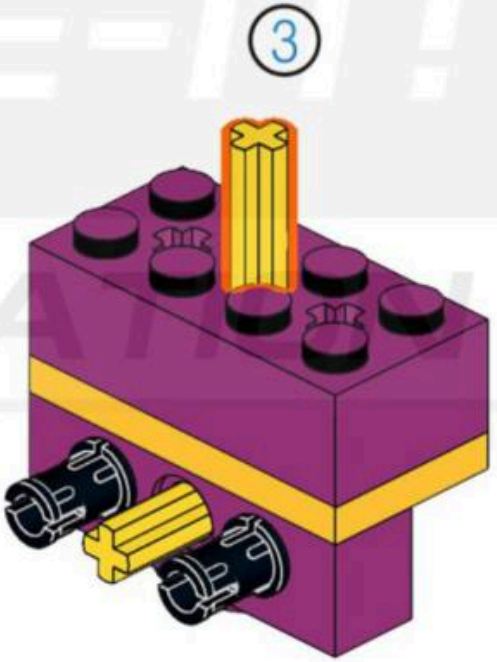


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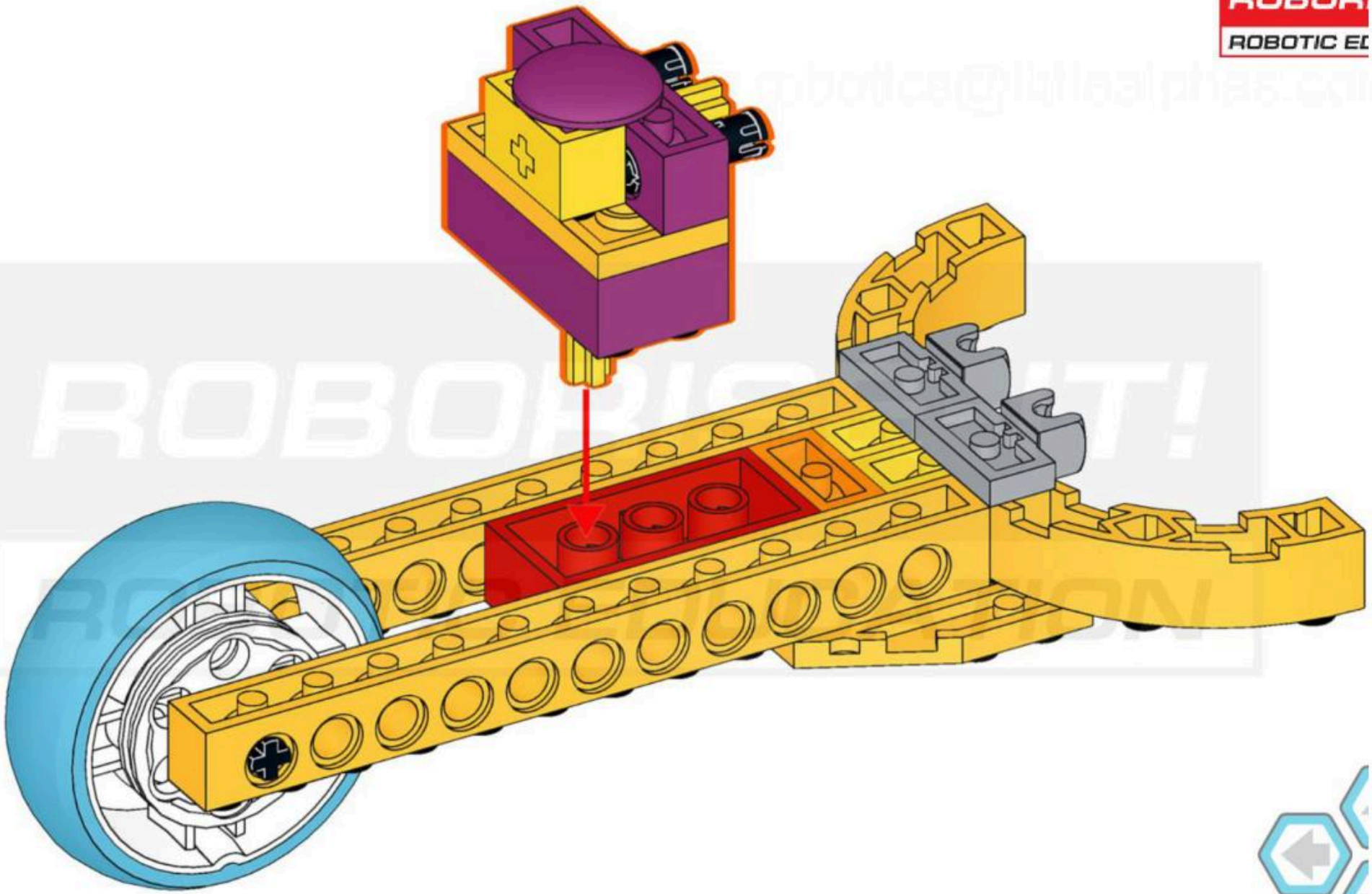


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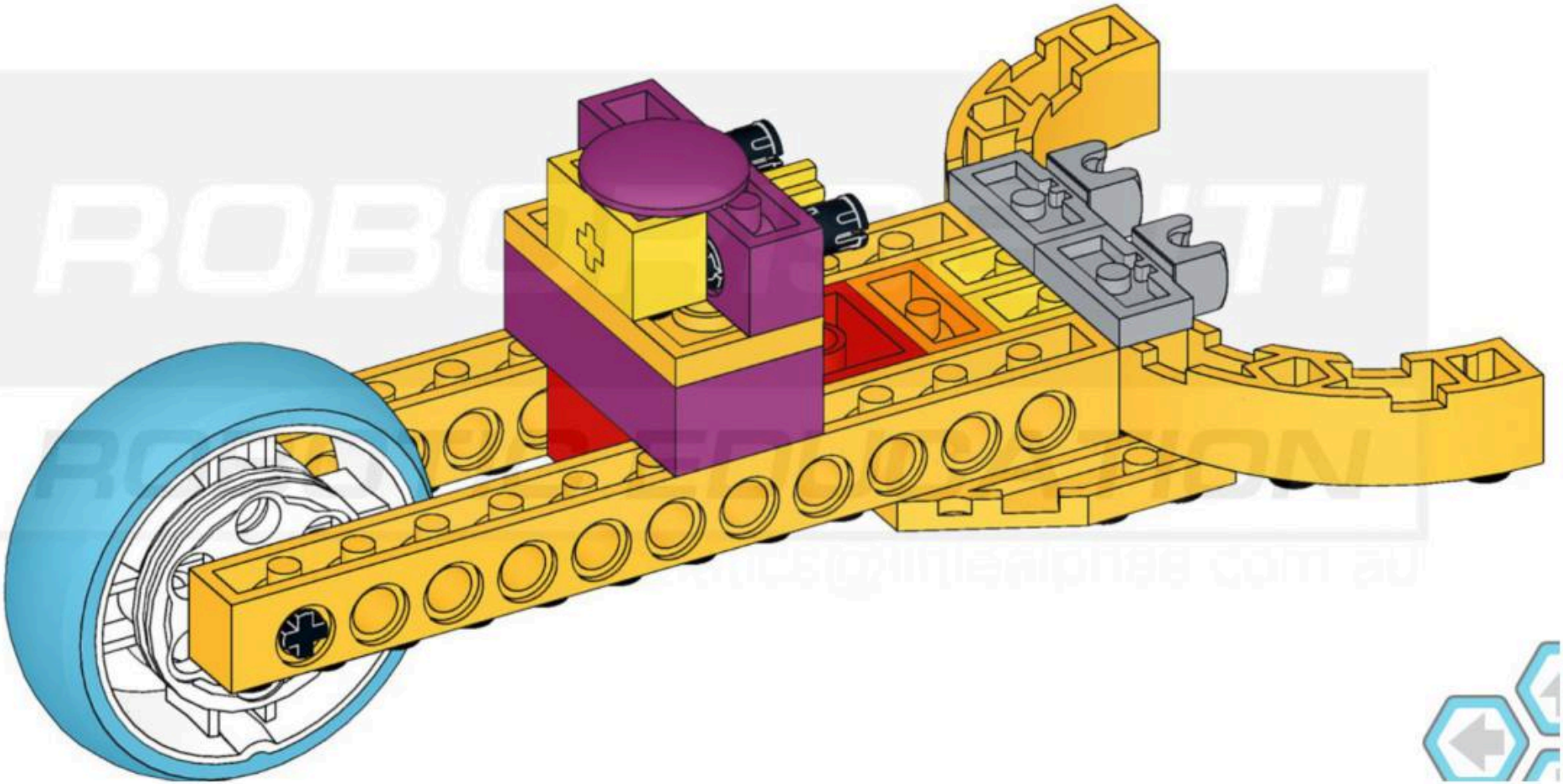


ROBORISE-IT!
ROBOTIC EDUCATION

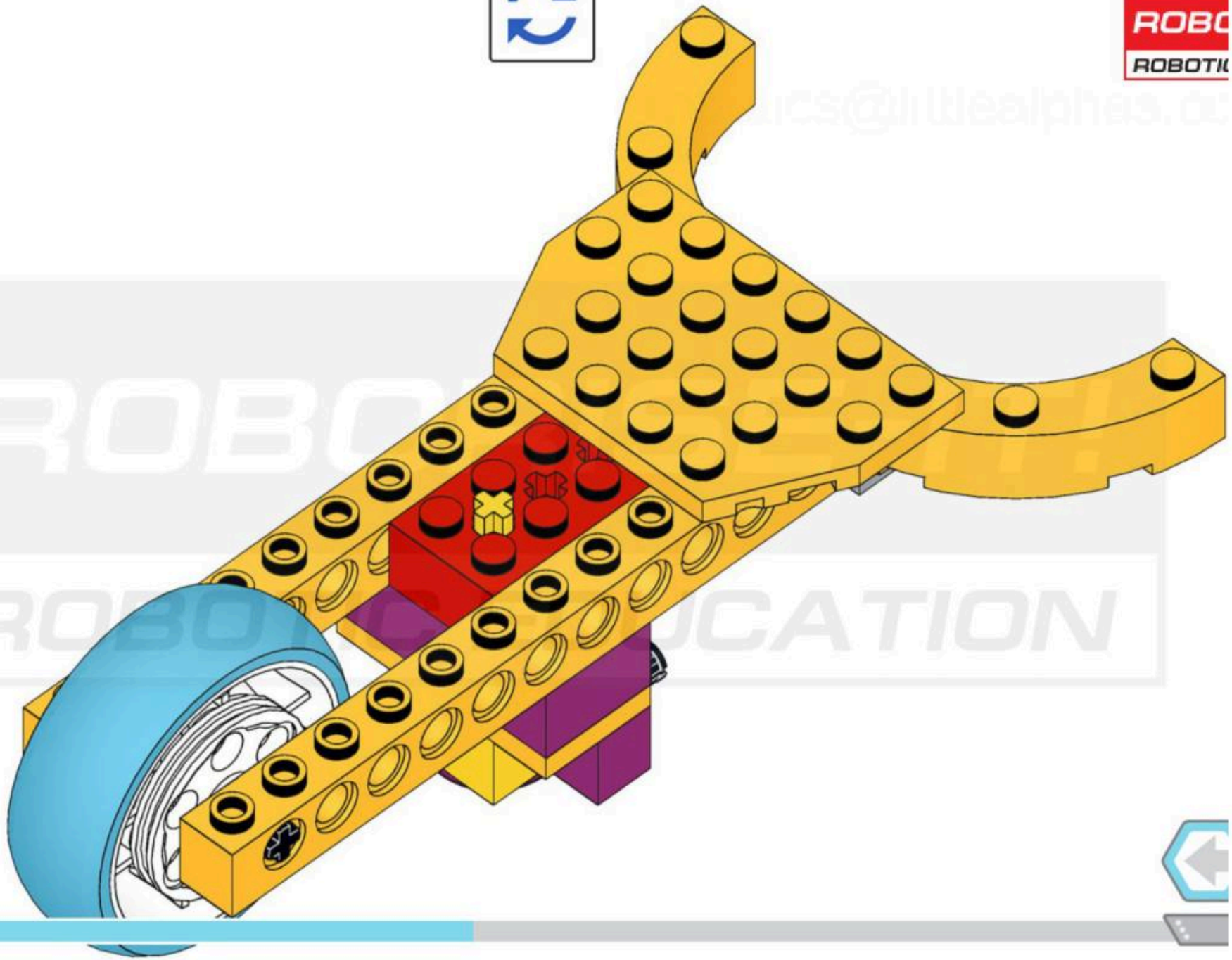
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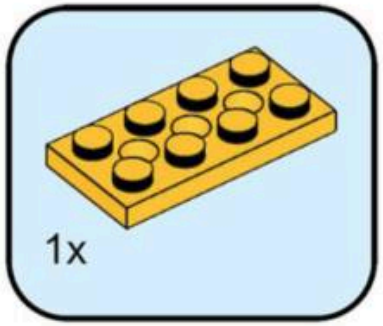


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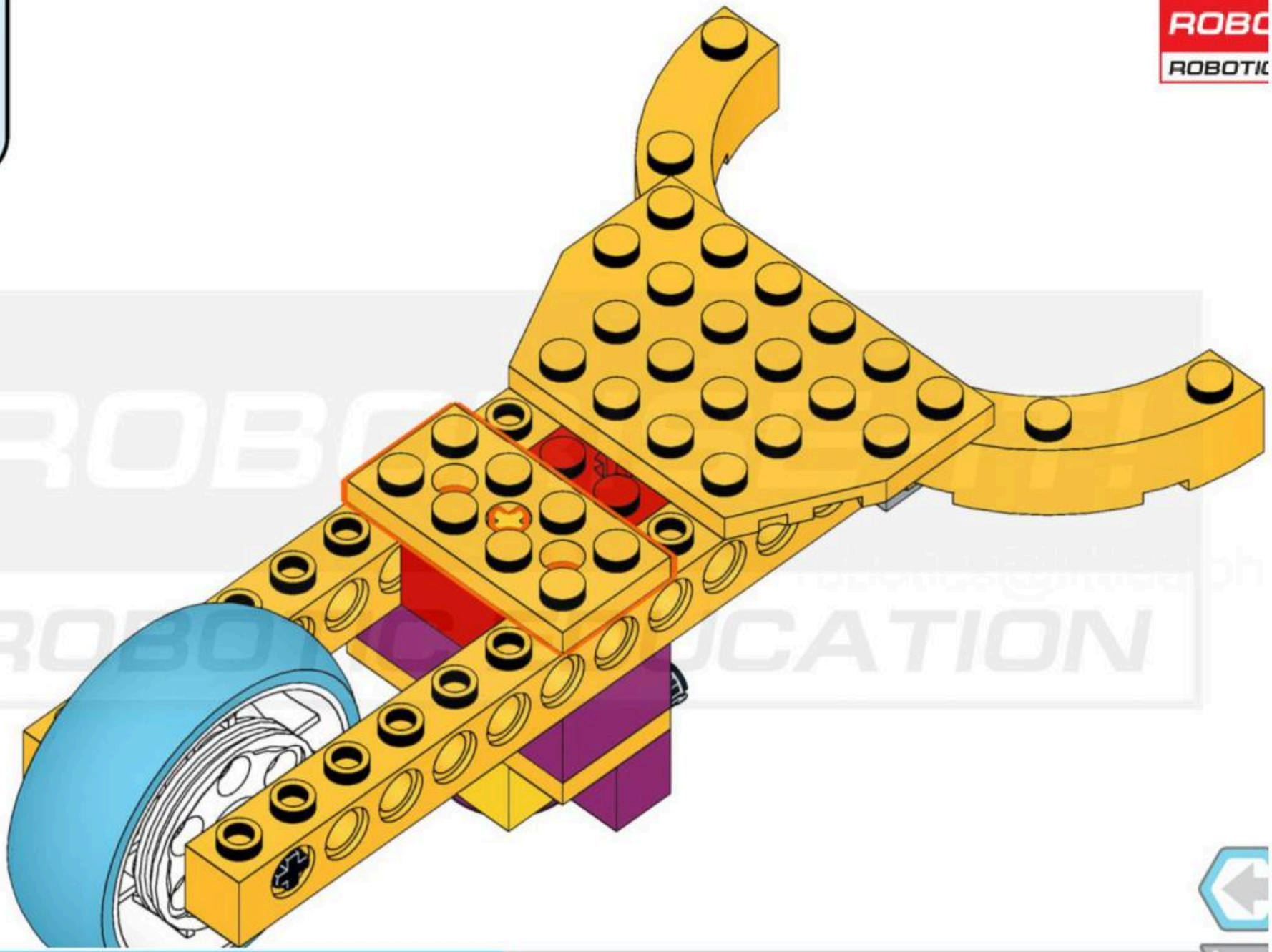


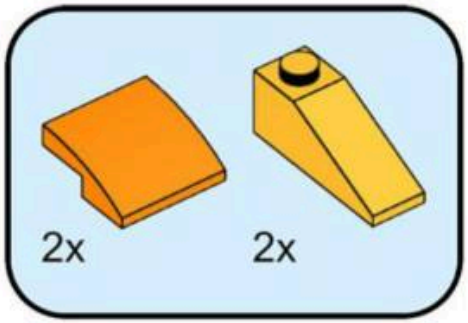
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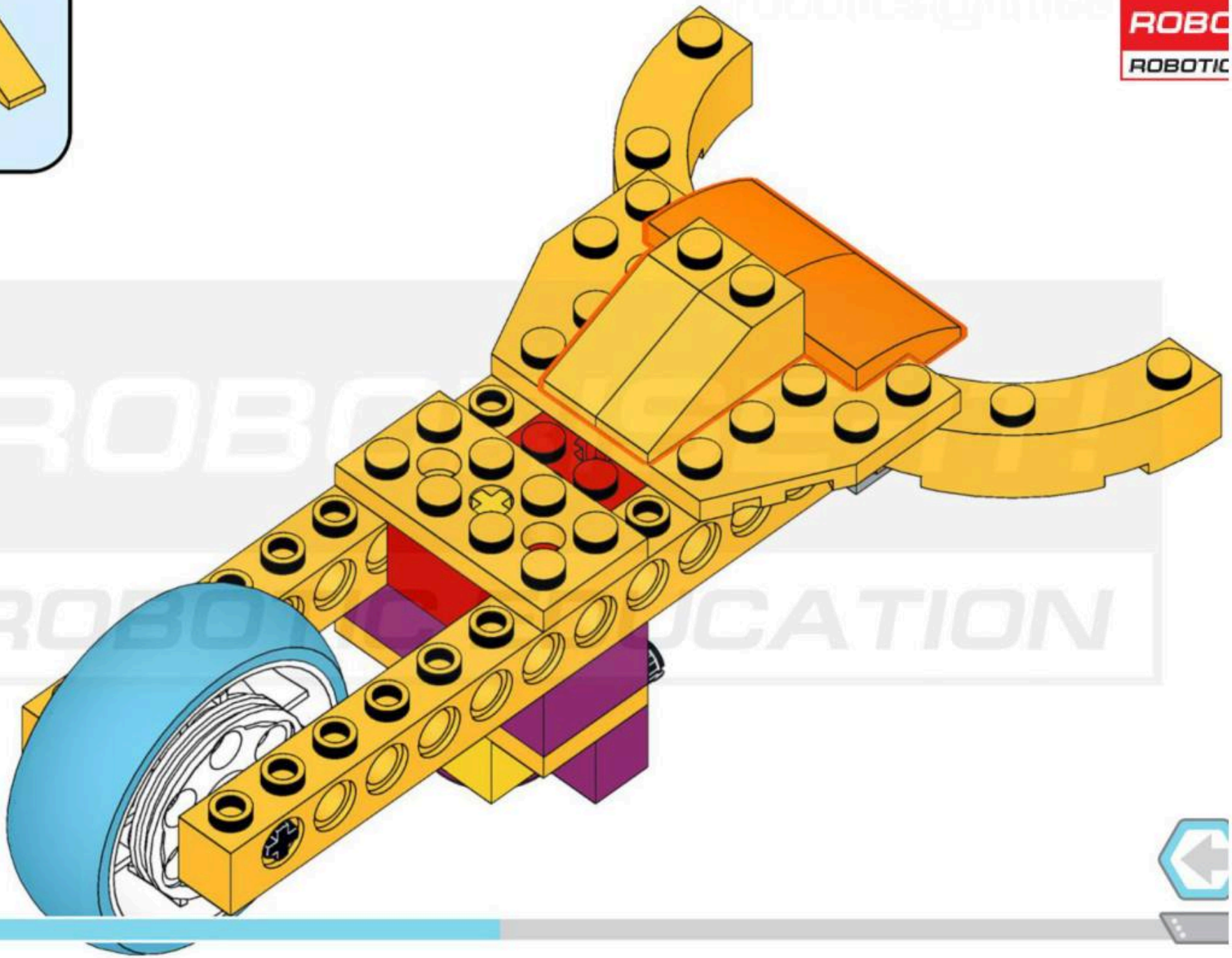


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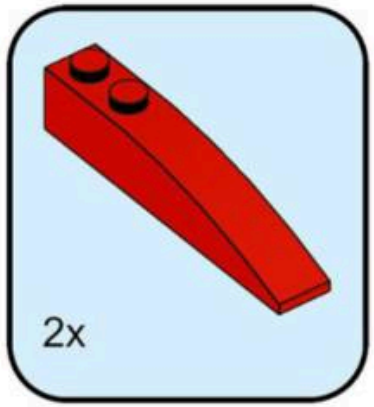




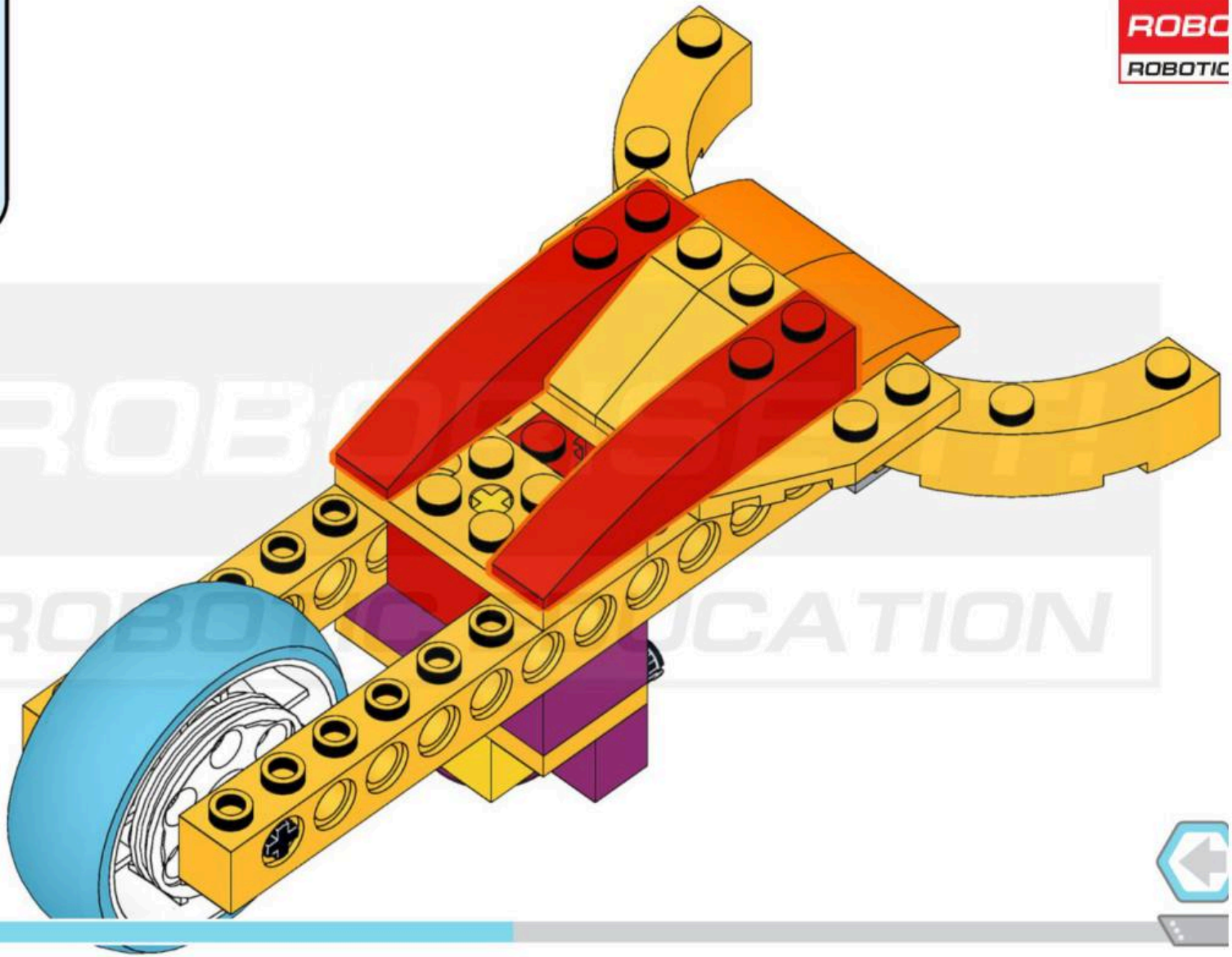
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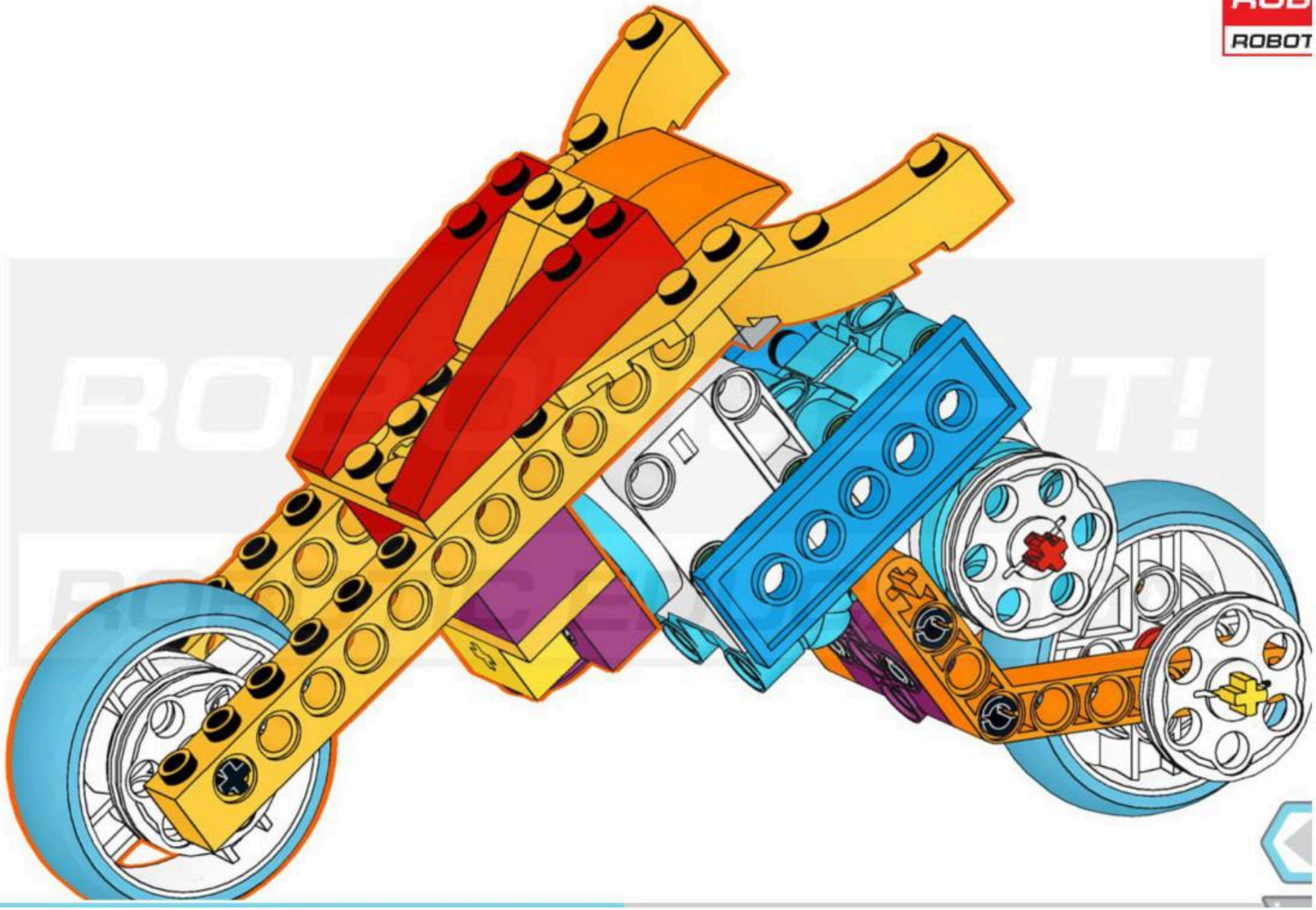
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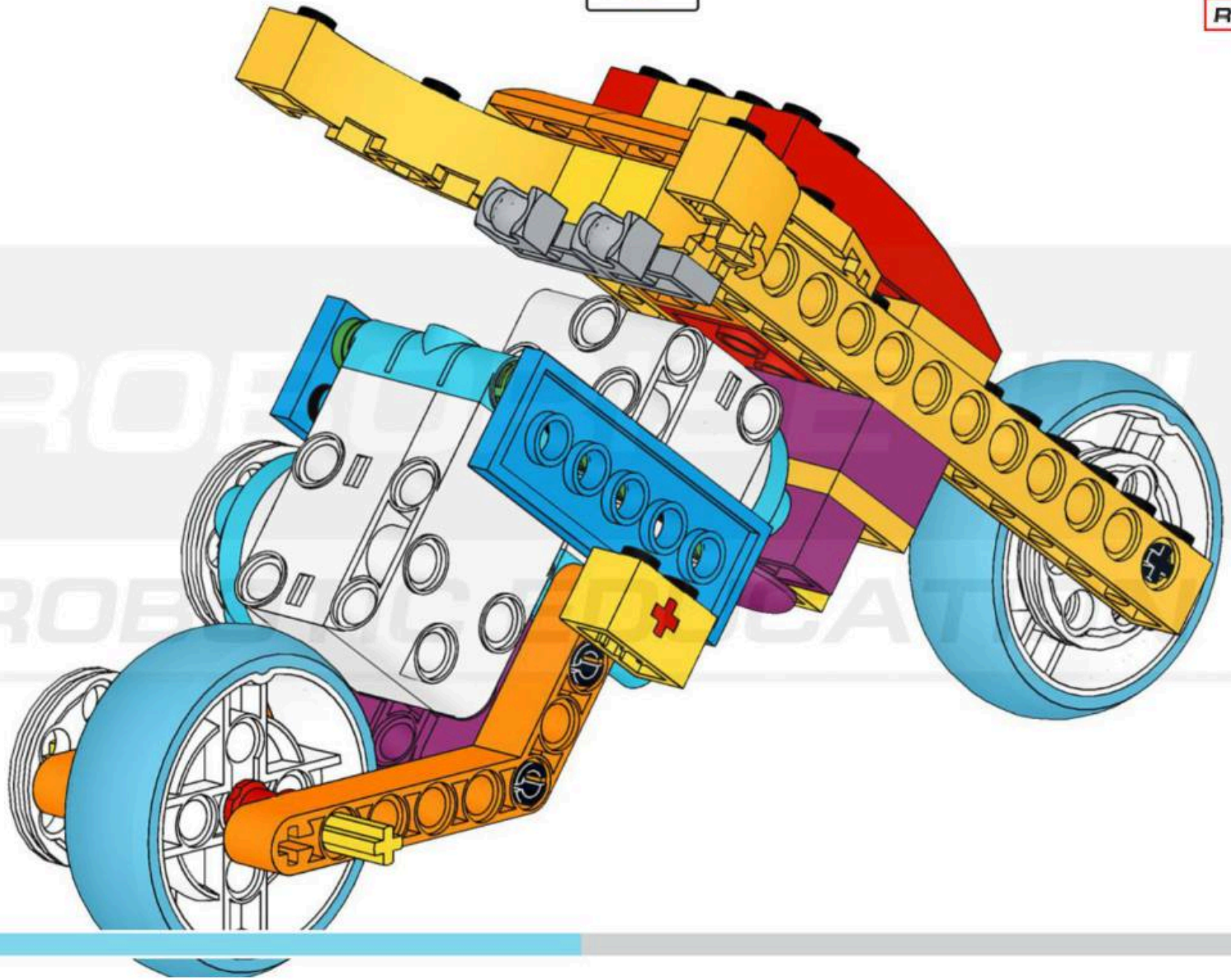
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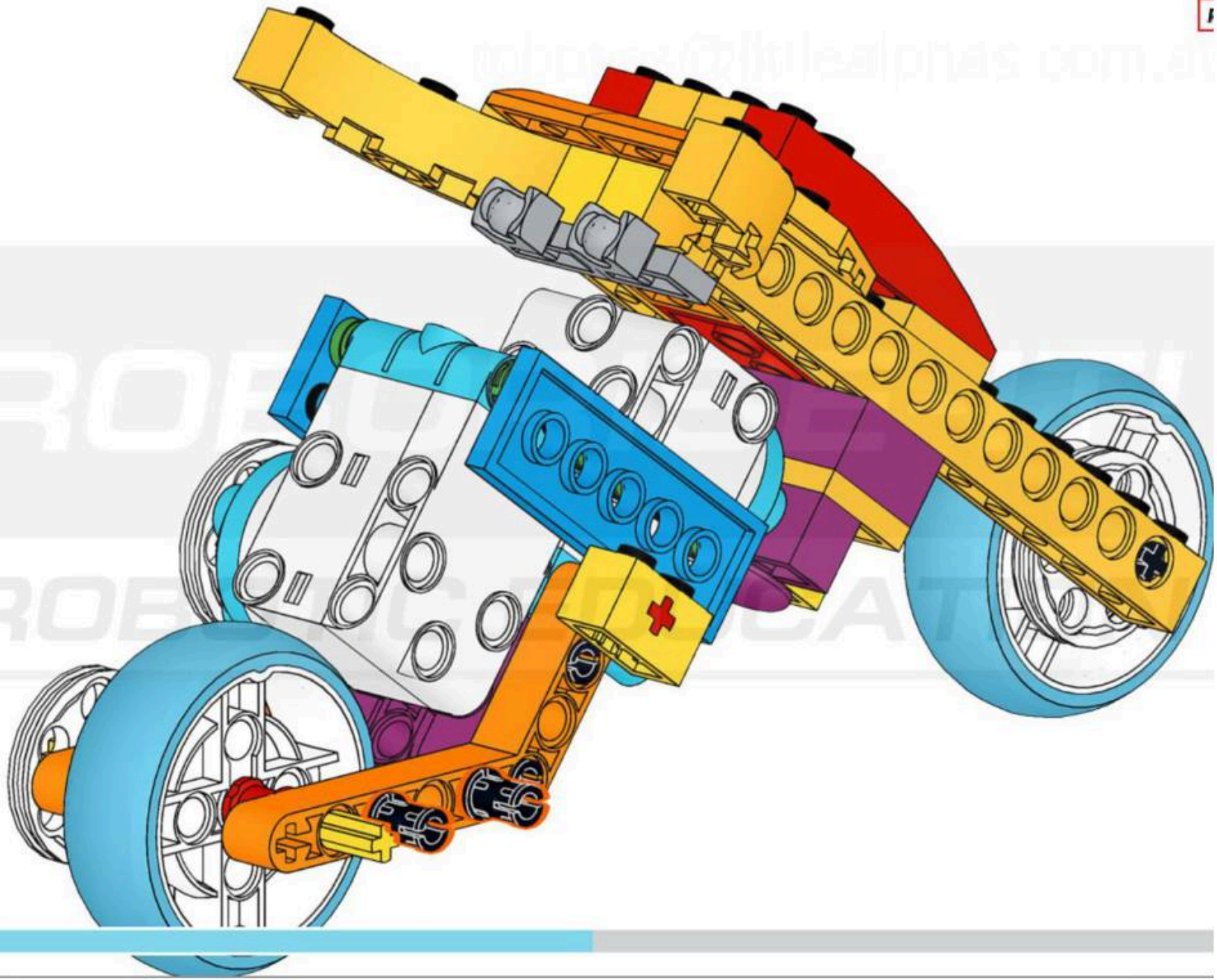


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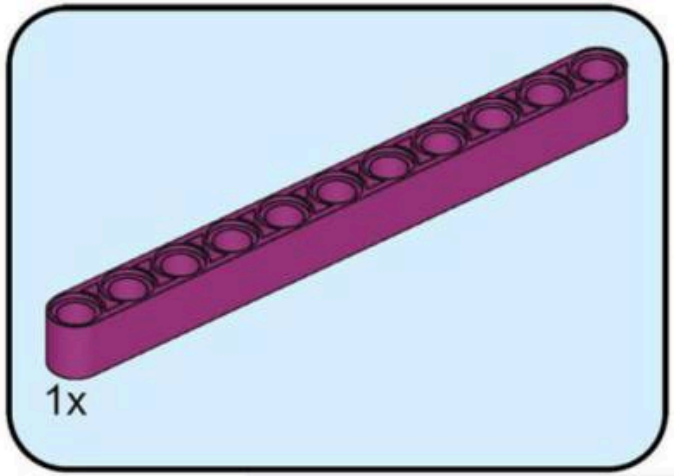




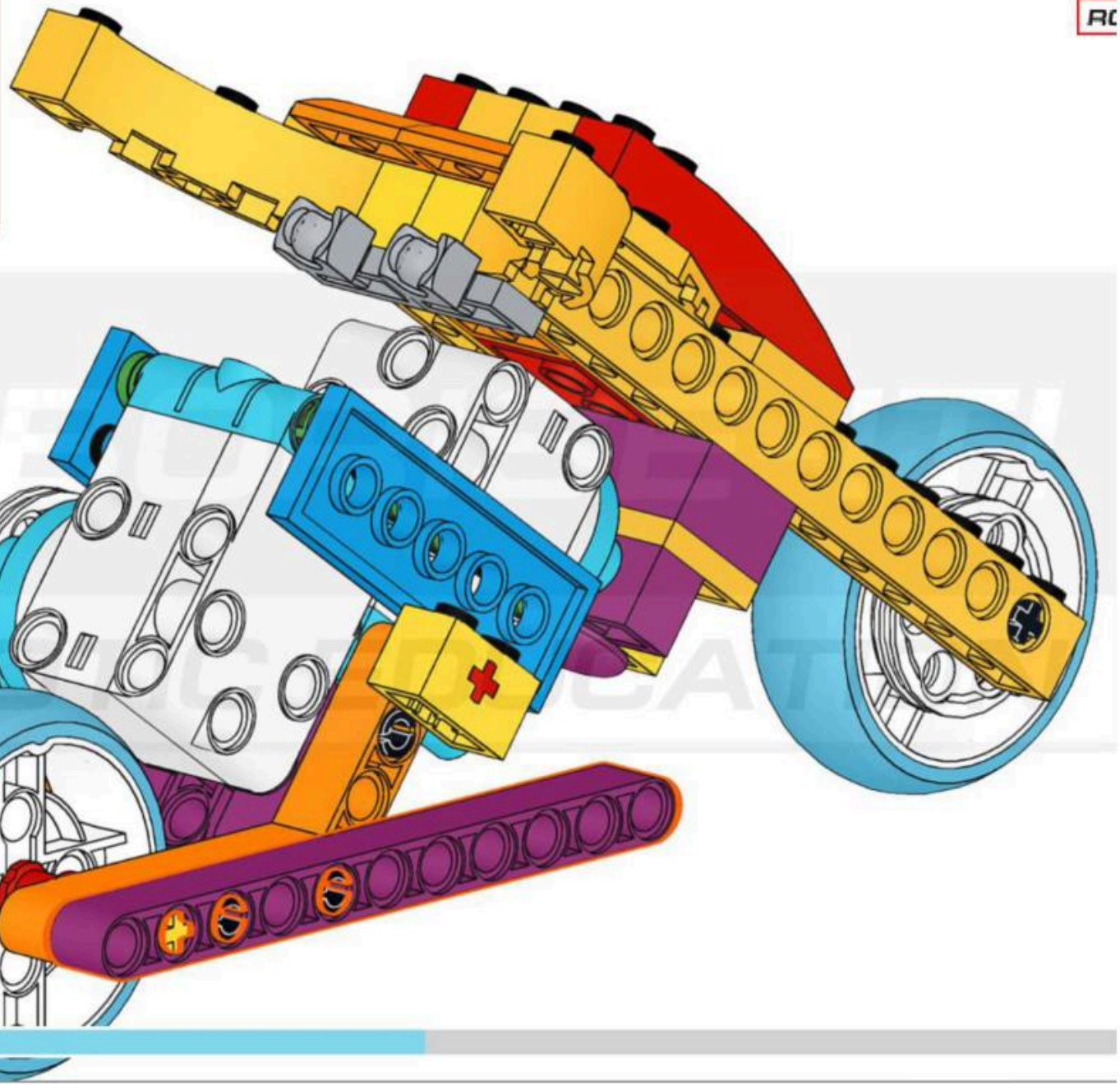
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ROBOTICS EDUCATION

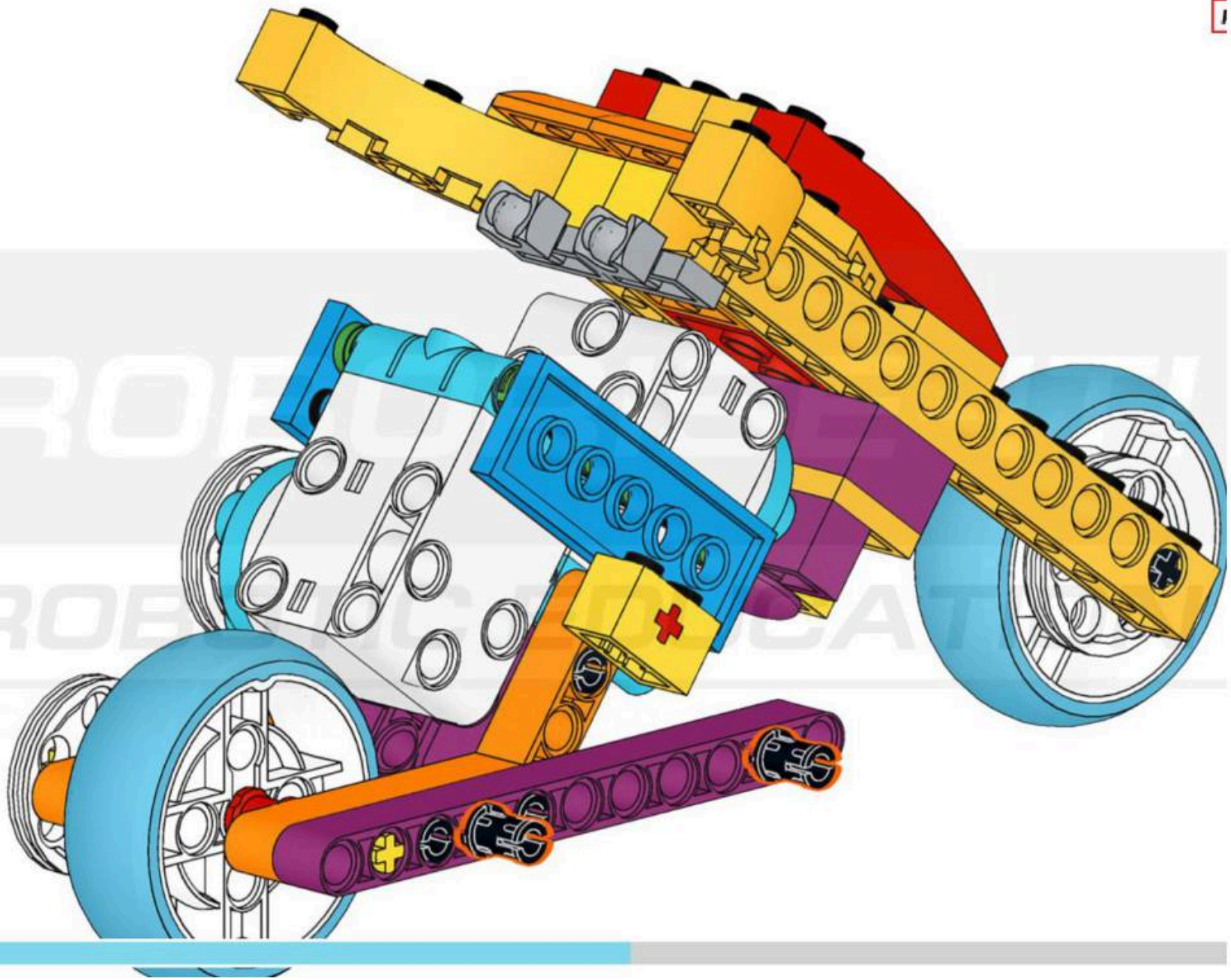


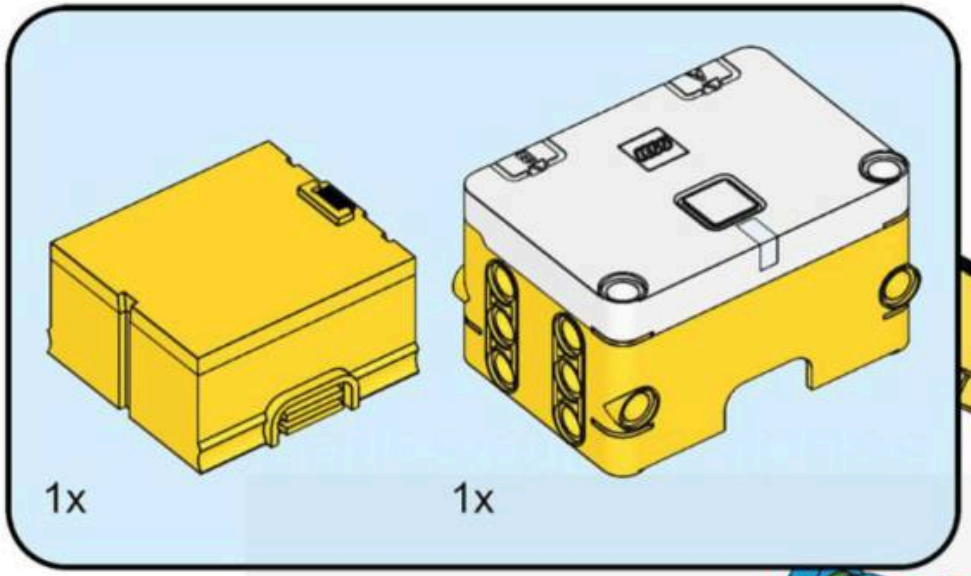
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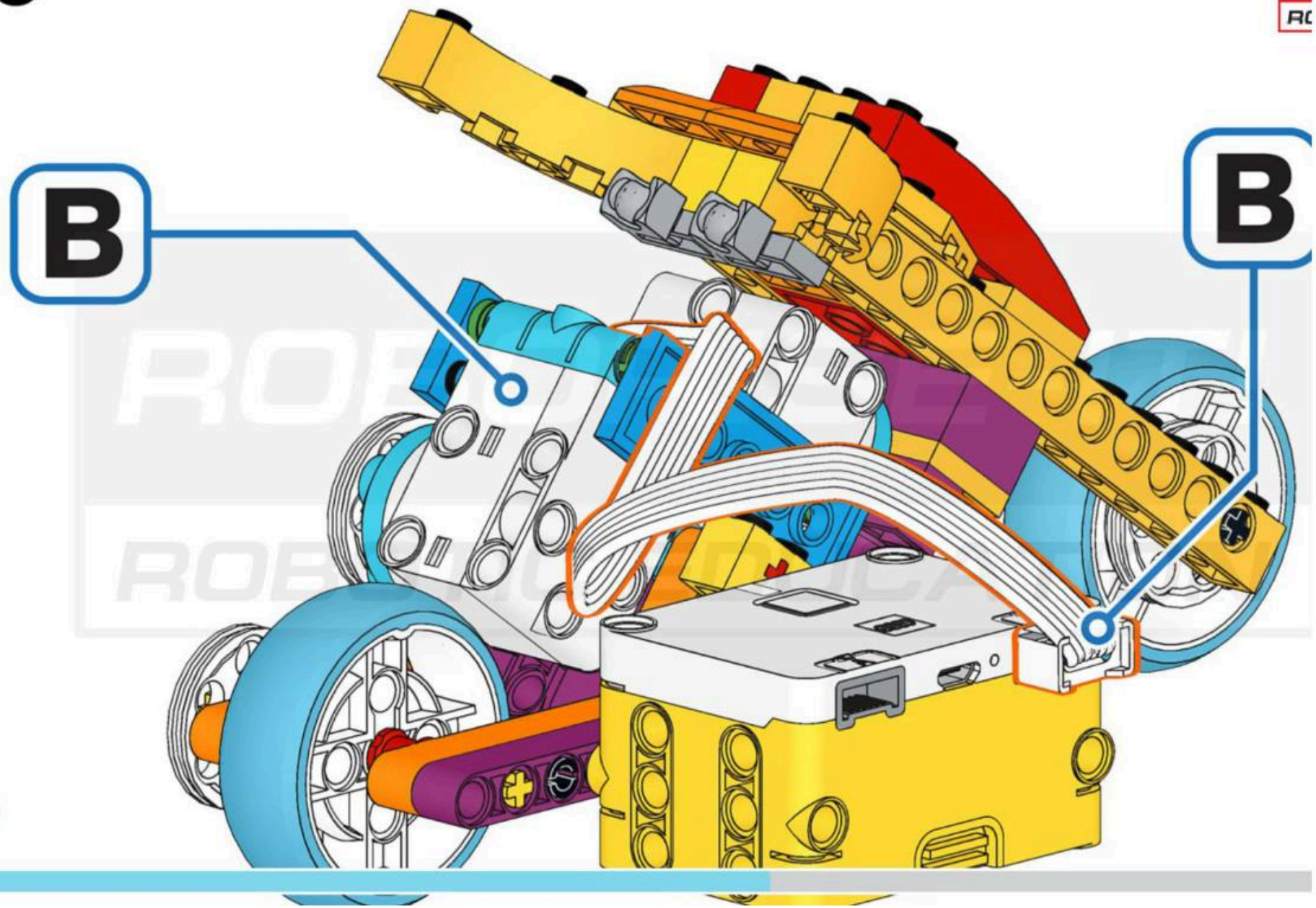
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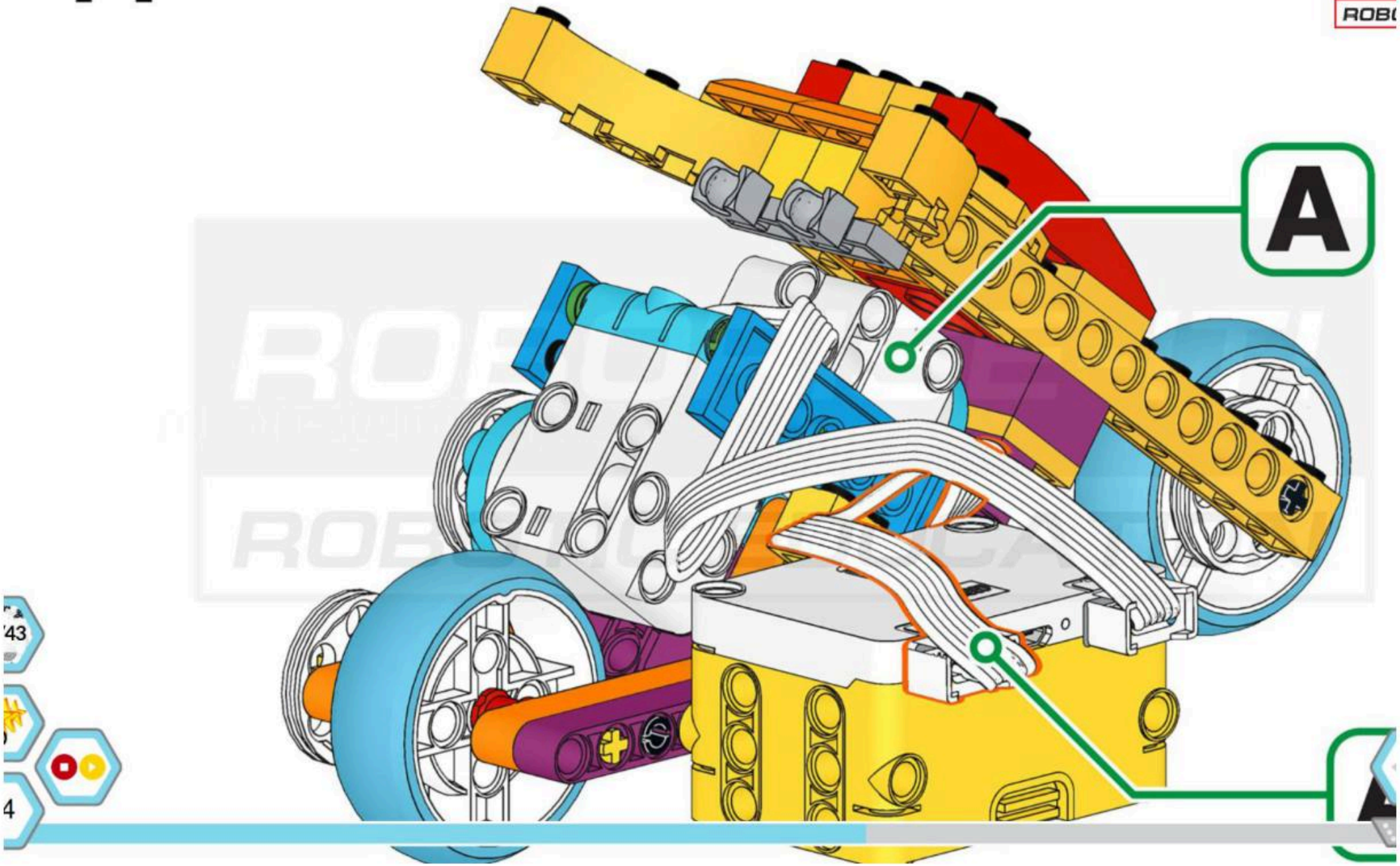


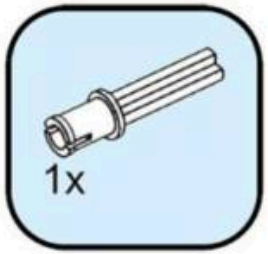
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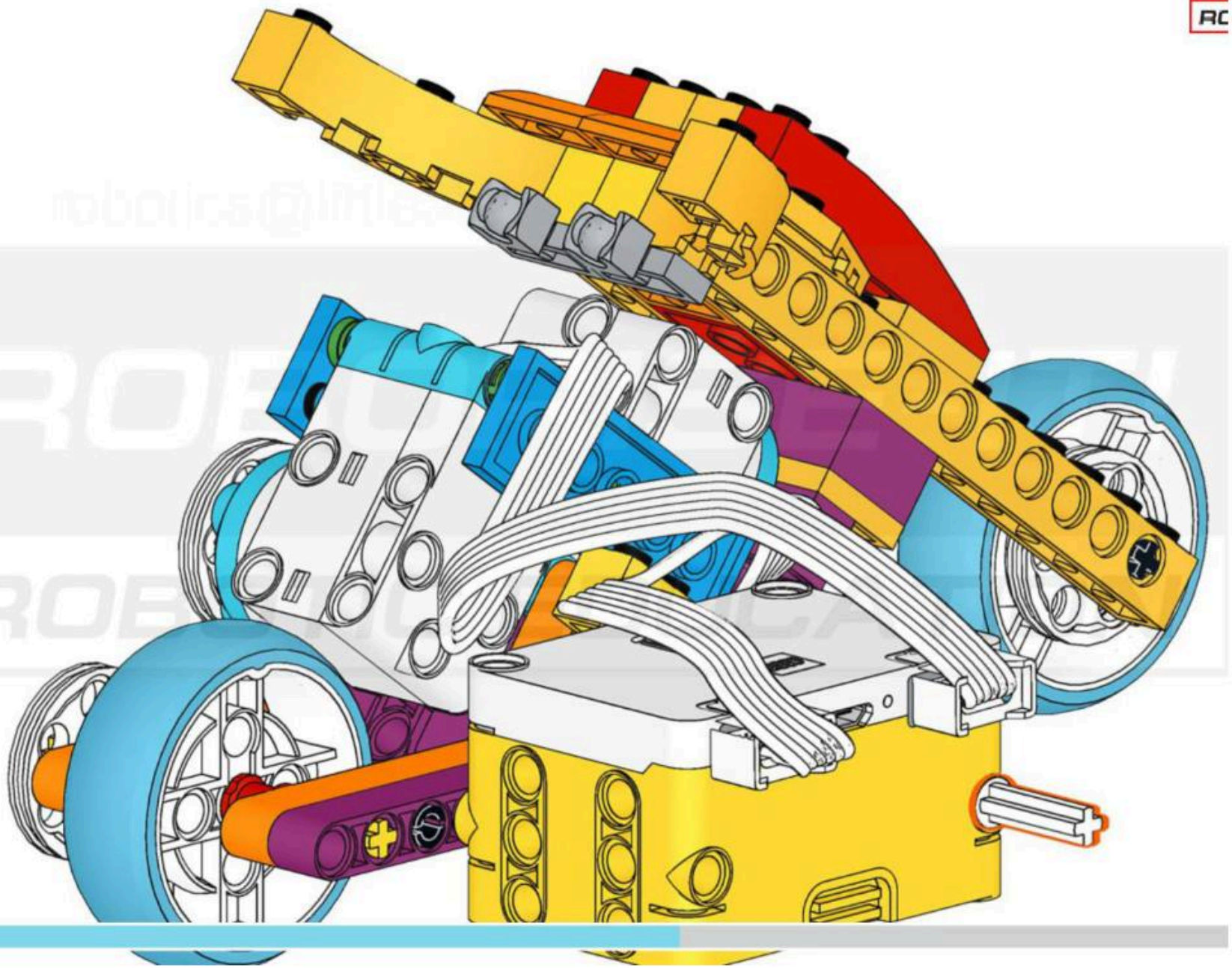


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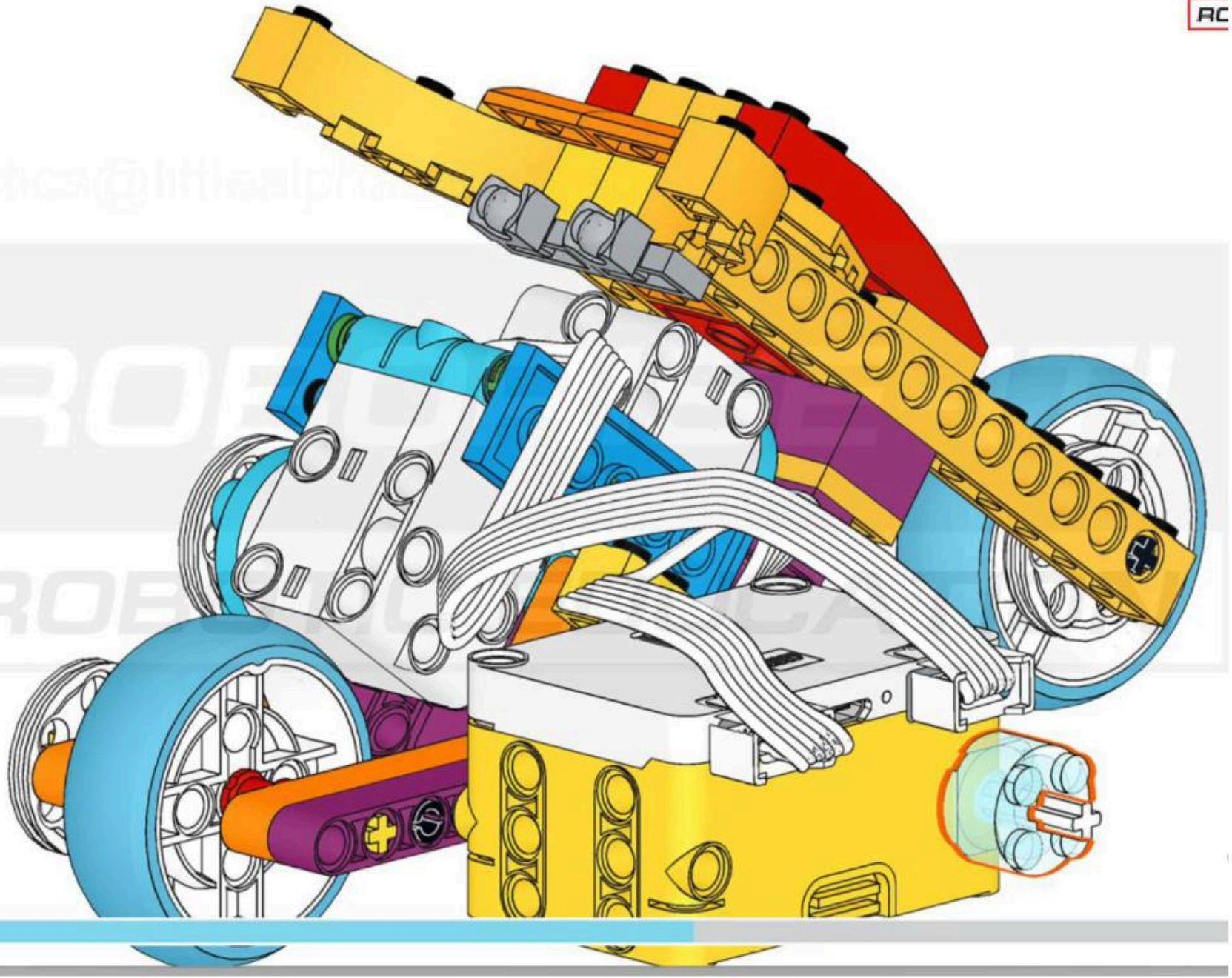


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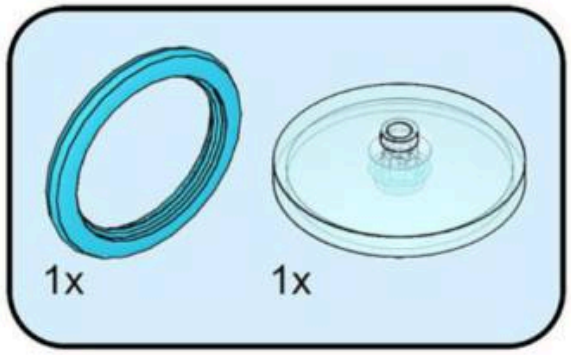




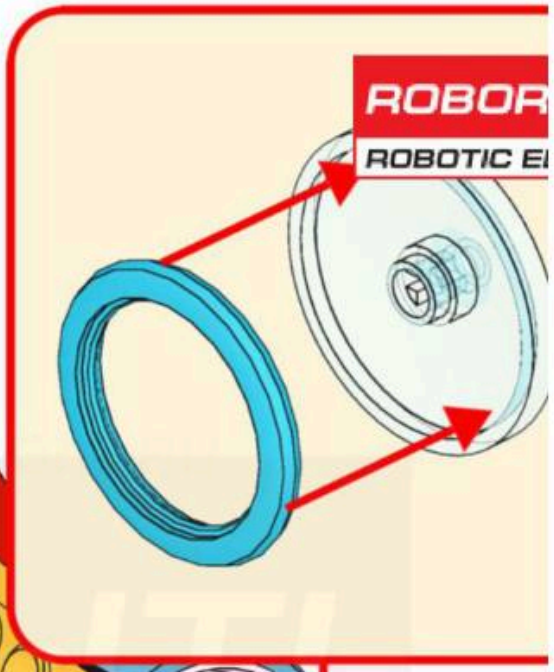
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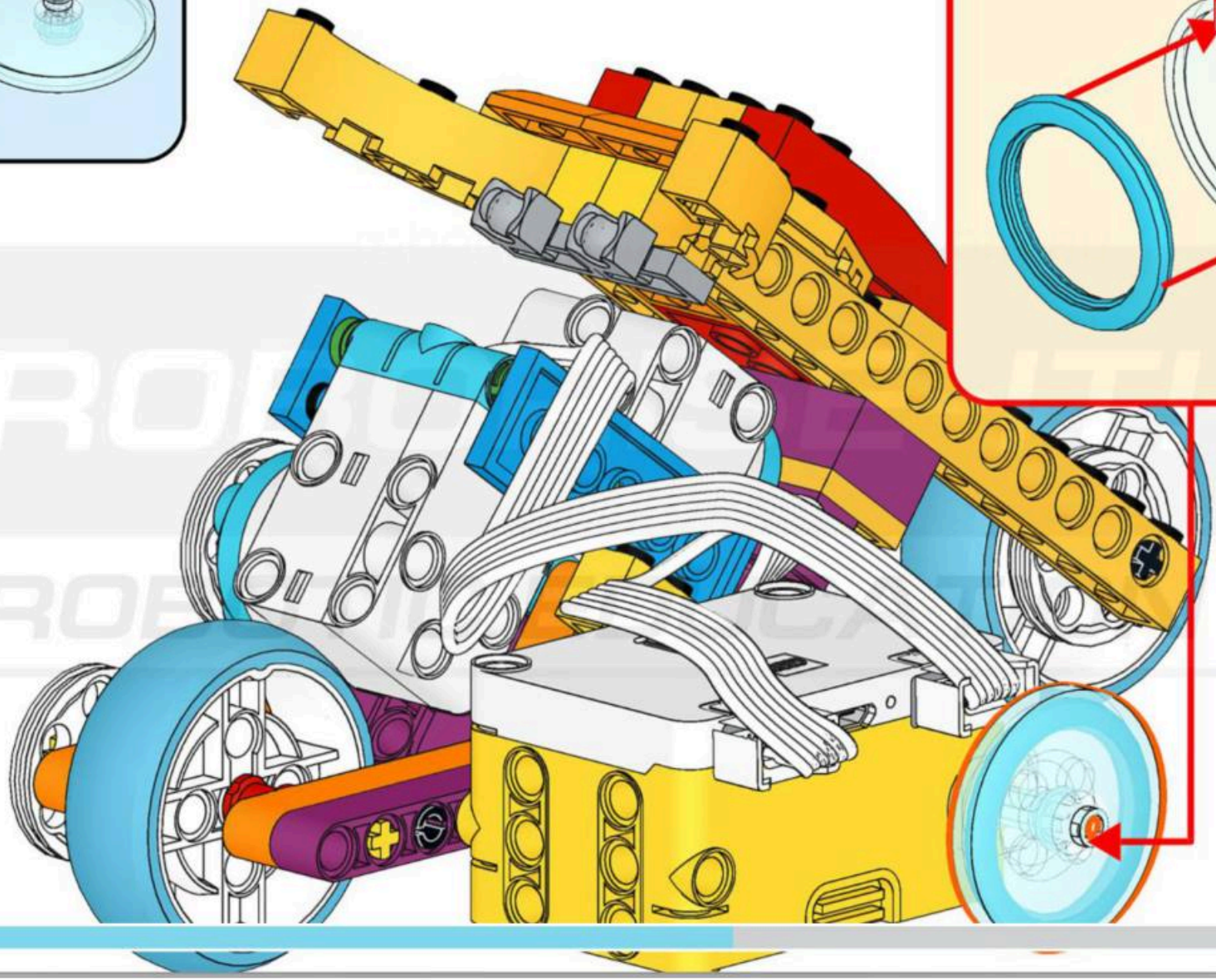


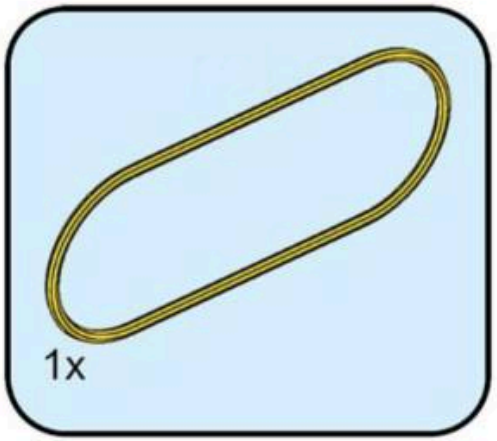
robotics@ithaalpha



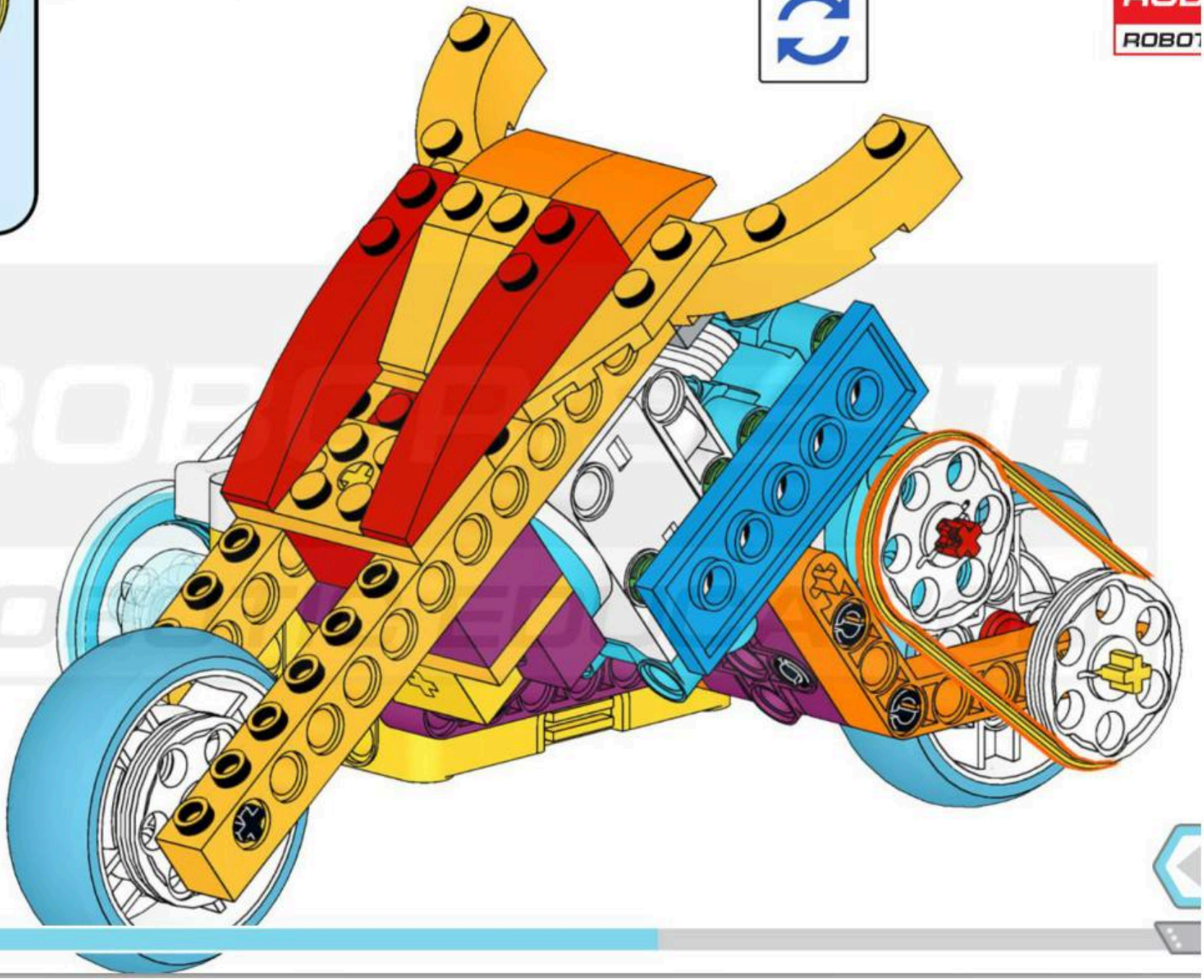
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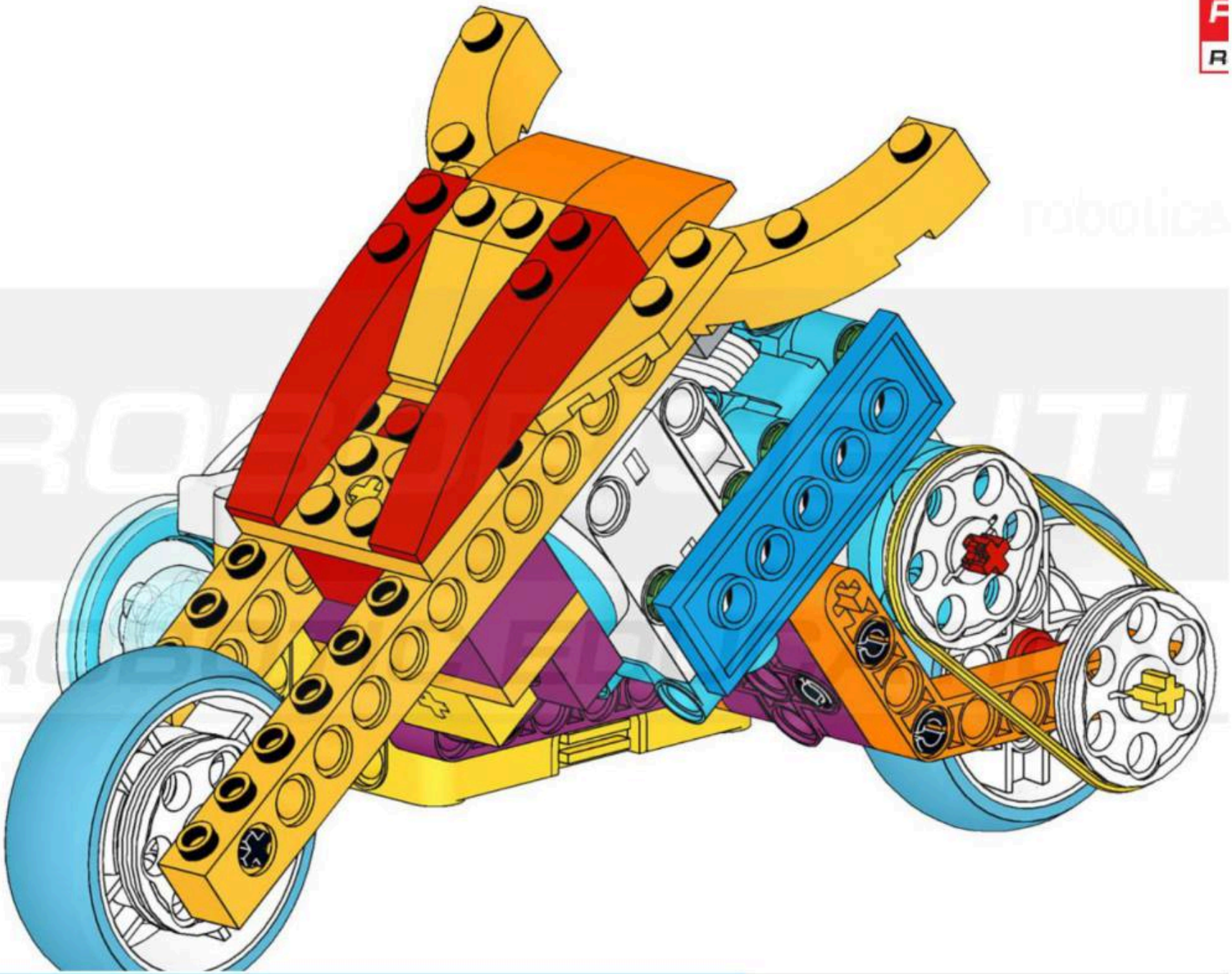


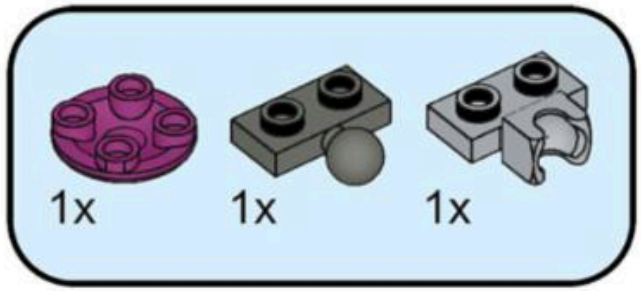


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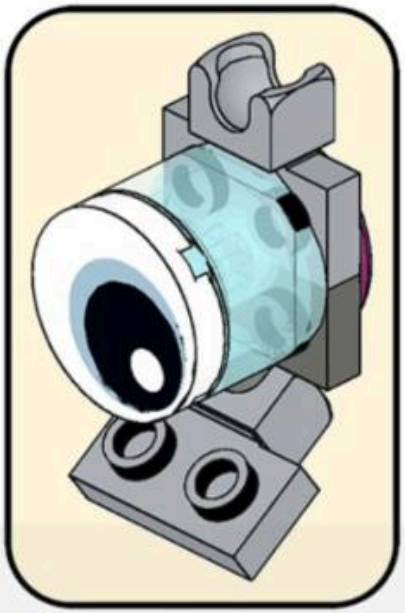


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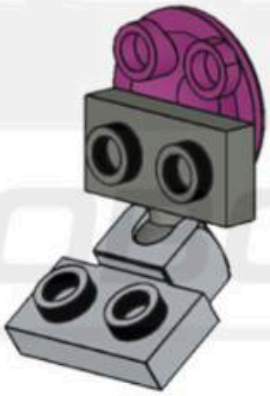


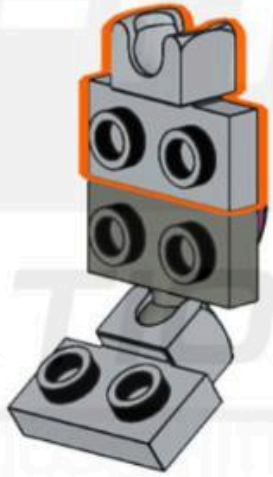
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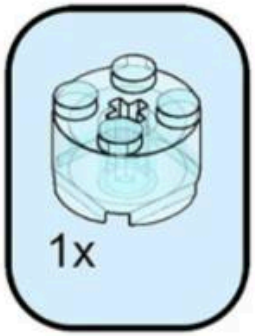


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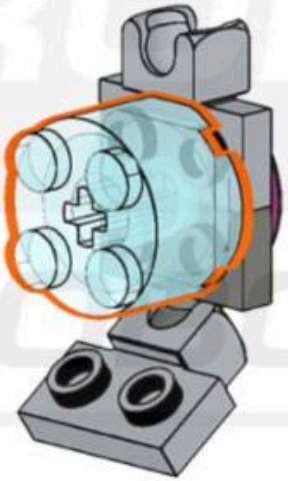


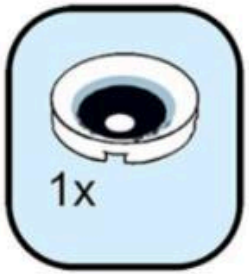


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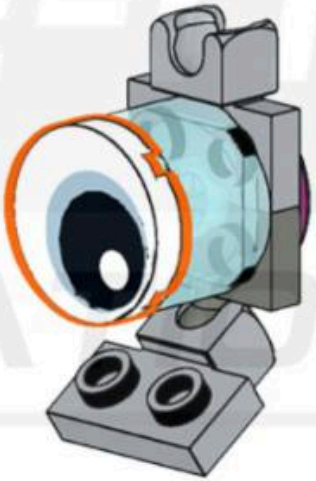


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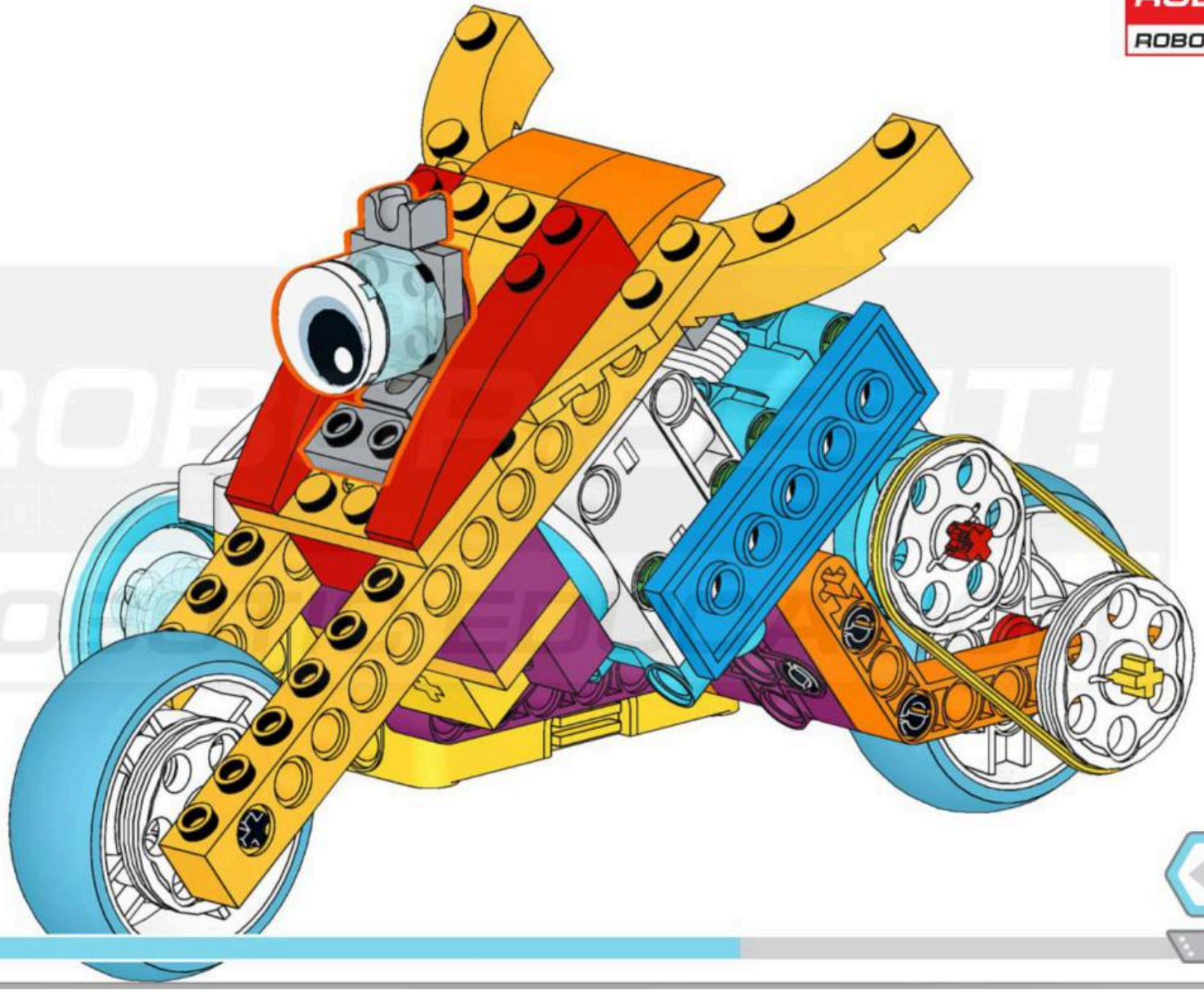


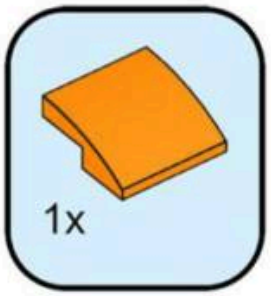


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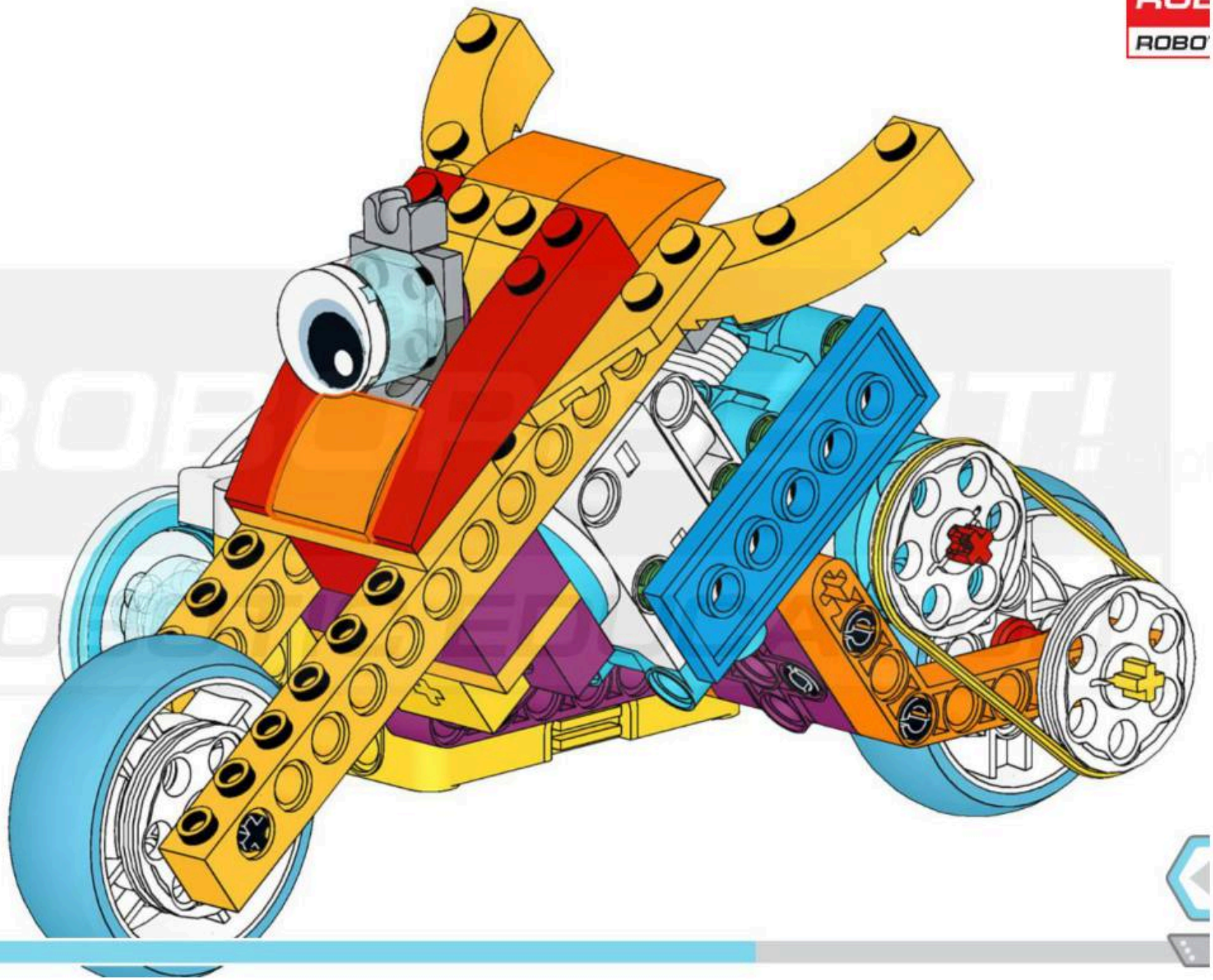


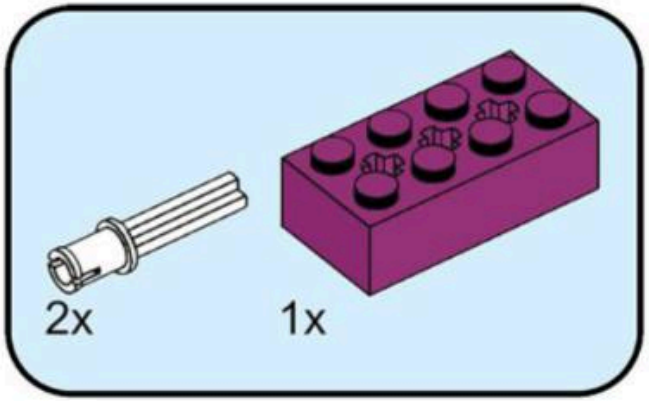
robotics@lit...as.com.au
ROBORISE-IT
ROBOTIC EDUCATION



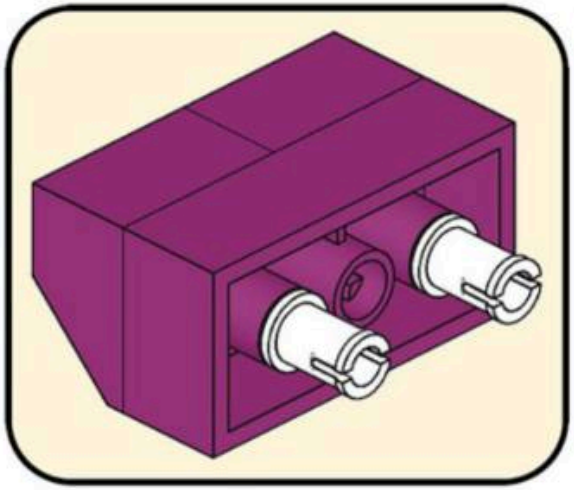


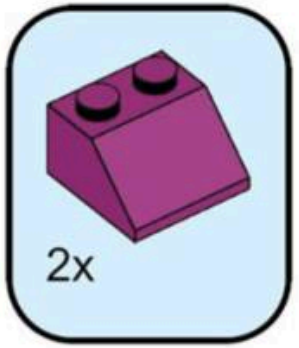
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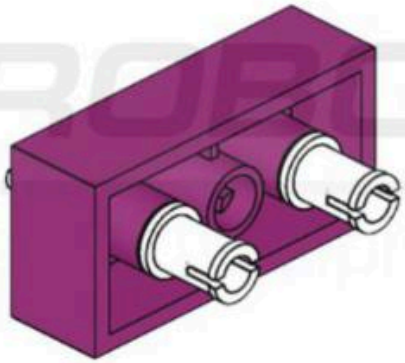


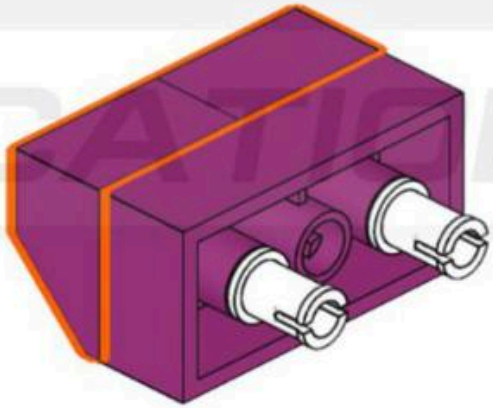
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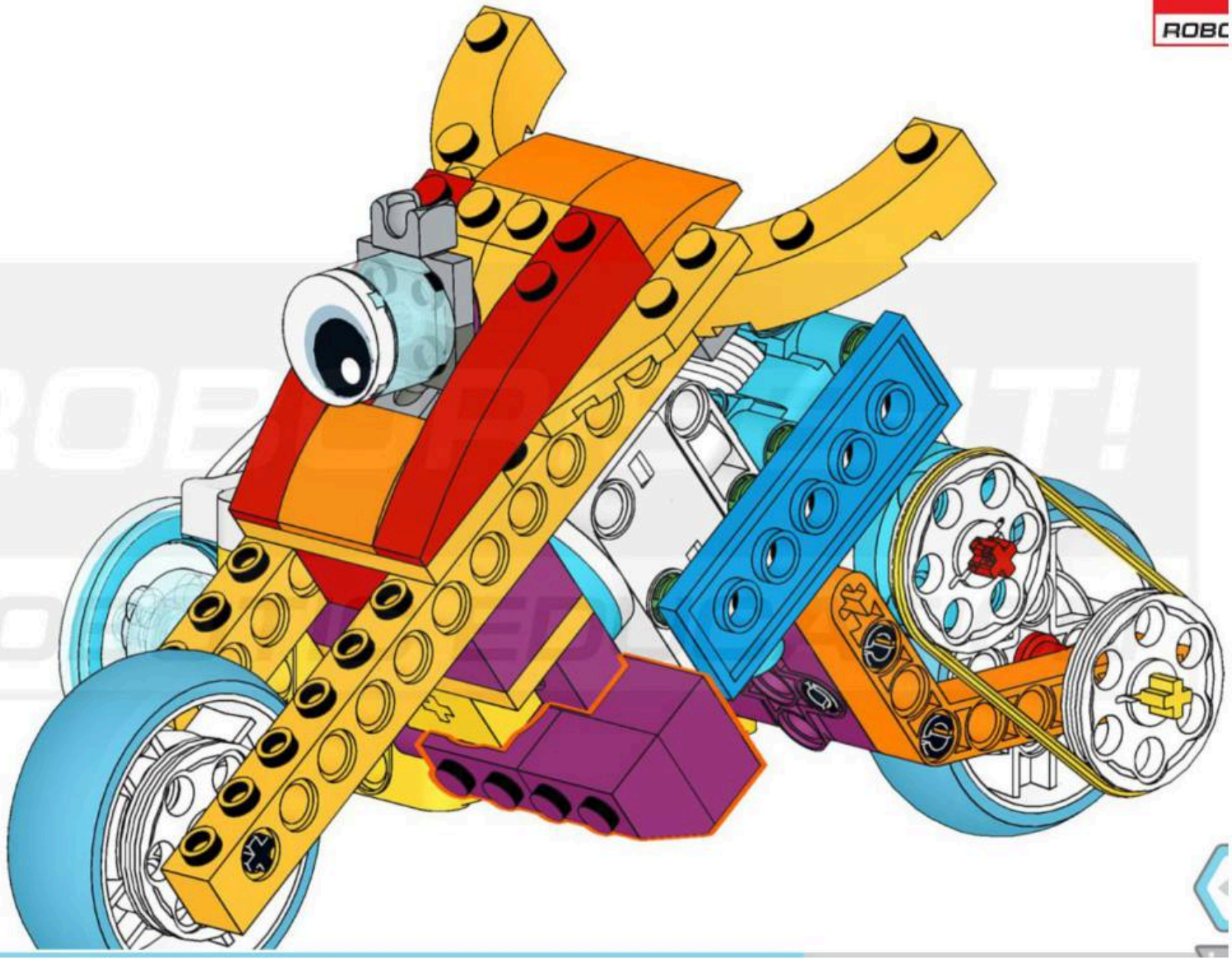
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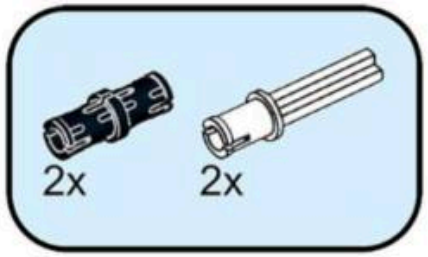




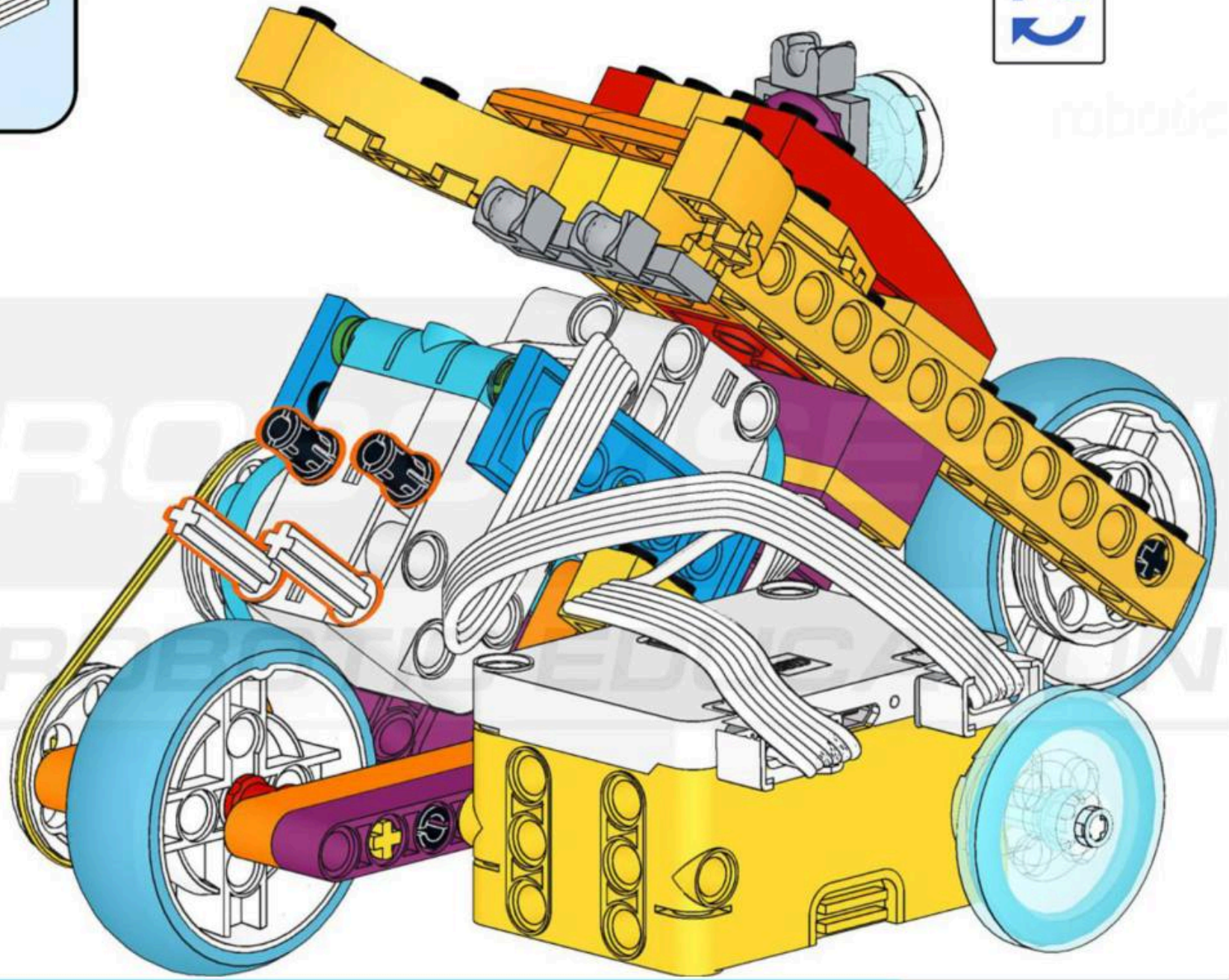
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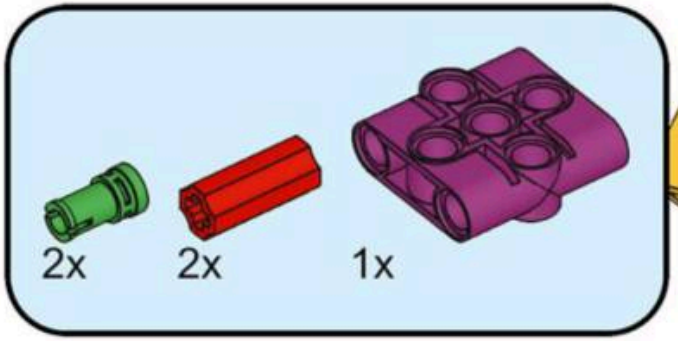
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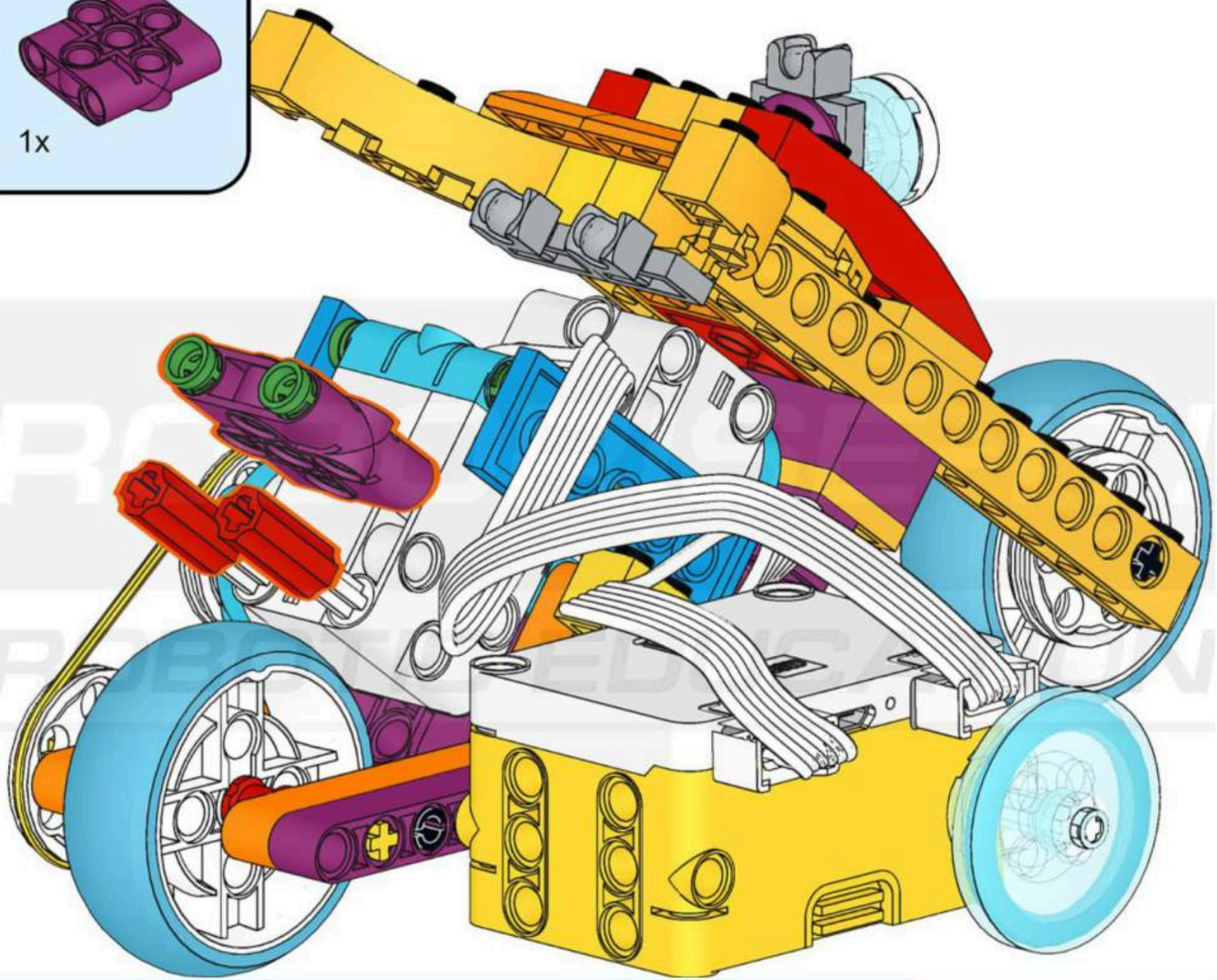


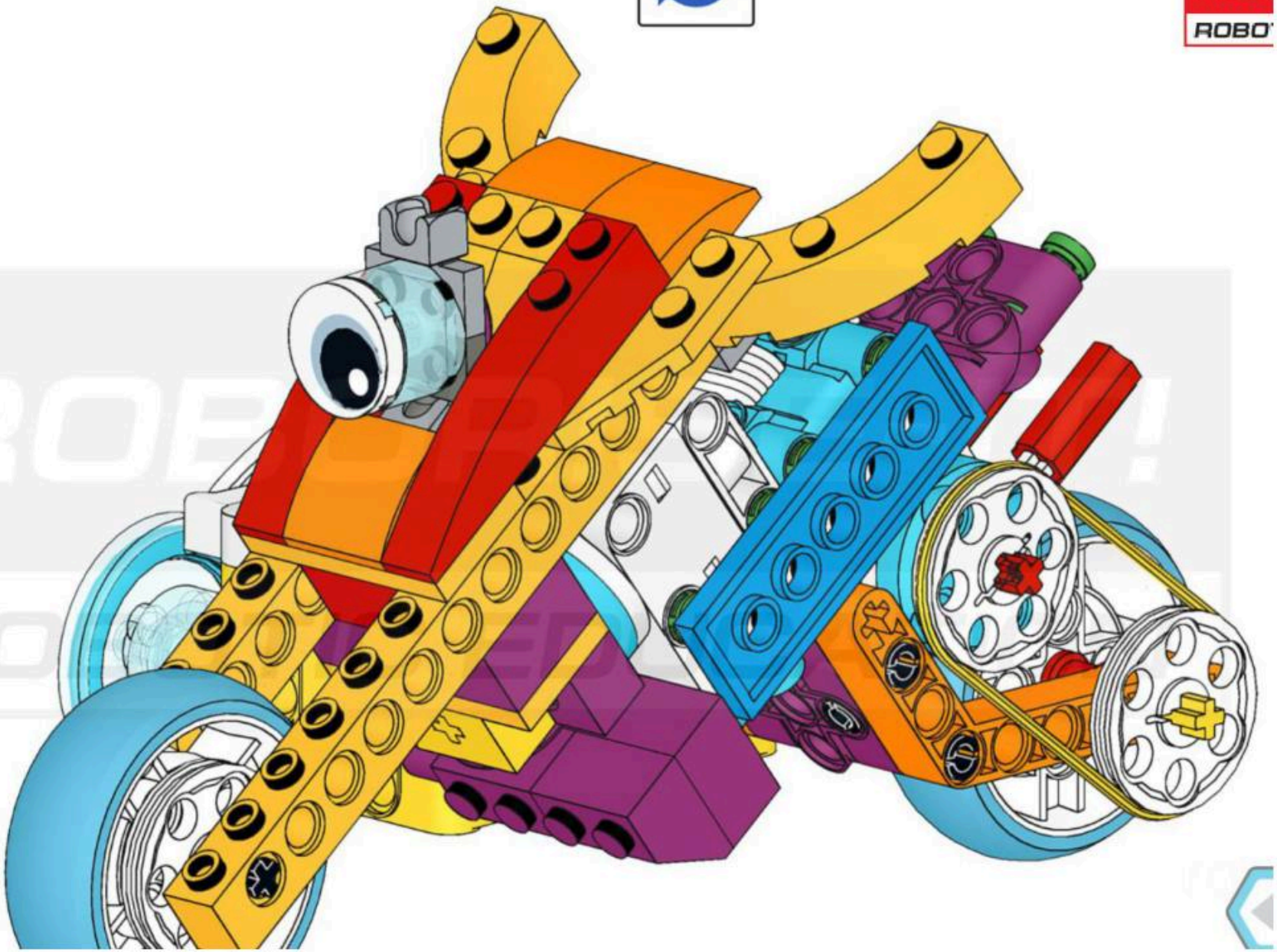
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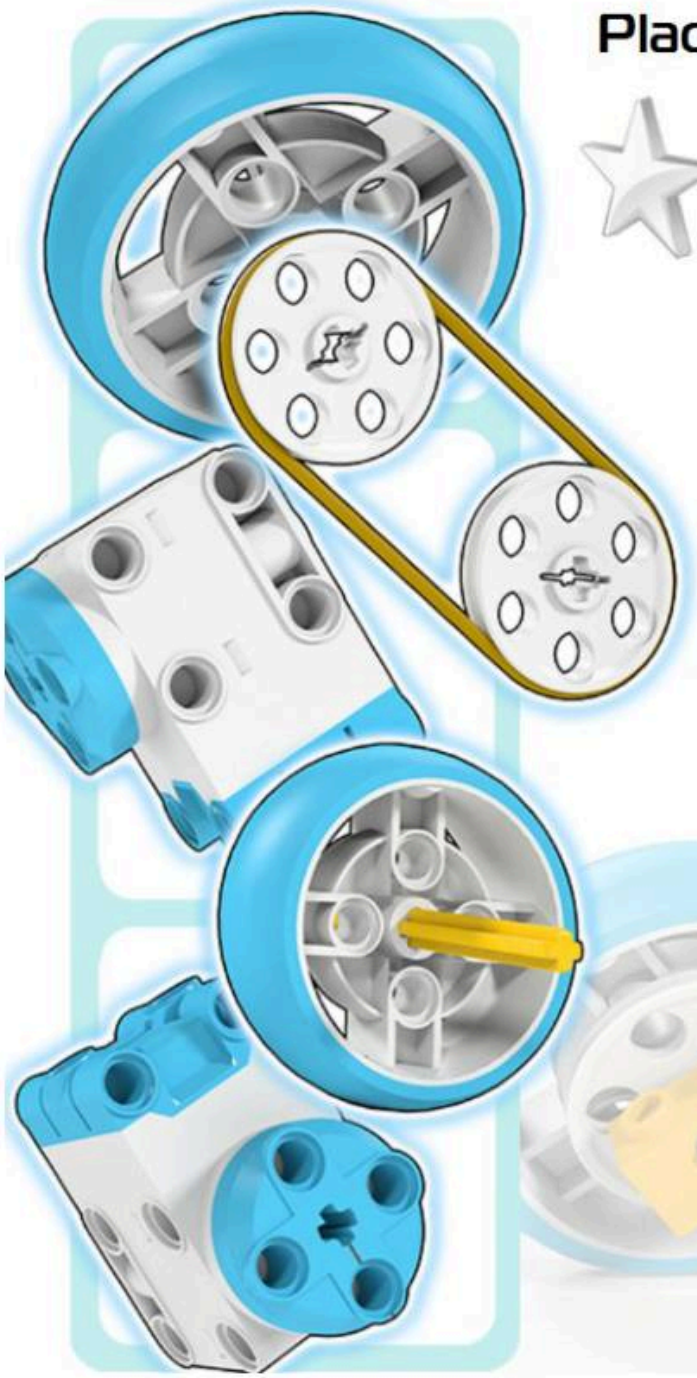


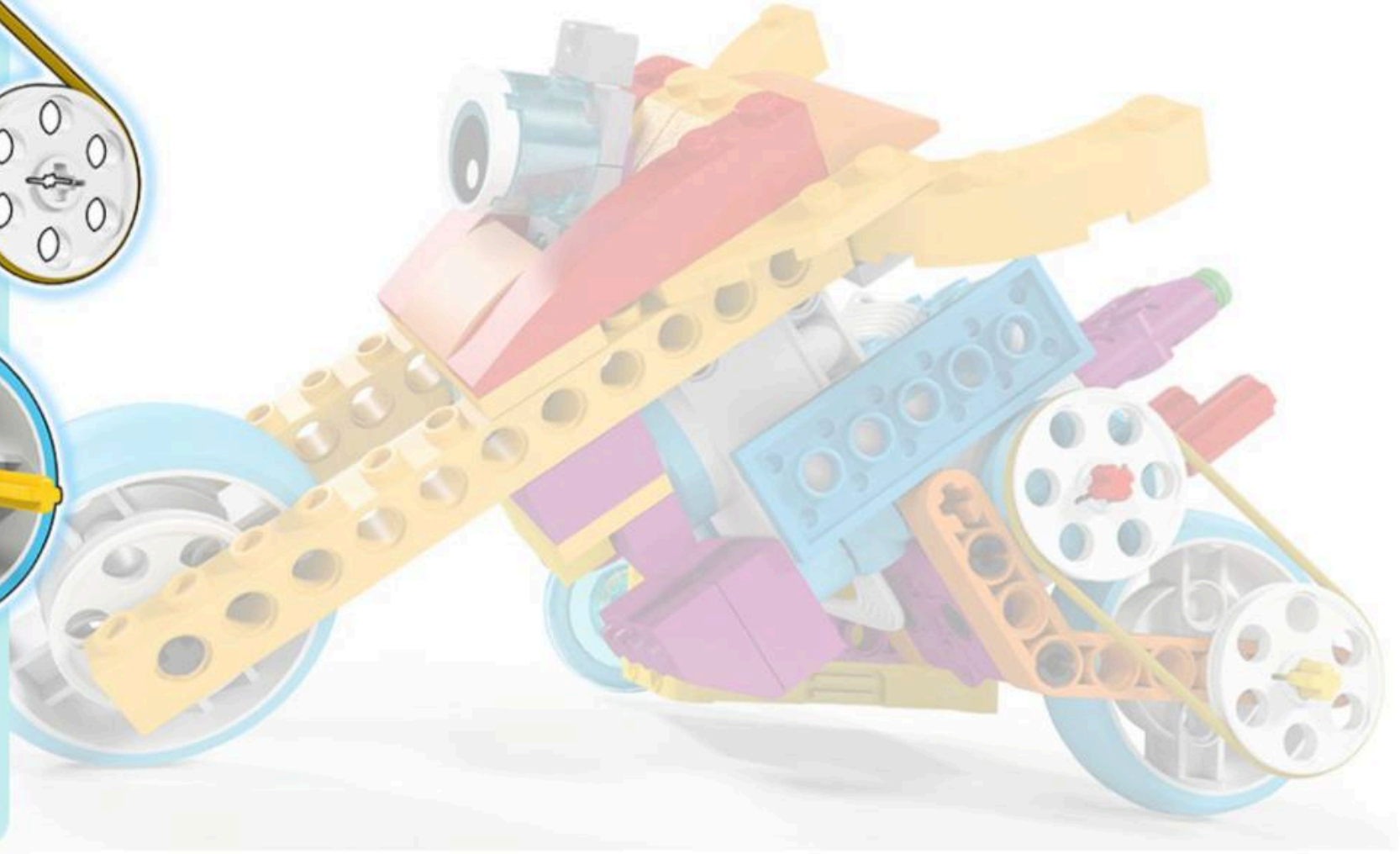
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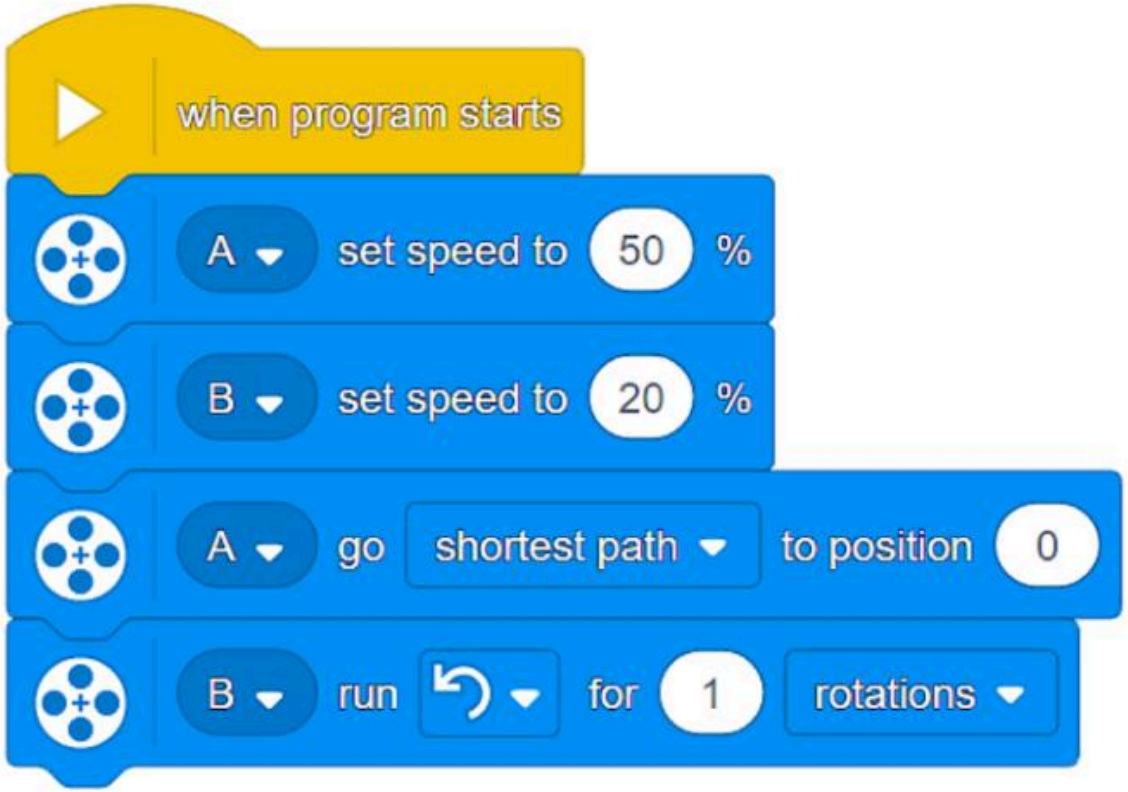
Place the parts of the robot in the correct places





Use the motor blocks to program the robot's movement:

In Word Blocks mode, the program looks like this. Thanks to the possibility of programming both motors separately, the robot independently sets the fork of the front wheel in the central position



Divide 50 cm by the distance the robot travels in one turn, and you will calculate the number of turns necessary for the robot to travel 50 cm. Do the test. Most likely, the robot will travel less than planned. Explain the result. Why did this happen?

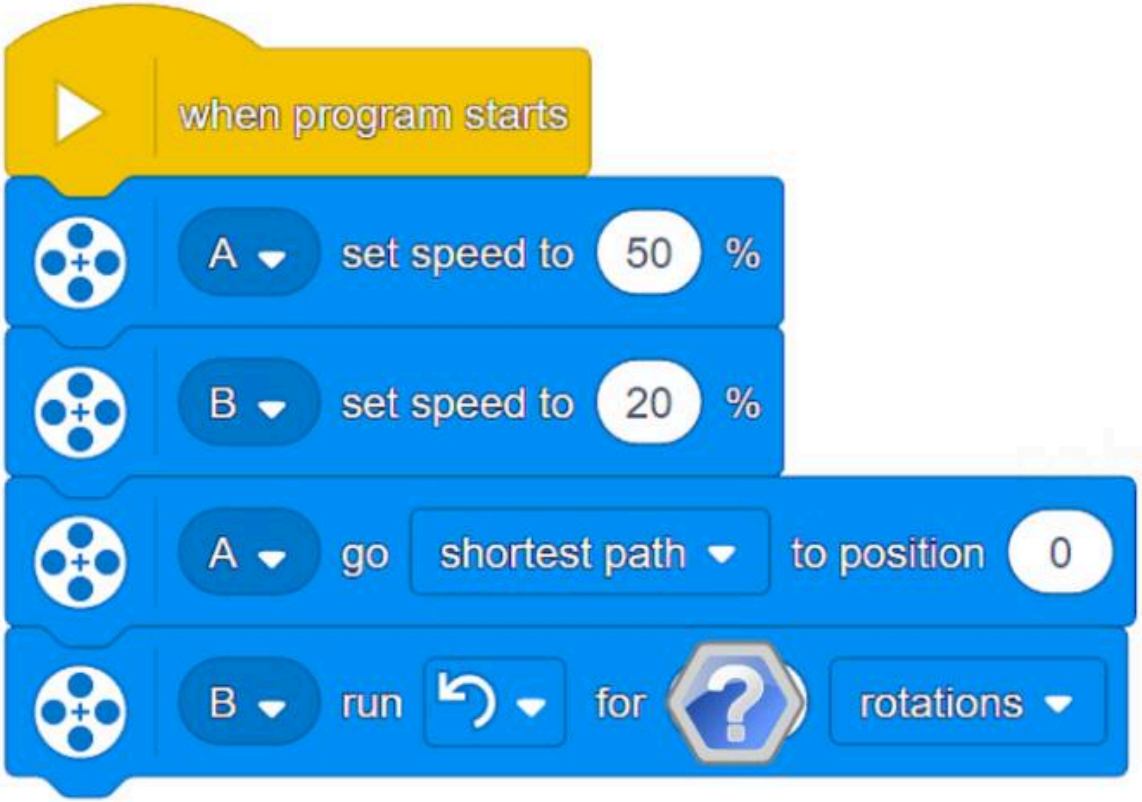
Use the motor blocks to program the robot's movement:

In ICON Blocks mode, the program looks like this:

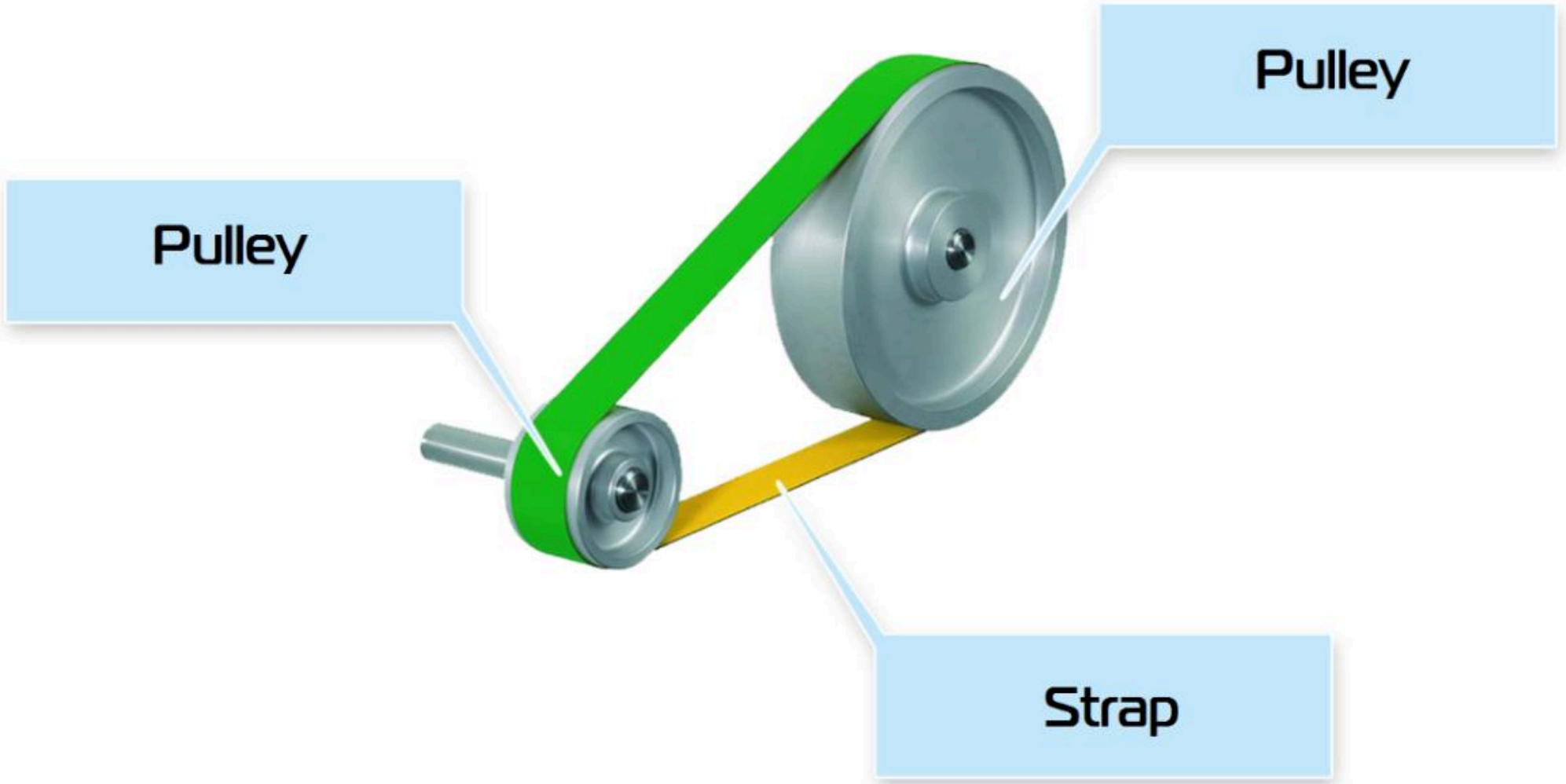


Use the motor blocks to program the robot's movement:

In Word Blocks mode, the program looks like this. Thanks to the possibility of programming both motors separately, the robot independently sets the fork of the front wheel in the central position

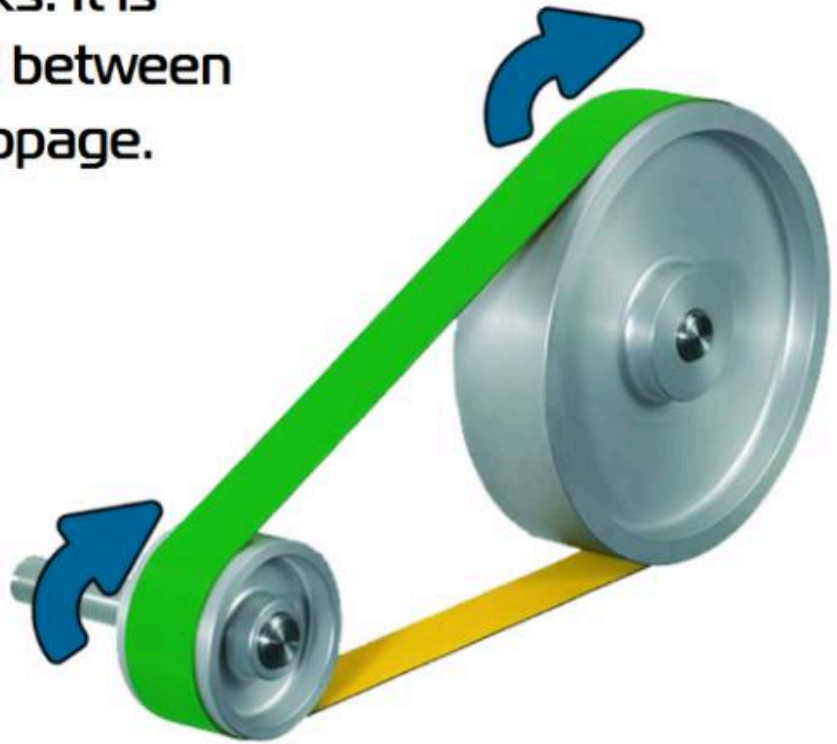


Belt transmission is widely used to transmit rotation. It consists of at least two pulleys and a belt. Such a transmission has minimal vibrations and noise, but cannot transmit much power. The speed of rotation of the pulley depends on the ratio of the diameter of the pulleys.



Pay attention! Because the belt can slip, the output shaft may not always rotate as you expect it to. For example, a heavy robot may not move at all, although if it is picked up, the wheels will turn.

The pulley has a recess in which the belt works. It is made in order to increase the area of contact between the belt and the pulley and thereby reduce slippage.



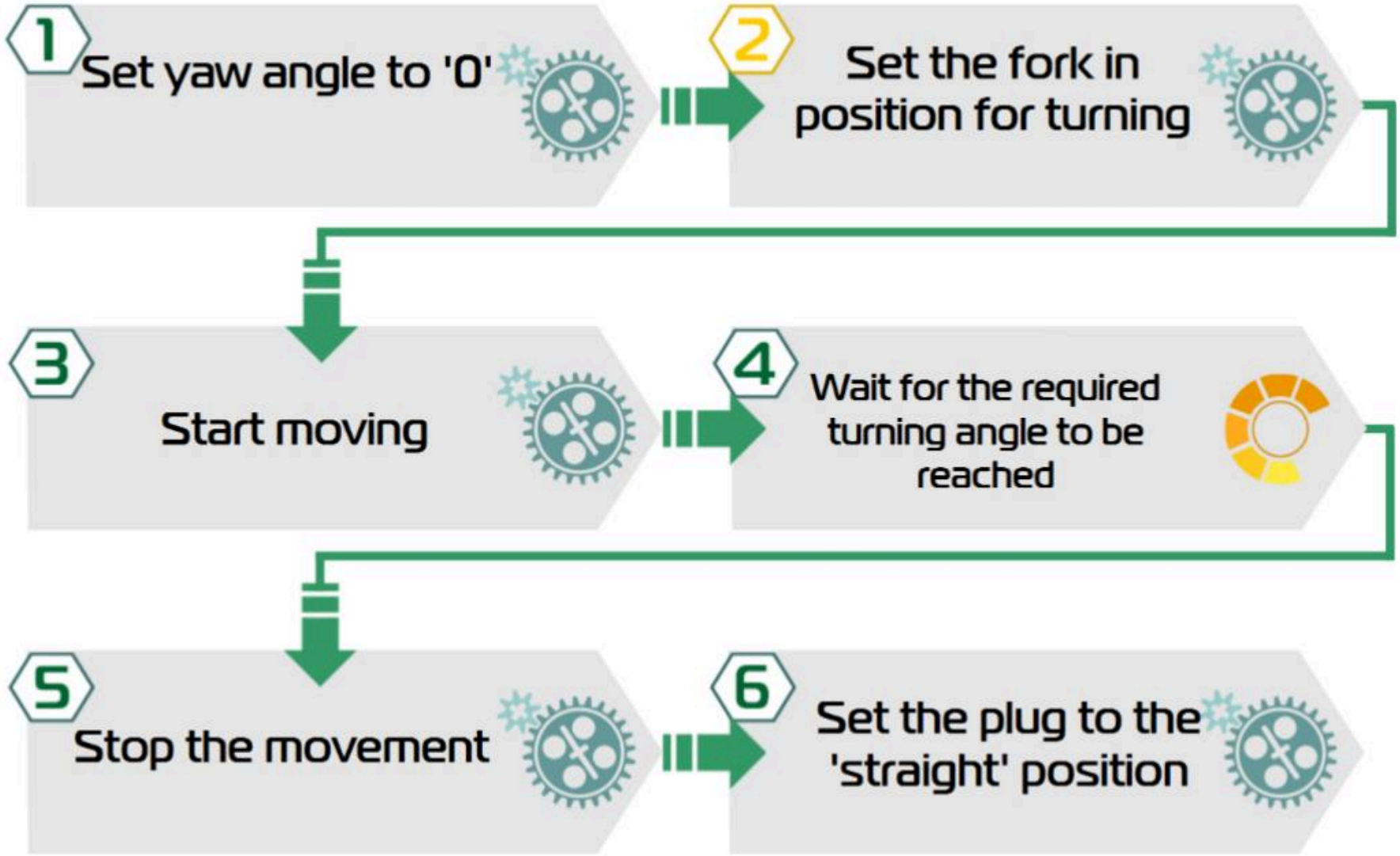
When using a belt transmission, it should be taken into account that the direction of rotation of the input and output axes coincide.
It is because of these features that your robot travels less than you expected during testing.

Estimate drive lengths in your field, taking into account corner offsets, write and test a program for your robot

In ICON Blocks mode, the program looks like this:

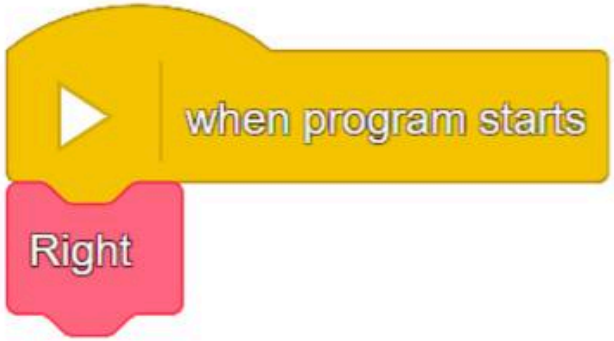


The program should work according to the following algorithm:

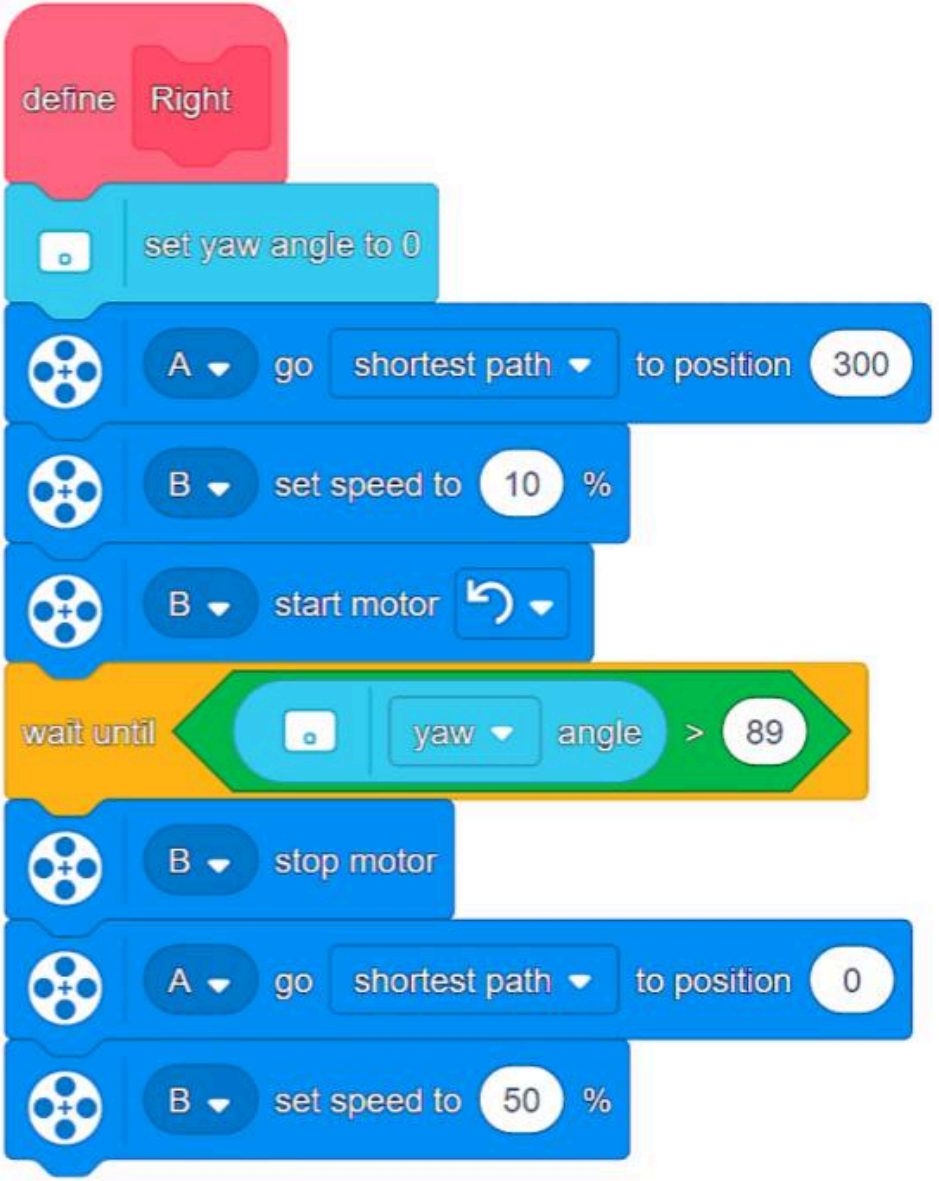


Program and test the right turn program.

In Word Blocks mode, programs will look like this.



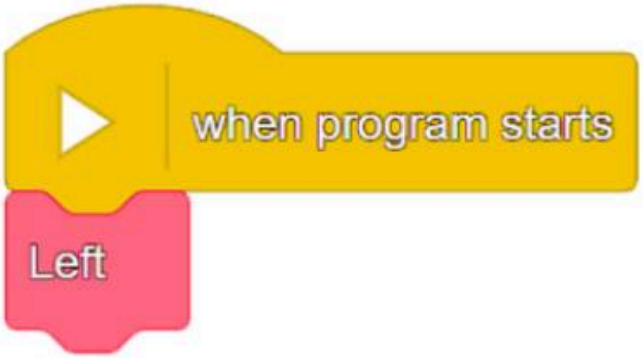
Use routines for this task. They allow you to allocate part of the program in a separate block and then use it in a simple and compact form whenever you need to perform the same actions.



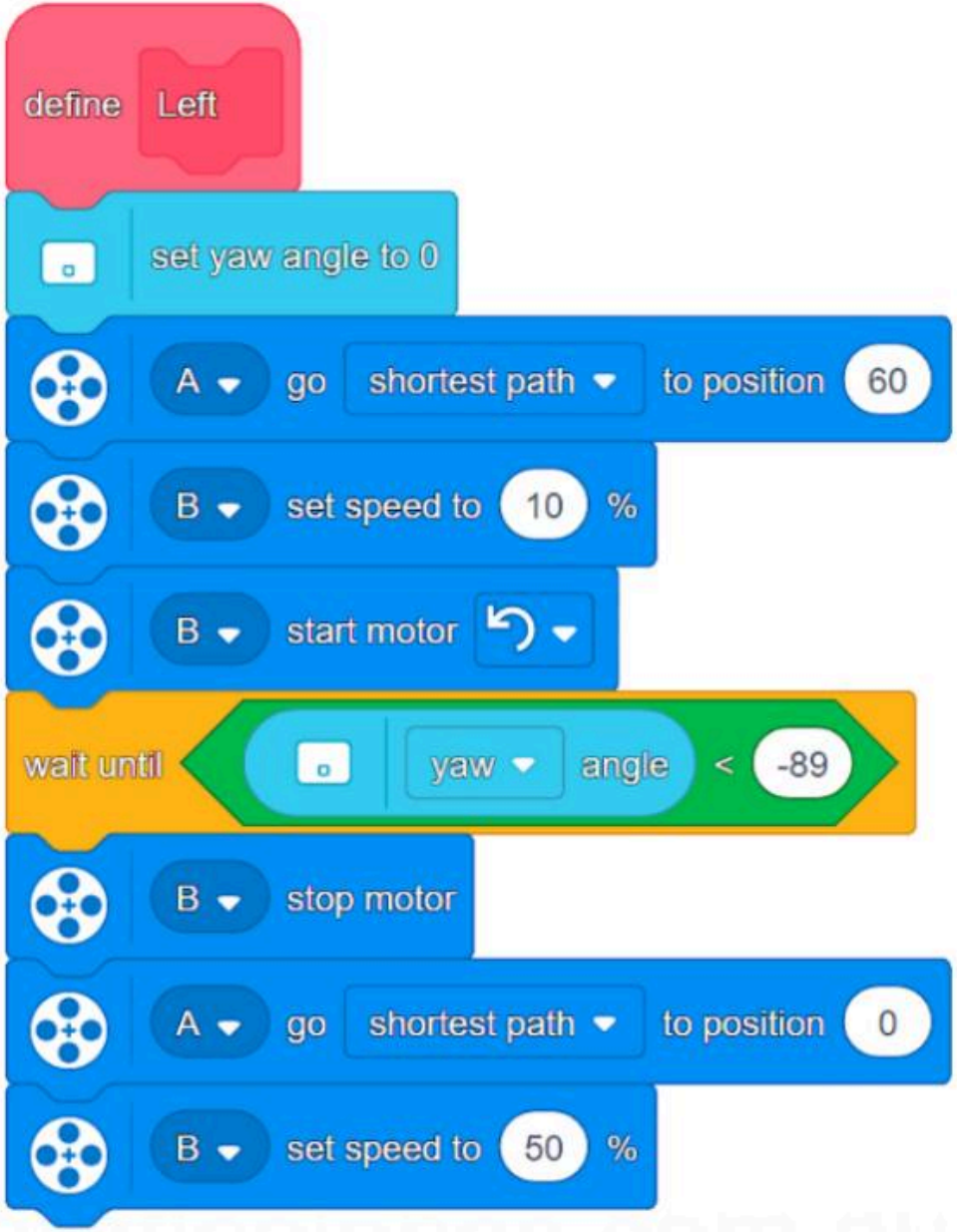
obotics@littlealphas.c

Program and test the right turn program.

In Word Blocks mode, programs will look like this.

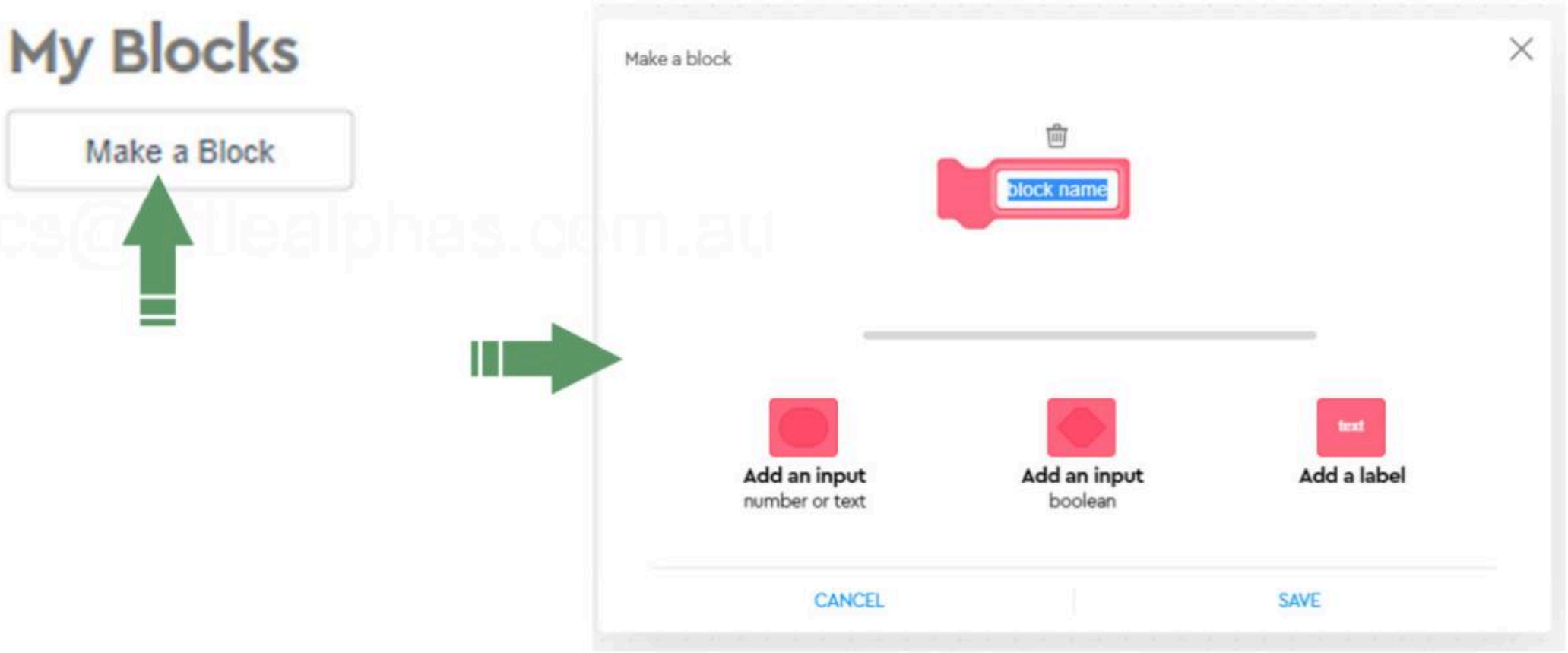


Also, subroutines are very convenient for configuring the robot. It is enough to make changes in one place - and they will be applied everywhere, where this routine was used



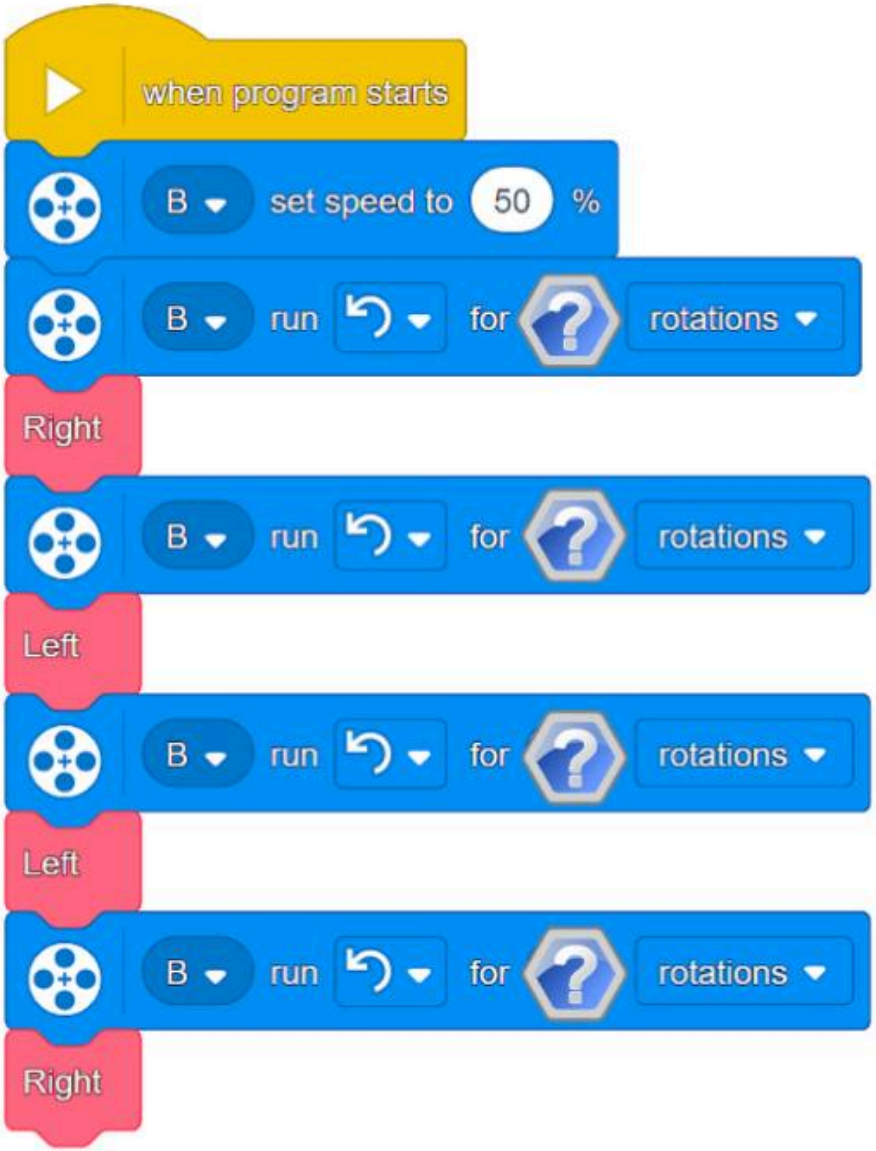
robotics@melbpriats.com.au

When creating a subroutine, you can choose its name and optionally add parameters that will be used to configure the operation of the blocks inside the subroutine.



Program and test your robot.

In Word Blocks mode, the program can look like this:



What transmission type does your robot have?



Worm drive

Gear drive

Belt drive